KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours | FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) As an application developer, why do you choose to run an application over UDP rather than TCP. (06) [CO1]
   b) Explain the concept of piggybacking in data communication. Discuss how it affects network congestion. (05) [CO2]
   c) Discuss how TCP detects congestion in a network. Describe the congestion control techniques used in TCP to manage and avoid congestion. (12) [CO3]
   d) An organization's network is connected to the internet through a 25 Mbps access link. Users request objects of average size 800 Kbits at 30 requests/second, with average internet delay 1.5 seconds. The organization's internal LAN has a speed of 1 Gbps. To reduce response time, the organization considers: (12) [CO4]
      Option 1: Upgrade the access link to 100 Mbps.
      Option 2: Install a local web cache that serves 40% of requests.
      Compare the average end-to-end delay for both options and recommend which provides better performance and is more cost effective.

2. a) Explain how a TCP connection is established, detailing the role of SYN and ACK flags. Justify why a two-way handshake is not sufficient for reliable communication. (08) [CO2]
   b) Describe why flow control is required in TCP implements flow control between sender and receiver, specifying the conditions that must be met, and illustrate your answer with a suitable example. Additionally, explain the concept of silly window syndrome. (13) [CO3]
   c) In TCP, *EstimatedRTT*, *DevRTT* and *TimeoutInterval* adapt to changing network conditions. Suppose, three successive *SampleRTT* measurements are recorded as 110 ms, 95 ms, and 120 ms. Given: initial $EstimatedRTT = 100\ ms$, initial *DevRTT* = 5 ms, $\alpha = 0.125$, and $\beta = 0.25$ (14) [CO3]
      i) Compute *EstimatedRTT*, *DevRTT* and *TimeoutInterval* after each *SampleRTT*
      ii) Explain why it is not advisable to use the RTT of retransmitted packets when updating *TimeoutInterval*

3. a) Discuss why DNS is used in networking. Illustrate the hierarchy of DNS servers and describe how a DNS query is resolved. (09) [CO2]
   b) Explain the working of the Selective Repeat (SR) protocol. Given a sequence number space of 4 and a sender window size of 3, determine whether the window size is appropriate for SR protocol. Discuss the problems that can occur if the window size exceeds the allowed limit with a timeline diagram. (13) [CO3]
   c) Are IP address alone sufficient for communication between applications? Suppose, you have a web server that hosts three web applications on ports 3000, 5000, and 8000, but for some reasons users can't access them via standard HTTP/HTTPS requests using the domain name. Identify the underlying issue and recommend an appropriate solution to enable access via standards ports. Illustrate your approach with a request-routing diagram. (13) [CO4]

4. a) While searching for new pair of shoes, you visited multiple e-commerce sites and may or may not have completed a purchase. Later, when browsing a social media platform, you noticed advertisements for the same types of shoes you had viewed earlier. Explain the underlying mechanism that enables such personalized advertisements across different websites. (10) [CO3]
   b) Explain the rules for generating acknowledgements in TCP and discuss the fast retransmit mechanism. Illustrate your answer with suitable examples. (14) [CO3]
   c) Describe how man-in-the-middle attack hinders the authentication process (ap5.0) that uses a nonce and public key cryptography. (11) [CO3]

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SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain the necessity of a trunk port in VLAN. Compare "Router-on-a-Stick" and "Layer 3 Switch (SVI)" based on following criteria: (10) [CO1]
   i) Scalability
   ii) Performance/Latency
   iii) Implementation Complexity
   iv) Cost

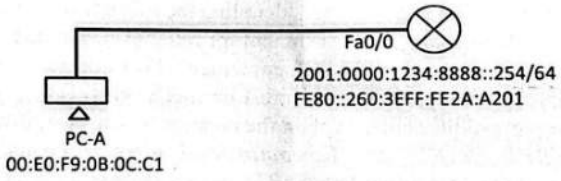
   b) Explain HOL Blocking. Analyze the major switching techniques with their benefits and drawbacks. (10) [CO1]

   c) Suppose a TCP message that contains 2048 bytes of data and 20 bytes of TCP header is passed to IP for delivery across two networks of the Internet (i.e., from the source host to a router to the destination host). The first network uses 14 bytes header and has an MTU of 1024 bytes; the second network uses 8 bytes header with an MTU of 512 bytes. Give the sizes and offsets of the sequence of fragments delivered to the network layer at the destination host. Assume all IP headers are 20 bytes. (15) [CO3]

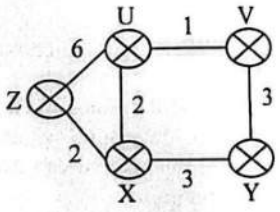
6. a) "IPv6 is faster and more secure than IPv4" – justify the statement based on IP datagram header. (05) [CO2]

   b) What are the differences between IP based forwarding and generalized forwarding? Explain simple packet handling rules for generalized forwarding based on OpenFlow protocol. (10) [CO3]

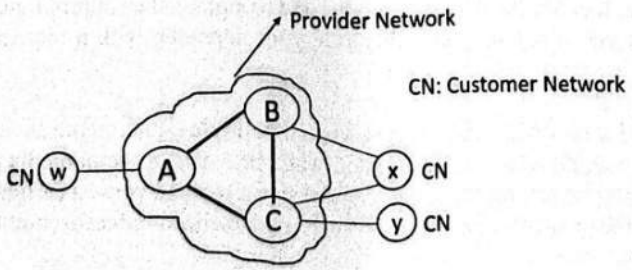
   c) What is link-local address in IPv6? Describe the process of Router Solicitation Message showing IPv6 header and ICMPv6 header values to get an IP using Stateless Auto Configuration mode. Calculate the IP of PC-A during this process shown in following figure. (20) [CO4]



7. a) Consider the network shown in following figure and assume that each node initially knows the costs to each of its neighbors. Consider the distance vector algorithm and show the distance table entries at node Z. (12) [CO3]



   b) What is Hot Potato routing? Describe the policy-based path selection for the following scenario. (10) [CO2]



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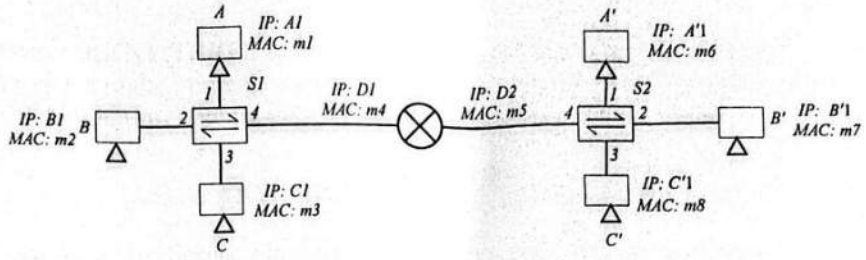
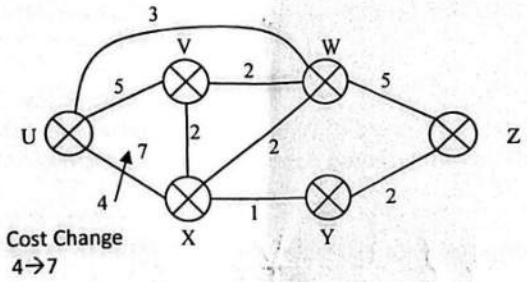
c) You have to look carefully the following figure. Computer A wants to send a frame to Computer C'. Describe the two switches' behavior when A connects to C' and after completing the communication, how the switch table of both switches look like? Initially switch table was empty. Finally, write down the self-learning algorithm of switch. (13) [CO2]



8. a) Explain Layer 2.5 protocol-based packet forwarding operation using an example. (10) [CO3]
   b) Emulate the ~~Software~~ Define Network (SDN) controller's interaction with local switches to modify the link-state changes occurred in the following network. Apply Dijkstra's link-state algorithm in the SDN controller to calculate the updated routing table for 'U'. (17) [CO4]



   c) State how packet filtering firewall works. (08) [CO1]

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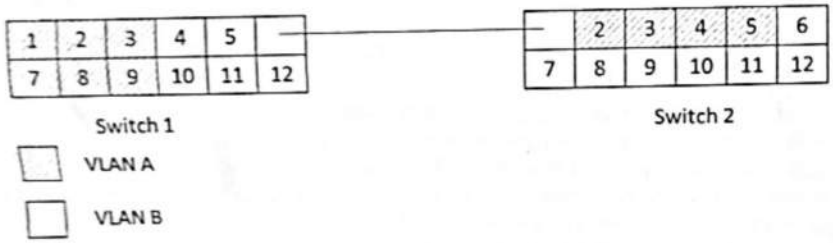
SECTION A
(Answer **ANY THREE** questions from this section in Script A)

1. a) Explain the purpose of OSI model. Do all devices present in a network implement all OSI layers? Justify your answer. (10) [CO1]

   b) In the network shown in following figure, the server 223.1.2.5 receives an IP packet with src: 0.0.0.0, 68 and dest: 255.255.255.255, 67. What would happen next? Complete the interaction. (12) [CO2]

   c) Explain the benefits of Port-based VLANs. Explain the process and switch table to transmit a frame from port 2 of switch 1 to port 4 of switch 2, shown in following figure. (13) [CO3]

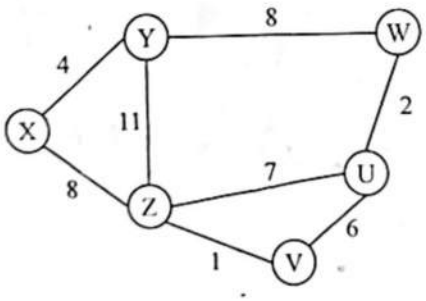


2. a) Consider the following 20-byte header (IPv4). Find the header checksum. Why does IPv4 header checksum cover only data portion? (10) [CO1]

4	5	0	28
1		0	0
4	17	0	
10.12.14.5			
12.6.7.9			

   b) Explain the hot potato routing protocol. Briefly explain the benefits and drawbacks of this protocol. (12) [CO3]

   c) Using a Link-State routing algorithm, determine the routing table at the router X from the topology given in following figure. (13) [CO2]



3. a) An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to 3 group of customers: (13) [CO2]
   i) 1st group: 52 customers, each needs 128 addresses.
   ii) 2nd group: 40 customers, each needs 256 addresses.
   iii) 3rd group: 100 customers, each needs 64 addresses.

   How many addresses are still unallocated? Note down the network addressed of each group.

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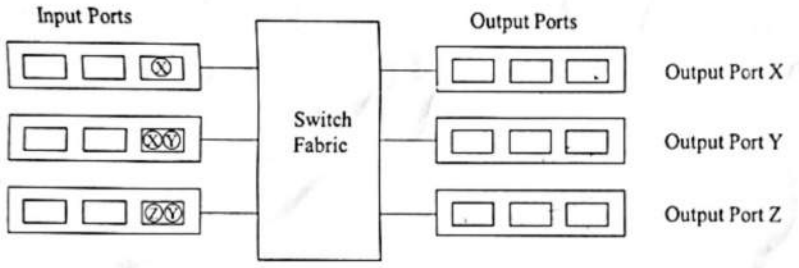
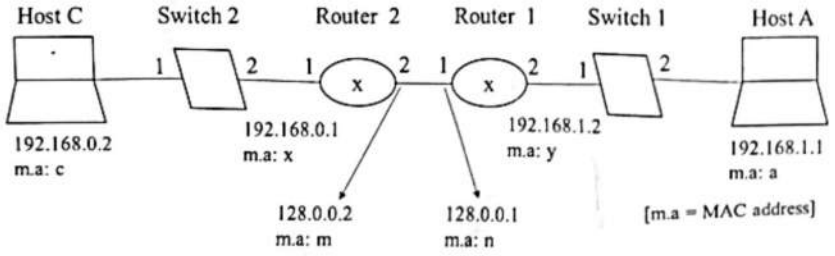
b) Suppose you have a broadcast channel in Ethernet CSMA/CD algorithm with transmission rate 10 Mbps where a node can transmit a frame of max 1500 bytes, and maximum propagation delay between 2 nodes is: 2ms. (10) [CO]
   i) Calculate the efficiency of the algorithm.
   ii) When would it reach its highest efficiency, and why?

c) In the switch shown in following figure, all datagrams have the same fixed length and the switch operates in a slotted, synchronous manner. Assuming a datagram can be transferred from input port to output port in one time slot, what is the minimal number of slots needed for the transfer of all datagrams when: (12) [CO1]
   i) The switching fabric is memory.
   ii) The switching fabric is bus.
   iii) The switching fabric is crossbar.

   Briefly explain the process for each of these switching fabrics (You can assume any input queue scheduling order).



4. a) Define SDN control plane. Briefly explain the characteristics of SDN. (10) [CO1]
   b) State router oscillation problem with proper example. When it occurs? (10) [CO3]
   c) In the network shown in following figure, a data packet is routing from A to C and C sends a response to A. Assuming all hosts know the necessary IP addresses, MAC addresses and first hop router addresses, show the data capsulation and decapsulation in each device with necessary diagram(s) and table(s). (15) [CO2]



SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) List down the steps involved in SSL handshaking. Also, present a pictorial view of packet transmission. (10) [CO2]
   b) Is SMTP enough to build a complete Email architecture? Justify your answer. (10) [CO2]
   c) Suppose a sender and receiver are using a pipelined protocol, i.e., Go Back-N or Selective Repeat. In this specific case, the rate at which the sender process pushes data is less than the rate at which then ~~sender~~ receiver consumes data. We want to introduce a buffer in the receiving side. Is it a wise decision? Explain the reasoning behind your answer. (05) [CO3]
   d) Let, host at *abc.poly.edu* wants IP address for *defg.cs.uxyz.edu*. Explain the procedure of getting IP address using recursive queries for various DNS servers. (10) [CO2]

6. a) State the advantage of P2P over client-server architecture. Despite those ~~dis~~advantages, why do we find Client-server architecture applied in most of the real-life cases? (10) [CO1]
   b) Consider the following figure where Mr. X wants to access a webpage hosted on a Google server. This webpage comprises a base HTML file along with 10 JPEG images, each with a length of 1024 bits. The transmission rate of the link connecting Mr. X to the server is 1 Mbps. Determine the total response time for both Non-persistent and persistent HTTP by real internet delay. Mr. X measures real internet delay by traceroute program. (15) [CO2]

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```
C:\Users\HP>tracert google.com
Tracing route to google.com [142.250.193.78]
over a maximum of 30 hops:
  1    <1 ms    <1 ms    <1 ms  10.24.241.1
                                  .
                                  .
                                  .
 14    82 ms    80 ms    80 ms  142.251.54.83
 15    80 ms    79 ms    78 ms  del11s16-in-f14.1e100.net
Trace complete
```

c) Consider a reliable data transfer protocol that uses only negative acknowledgements. Suppose the sender sends data only infrequently. Would a NAK-only protocol be preferable to a protocol that uses ACK? Why? Now suppose the sender has a lot of data to send and the end-to-end connection experiences few losses. In this second case, would a NAK only protocol be preferable to a protocol that uses ACKs? Why? Justify your answer. (10) [CO3]

7. a) Suppose, a HTTP client is sending 1 bytes of data every 5 minutes. Let, 1 HTTP packet contains 1 byte of data. The HTTP server uses the data to update its database in real time. Help the network administrator to choose between persistent and non-persistent HTTP connections. Also, advise him to use or avoid pipelining. Explain your reasoning. (15) [CO3]

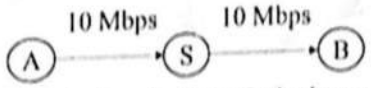
b) Suppose, in a TCP connection, the receiver process is slow at data consumption. Eventually, the receiver buffer is full and the receiver advertises zero window size. The sender halts sending. Whenever the receiver process consumes a packet, receiver sends an acknowledgement to the sender. Sender sends the next packet. Will this result in a problem? If yes, discuss about the reasoning and a possible solution. (11) [CO3]

c) State how packet filtering firewall works. (09) [CO1]

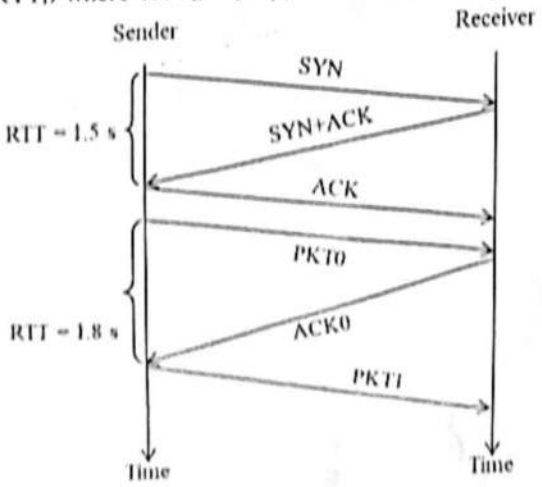
8. a) Let, a proxy server is used for web caching in case of accessing the URL *http://kuet.ac.bd/cse/home.html* and the browser of client is configured accordingly. Now, explain the scenario of accessing the web page. (10) [CO2]

b) Host A and B are each connected to a switch S via 10 Mbps link as shown in following figure. (10) [CO1]

The propagation delay on each link is 20μs. Assume signaling message for circuit setup and acceptance are 100 bit long and processing time of a signaling message at switch S, hosts A and B is 15μs. These signaling messages are transmitted over the 10 Mbps link. Calculate the total time required to transmit a 10,000-bit message from A to B for circuit switching.



c) In the following figure, some desired packet transfer is shown. Calculate Retransmission Timeout time (RTO) where applicable. Let, initial RTO=6s. Recall that, $RTO = RTT_s + 4\,RTT_D$ where $RTT_D = (1-\beta)\,RTT_D + \beta\,|RTT_s - RTT_M|$ (15) [CO3]



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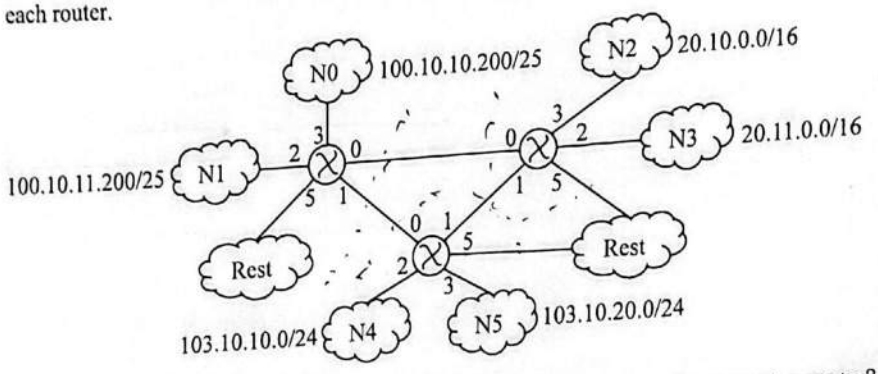
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define protocol with proper example. Do you think internet standards are necessary for defining computer network protocols? Why or why not? (10)
b) Let you are a network engineer at XYZ Inc. Your task is to design the forwarding tables of each router in a network. Consider the following network and design the forwarding tables of each router. (15)



c) Are there any relations between queuing and switching fabric speed in case of a router? Explain it. (10)

2. a) Define router oscillation problem with proper example. When it occurs? (11)
b) You are given an IPv4 datagram described in hexadecimal format: "4640 001C 00D1 00B9 1E11 DE0B 4AFA 0A14 C878 321E 0000 000F 0272 6F9A". Answer the following questions based on the given datagram (in decimal). (15)
i) Header length (in bytes) = ?
ii) Identifier = ?
iii) Fragment offset = ?
iv) Payload = ?
v) Is it a fragmented packet? If yes, does the packet has any specialty?
vi) Which fields will get updated if the packet is sent to the next hop router?
c) How does classless addressing solve the problem with classful addressing? (09)

3. a) Describe the services offered by network layer. Discuss the necessities of IPv6 over Ipv4. (10)
b) Why does collision occur in CSMA when we only send the packets if the channel sensed idle? Also, demonstrate the collision detection mechanism and algorithm to resolve the issue. (16)
c) "In BGP, packets are not routed to a specific destination address, but instead to CIDRized prefixes"—justify the statement. (09)

4. a) State how packet filtering firewall works. (06)
b) Demonstrate how NAT operates. What are the controversial things behind NAT? (09)
c) Explain how TLS functions. Mention at least one method of obtaining the four keys from the master key. (08)
d) Let you want to send an email to your friend securely. By ensuring the three most important properties of secure communication, demonstrate the scenario for safeguarding email. (12)

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SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) State the advantages of Peer-to-Peer architecture over Client-Server architecture. Despite those advantages, why do we find Client-Server architecture applied in most of the real life cases? (10)

   b) Should your ISP have its own cache server? Justify your answer. Explain how a cache server acts as both a client and a server. (10)

   c) Say, there is a HTTP client sending 1 byte of data every 5 minutes. Let, 1 HTTP packet contains 1 byte of data. The HTTP server uses the data to update its database in real time. Help the network administrator to choose between persistent and non-persistent HTTP connections. Also, advise him to use or avoid pipelining. Explain your reasonings. (15)

6. a) Does rdt 3.0 use same procedure in case of lost packet and lost acknowledgement? Explain with proper illustrations. (10)

   b) Is SMTP enough to build a complete Email architecture? Justify your answer. (10)

   c) Suppose, a client wants to start an online service. He has five computers ready for this job. These are connected to the internet with IP addresses 212.212.212.101–105. He picked "Network utopia.com" as the domain name for the online service. Your job is to make the service accessible by anyone on the internet. What will be your steps to accomplish that? Explain briefly. (15)

7. a) What are port addresses used for? Is there any constraint on how to assign port addresses or how many of them can be assigned at a time? If yes, then discuss about them. (10)

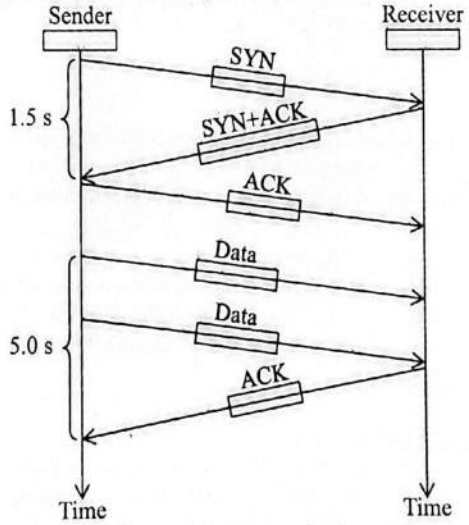
   b) Suppose a sender and a receiver are using a pipelined protocol, i.e., GBN or SR. In this specific case, the rate at which the sender process pushes data is less than the rate at which the receiver process consumes data. We want to introduce a buffer in the receiving side. Is it a wise decision? Explain the reasoning behind your answer. Also, say, the speed of transport layer at sender side is greater than the sender process pushing data to transport layer. In that case, is there a need for a sender buffer at transport layer? Justify your answer. (13)

   c) Let, a proxy server is used for web caching in case of accessing the URL *http://kuet.ac.bd/cse/home.html*, and the browser of client is configured accordingly. Now, explain the scenario of accessing the web page. (12)

8. a) Differentiate between flow control and congestion control. Explain the necessity of either of those in a reliable data transfer protocol. (10)

   b) In the following figure, some desired packet transfer is shown. Introduce a retransmission timer and then calculate and show Retransmission Timeout time (RTO) where applicable. Let, the initial RTO = 6 s. (12)

   Recall that, $RTO = RTT_s + 4RTT_D$, where $RTT_D = (1-\beta)\, RTT_D + \beta \,|\, RTT_s - RTT_M|$. Let, the value of α to calculate RTT_s is 0.125 and β is 0.25.



   c) Suppose, in a TCP connection, the receiver process is slow at data consumption. Eventually, the receiver buffer is full and the receiver advertises zero window size. The sender halts sending. Whenever the receiver process consumes a packet, receiver sends an acknowledgement to the sender. Sender sends the next packet. Will this result in a problem? If yes, discuss about the reasoning and a possible solution. (13)

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SECTION A

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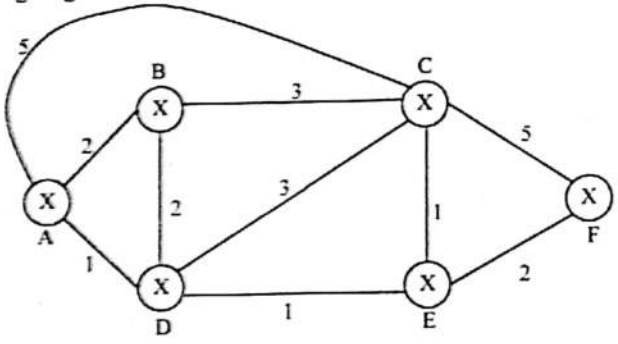
1. a) Briefly discuss about client-server architecture. Discuss about the major challenges of P2P architecture. (12)
   b) "Socket is analogous to door" – Justify the statement. What transport service does an app need? (08)
   c) What are the components of cookies? Describe how cookies keep state using database with proper examples. (10)
   d) FTP control connection is "out of band". Explain. (05)
2. a) Let, host at *abc.poly.edu* wants IP address for *defg.cs.uxyz.edu*. Show, how it is done using recursive queries for various DNS servers. (09)
   b) Explain the necessity of mail access agent in email system. Compare and contrast the functionalities of POP and IMAP. (10)
   c) Consider the following 20-byte header (IPv4). Find the header checksum. Why does IPv4 header checksum cover only data portion? (10)

4	5	0	28	
1			0	0
4		17	0	
10.12.14.5				
12.6.7.9				

   d) How does conditional GET command improve performance of web proxy and local cache mechanism? (06)
3. a) Explain how DHCP works. What are the main benefits of DHCP? (09)
   b) Apply fragmentation on following information for IPv4 datagram. (13)
      i) Original Datagram: 5000 bytes. Header: 20 bytes.
      ii) MTU: 1st level 2000 bytes & 2nd level: 800 bytes.
   c) An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to 3 groups of customers: (13)
      i) 1st group: 52 customers, each needs 128 addresses.
      ii) 2nd group: 40 customers, each needs 256 addresses.
      iii) 3rd group: 100 customers, each needs 64 addresses. How many addresses are still unallocated? Note down the network addresses of each group.
4. a) Briefly explain how PAT works. Why is PAT used over NAT? (10)
   b) Determine the routing table/forwarding table at the router B from the following topology using a Link-state Routing Algorithm. (17)



   c) Concisely describe the error reporting messages of ICMP. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Describe the working principle of Ethernet's CSMA/CD protocol with a detail look on exponential back-off algorithm. (10)
   b) Write filtering/forwarding algorithm of a switch. List the actions with corresponding conditions that a switch may perform when it receives a frame. (09)
   c) What is multiprotocol label switching (MPLS)? How does it work? Describe MPLS functionality with a simple example. (12)
   d) Describe silly windows syndrome created by the sender. (04)

6. a) How does TCP handle congestion? Describe with proper state diagram and graph. (15)
   b) Write down the connection establishment and half close in TCP. (10)
   c) How retransmission timeout is calculated in TCP? In which situation Karn's algorithm is applied? (10)

7. a) Briefly explain how Secure Socket Layer (SSL) works on top of TCP and what services it provides. (10)
   b) What is the difference between RDT 2.2 and RDT 3.0? Describe the FSM of RDT 3.0. (12)
   c) Write short notes on (i) Persistent Timer and (ii) Keepalive Timer. (08)
   d) What should be the relation between window sizes and sequence number in case of Selective Repeat? Why do we need the relationship? (05)

8. a) What are the shortcomings of symmetric key cryptography? How are those solved in public key cryptography. (06)
   b) In symmetric key system, how do two entities establish shared secret key over a network? Explain a solution with suitable example. (10)
   c) How does Multiple Access with Collision Avoidance (MACA) solve hidden terminal problem in wireless network? Explain with example. (07)
   d) Suppose, A wants to send a packet to B. If A does not know B's MAC address then how does it determine this MAC address in the following two situations: (12)
      i) A and B are in the same network.
      ii) A and B are in the different network.

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2020
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 1.5 hours FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) What does protocol achieve? Differentiate between intranet and extranet. (10)
   b) Why do we need Web caching? Discuss about the consistency of Web caching. (10)
   c) What is P2P architecture. Describe what transport service does an app need? (10)

2. a) Compare among WWW, HTTP and Internet. Show the HTTP request and response message format. (10)
   b) Suppose, *Alice* (ID – "alice@net.edu") wants to send message to *Bob* (ID – "bob@denza.edu"). Show SMTP interactions between SMTP client and SMTP server for this communication. (20)

3. a) Consider the data bits $D = 11011011$ and Generator $G = 10110$. Calculate the Cyclic Redundancy Check bit R. (14)
   b) How does RSA provide digital signature mechanism? (10)
   c) Justify that – "Switch is more intelligent than hub". (06)

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Why do we need Network Address Translation (NAT)? Briefly explain how does NAT work. (10)
   b) Compare the contrast link-state and distance vector routing algorithms. (09)
   c) Suppose a source and a destination are connected through a switch and a router. If the source sends a message to the destination, then how do encapsulation and decapsulation happen in each layer of TCP/IP model of above-mentioned devices. Explain with necessary figure. (11)

5. a) Mention different components of Domain Name System (DNS). Explain various types of name server used in DNS by presenting their architecture and working mechanism. (11)
   b) What is the relation between window size and sequence number in case of selective repeat? Why this relation is needed? (09)
   c) List down the steps involved in SSL handshaking. Also present a pictorial view of packet transmission. (10)

6. a) Describe the technologies that are used for transitioning from IPv4 to IPv6. (06)
   b) What is Ethernet? Describe how does CSMA/CD protocol work with a detail look on exponential back-off algorithm. (12)
   c) Suppose, a company has 200 personal computers and 10 servers. The network administrator wants you to dynamically assign the IP address of those personal PCs. How personal computer get the IP address if you have a network address: 172.16.0.0/24. Also, state the dynamic range of IP addresses. (12)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

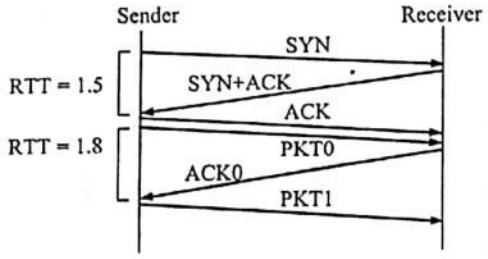
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is access network? Describe the enterprize access network with diagram. (08)
   b) How does data pass through network in case of circuit switching? Explain in brief. (08)
   c) What is troughput? What will be the troughput in case of Internet scenario? (06)
   d) What is the differences between RDT 2.1 and RDT 2.2? Draw RDT 2.2 finite state machine for sender and receiver. Briefly describe this. (13)

2. a) Why is UDP used in transport layer though it is unreliable? (05)
   b) In GB3, if every 5th packet is being transmitted is lost and if we have to send 10 packets, then how many transmissions are required? (11)
   c) What is the relation between window size and sequence number in case of selective repeat? Why is this relation needed? (10)
   d) Write short notes on the following terms: (09)
      (i) Demultiplexing, (ii) Characteristics of UDP, and (iii) UDP segment format.

3. a) What is "syn-flooding" attack in case of connection establishment in TCP? Explain. (06)
   b) How many ways are there to terminate a connection in TCP? Explain the ways in proper diagram. (12)
   c) Suppose receiver application program consumes data as 1byte at a time. For this, the receiver TCP announces the window size as 1byte. So, the sending TCP sends 1byte at a time and the traffic intensity will be high. What has to be done to solve the problem? Explain two solutions. (06)
   d) Explain Karn's algorithm in case of TCP timer. What will be the RTO of PKT1 in the following figure? (11)



4. a) How does TCP recognize the network congestion? Explain the Reno approach of managing this congestion by using necessary illustration. (13)
   b) Explain block cipher and cipher-block chain algorithm in symmetric key cryptography. Why will we use cipher-block chain instead of block cipher? (14)
   c) In case of RSA, if primes are $p = 5, q = 7$, public key is (35,5) and cipher text $c = 18$, then decrypt the cipher text to get plain text. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Suppose two nodes start to transmit at the same time a packet of length L over a broadcast channel of rate R. Denote the propagation delay between the two nodes as d_{prop}. What there be a collision if $d_{prop} > \frac{L}{R}$? Why or why not. (08)
   b) Briefly describe that two dimensional parity checks can correct and detect a single bit error but a double bit error that can be detected but not corrected. (08)

c) Why is an ARP query sent within a broadcast frame? Why is an ARP repose sent within a frame with a specific MAC address? (07)

d) You have to look carefully Figure 5(d). Computer A wants to send a frame to computer C'. Describe the two switches' behavior when A connects to C' and after completing the communication, how the switch table of both switches look like? (Initially switch table was empty). Finally, write down the self-learning algorithm of switch. (12)

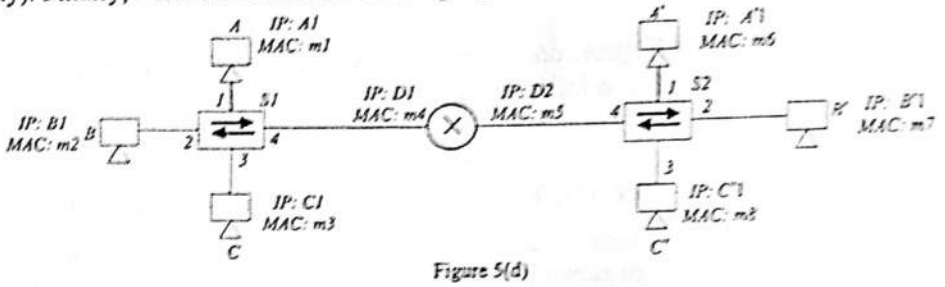


Figure 5(d)

6. a) Suppose you wanted to do a transaction securely from a remote client to a server as fast as possible. Would you use UDP or TCP? Why? (07)

b) What is meant by handshaking protocol? Why is it said that FTP sends control information "out of band"? (07)

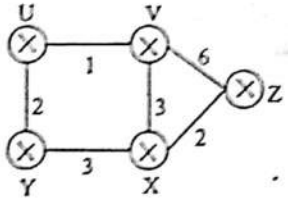
c) What are the advantages and disadvantages of a circular DHT (with no shortcuts)? (08)

d) Suppose the CSE department of KUET has 200 personal computer and 10 servers. The network administrator wants you to dynamically assign the IP address of those personal PC. How personal computer get the IP address if you have a network address 172.16.0.0/24. Also, state the dynamic range of IP addresses. (13)

7. a) Briefly describe different types of switching fabrics. Which, if any, can send multiple packets across the fabric in parallel? (06)

b) Define HOL blocking and longest prefix matching with appropriate example. (08)

c) Consider the network shown below and assume that each node initially knows the costs to each of its neighbors. Consider the distance vector algorithm and show the distance table entries at node Z. (12)



d) Differentiate GET and conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cache mechanism? (09)

8. a) What is NAT traversal problem? How can you solve it? Explain briefly with appropriate figure. (11)

b) Compare and contrast link-state and distance vector routing algorithms. (08)

c) Describe the technologies that are used for transitioning from IPv6 to IPv4 to IPv6. (07)

d) Consider the generator G = 10011 and suppose that D has the value 1010101010. What is the value of R? How can you detect the error using Cyclic Redundancy Check method? (09)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) State the problems of a stop-n-wait type reliable data transfer protocol. Design a reliable, pipelined data transfer protocol that uses no buffer for out of order packets. (15)
   b) Briefly describe the services provided by TCP. (09)
   c) Show the format of HTTP request line. Mention at least two request type used by HTTP version 1.1 and explain. (06)
   d) Explain common FTP commands. (05)

2. a) Consider transferring an enormous file of L bytes from host A to host B. What is the maximum value of L such that TCP sequence numbers are not exhausted? Assume an MSS of 1460 bytes. (05)
   b) The TCP of a host receives five segments at the time shown (HH:MM:SS.mmm) below. If it has no data to send, show the acknowledgements sent by the host using line diagram. (09)
   (i) at 0:0:0.000 (ii) at 0:0:0.027 (iii) at 0:0:0.400 (iv) at 0:0:1.200 (v) at 0:0:1.208
   c) Show a sample SMTP interaction between SMTP client and server to transfer the message "Good morning my friend" from alice@xxx.com to bob@yyy.com. (11)
   d) In TCP, what problems are solved by the following timers and how? (10)
   (i) Persistence timer (ii) Time-waited timer

3. a) Suppose a DNS client is looking for the IP address of www.kuet.ac.bd. It already sent a query message to the DNS server. Show the response message sent by the server. Use necessary DNS compression. Assume the IP address is 128.175.13.63. (10)
   b) Explain the necessity of mail access agent in email system. Compare and contrast the functionalities of POP and IMAP. (09)
   c) Since UDP is unreliable, why is it used at transport layer? (05)
   d) What do you mean by digital signature? How does RSA provides digital signature mechanism? (11)

4. a) (15)

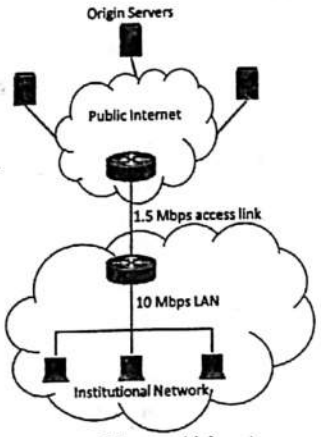


Figure 4(a)

Consider Figure 4(a), for which there is an institutional network connected to the internet. Let the average object size is 900 kbit and the average request rate from institutional browsers to the origin servers is 1.5 request per second. Also suppose that the amount of time it takes from when the router on the internet side of the access link forwards an HTTP request until it receives the response in two seconds on average. Model the total average response time as a sum of the average access delay and the average internet delay. For average access delay, use $\Delta/(1-\Delta\beta)$, where Δ is the time required to send an object over the access link and β is the arrival rate of objects to the access link.

i) Find the total average response time.
ii) Now suppose a web cache is installed in the institutional LAN. Let the hit rate is 0.4. Find the total response time.

b) In symmetric key system, how do two entities establish shared secret key over a network? Explain a solution with suitable example. (10)

c) List down the steps involved in SSL handshaking. Also present a pictorial view of packets transmission. (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) How does header checksum can be calculated in IPv4 datagram? Why does it cover only data portion? Explain with example. (12)

b) Define Computer Network. What are the major functionalities of Computer Network? (08)

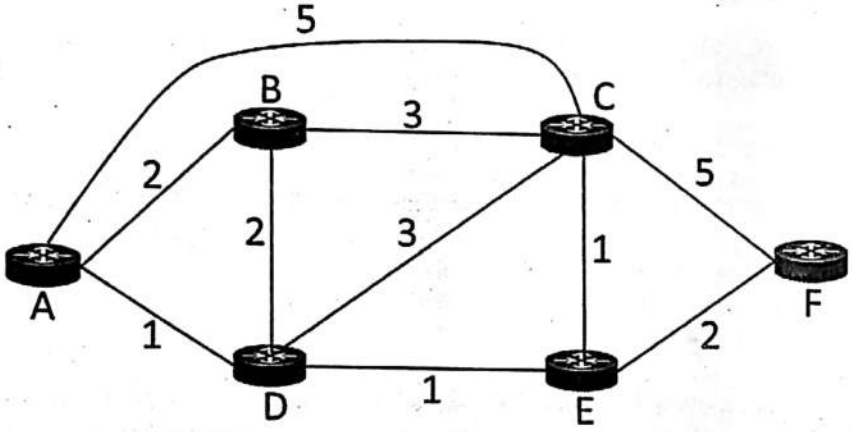
c) Apply Fragmentation on following information for IPv4 Datagram. (15)

i) Original Datagram : 5000 bytes, Header : 20 bytes.

ii) MTU : First level 2000 bytes and Second level 800 bytes.

6. a) Concisely describe the error reporting messages of ICMP. (10)

b) Determine the routing table/forwarding table at the router B from the following topology using a Link-State Routing Algorithm. (17)



c) "Switch is more intelligent than hub" – justify the statement. (08)

7. a) A router inside an organization receives a packet with the destination 190.240.33.91. How does it find the subnetwork address to route the packet? Assume the subnet mask is /19. (05)

b) Write a short note on Address Resolution Protocol (ARP). (09)

c) What is count to infinity problem? Use a suitable example to show that the poisoned reverse technique can solve the problem. (12)

d) Why do we need Network Address Translation (NAT)? Briefly explain how NAT works. (09)

8. a) Consider the data bits $D = 11011011$ and Generator $G = 10110$. Calculate the Cyclic Redundancy Check bit R. (12)

b) How does Slotted ALOHA random access MAC protocol work? Show the efficiency of it. (13)

c) Write down the services provided by the data link layer. (10)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2017
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

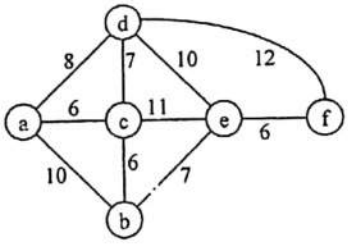
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define statistical multiplexing. Which one is better between statistical multiplexing and Time Division Multiplexing? Explain with an appropriate example. (12)
   b) Suppose source and destination are connected through a switch and a router. If source sends a message to destination then how do encapsulation and decapsulation happen in each layer of TCP/IP model of above mentioned devices? Explain with necessary figure. (11)
   c) Suppose A is transmitting a packet to B. They are directly connected through a physical link. The length of the link is 1 km, the length of the packet is L bits and the transmission rate is 1 Mbps. (i) Now calculate how many bits will be arrived at B in 50 μsec? (ii) If the packet length is 100 bytes then at what transmission rate R, the propagation delay would be equal to the transmission delay of packet? (12)

2. a) Consider sending a 2400 byte datagram into a link that has an MTU of 700 bytes. Suppose the original datagram is stamped with the identification number 422. How many fragments are generated? What are the values in the various fields in the IP datagram(s) generated related to fragmentation? (10)
   b) Suppose the CSE department of KUET has 200 personal computers. The network administrator does not want to set the IP addresses statically. So, how the personal computers can get IP addresses? Explain steps of getting IP. (11)
   c) Consider the following network. With the indicated link costs, use Dijkstra's shortest path algorithm to compute the shortest path from *b* to all network nodes. Show how the algorithm works by computing a table. (14)



3. a) How does private IP addressing concepts resolve IPv4 address space problem? What are the new problems arises? Explain. (09)
   b) Suppose A wants to send a packet to B. If A does not know B's MAC address then how does it determine this MAC address in the following two situations: (13)
      i) A and B are in the same network.
      ii) A and B are in the different network.
   c) What is Ethernet? Describe the working principle of Ethernet's CSMA/CD protocol with a detail look on exponential back-off algorithm. (13)

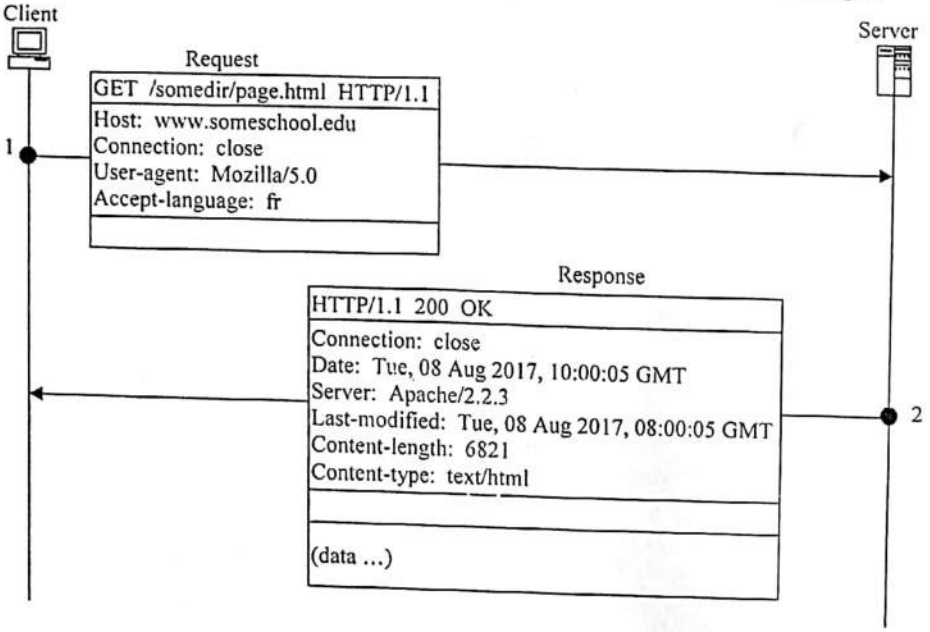
4. a) Describe the technologies that are used for transitioning from IPv4 to IPv6. (06)
   b) How does Multiple Access with Collision Avoidance (MACA) solve hidden terminal problem in wireless network? Explain with example. (08)
   c) How does IEEE 802.11 deal with data fragmentation? Discuss. (10)
   d) The CSE department of KUET just bought a Cisco switch which has 50 ports. The network administrator connected 40 personal computers to that switch. Now explain the learning mechanism of it. (11)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

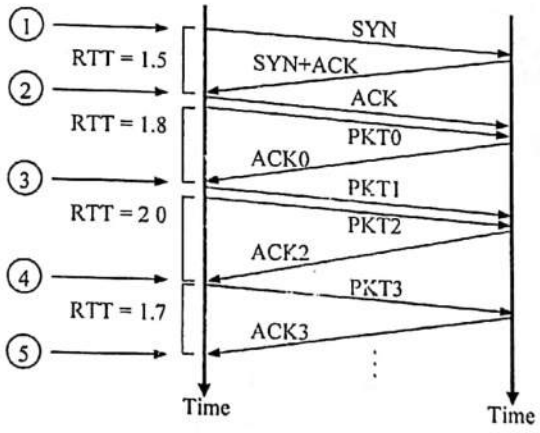
5. a) Compare among the Internet, WWW and HTTP. (09)
   b) What are the limitations of SMTP? How can these be solved by Multipurpose Internet Mail Extensions (MIME)? Briefly explain the working process and *Base64* encoding of MIME. (11)

c) Consider the following communication between a client and a server. The figure shows the different messages transferred between them. Explain each section as well as the statements that are presented in the messages. Also provide a general format of those messages. (15)



6. a) Mention different components of Domain Name System (DNS). Explain various types of name server used in DNS by presenting their architecture and working mechanism. (11)
   b) Design a stop-n-wait type reliable data transfer protocol by showing a finite state machine. Also present the line diagrams to explain its operation for various packet transmission scenarios. (14)
   c) How does selective repeat protocol work? Explain. (10)

7. a) How does TCP recognize the network congestion? Explain, with necessary illustration, the Reno approach of managing this congestion. (14)
   b) What is silly window syndrome? How does TCP handle this issue? (08)
   c) How does TCP set the timeout interval for retransmission timer? What are the Jacobson/Karels proposals for setting that interval? Consider the following communication scenario of TCP. Figure shows the sample RTTs in seconds. Compute the timeout intervals at the points shown by arrows i.e. at initial position and at the end of each acknowledge received. (13)



8. a) Design a protocol to authenticate a live entity. Can it be used for public key system? Why or why not? (12)
   b) Let you are a network administrator, design a rule set for your packet filter firewall to allow your local users only to browse outside and no other traffic except the locally initiated return traffic from outside is allowed to enter your network. (07)
   c) Briefly explain how Secure Socket Layer (SSL) works on top of TCP and what services it provides. (09)
   d) What are the basic differences between TCP and UDP? (07)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2016
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

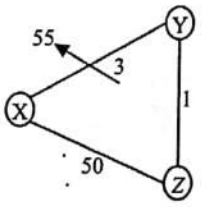
(Answer **ANY THREE** questions from this section in Script A)

1. a) What is layering in computer networks? Write down the major functions of network layer. (07)
   b) What are the advantages of packet switching over circuit switching? (06)
   c) Suppose a TCP message that contains 2048 bytes of data and 20 bytes of TCP header is passed to IP for delivery across two networks of the Internet (i.e., from the source host to a router to the destination host). The first network uses 14-bytes headers and has an MTU of 1024 bytes; the second user 8-byte header with an MTU of 512 bytes. Give the sizes and offsets of the sequence of fragments delivered to the network layer at the destination host. Assume all IP header are 20 bytes. (12)
   d) Differentiate between IPv4 and IPv6 header format. Explain the technologies that are used for transition from IPv4 to IPv6. (10)

2. a) Write down the pros and cons of static and dynamic routing. Explain link-state routing algorithm with a proper example. (13)
   b) What is DHCP? How does DHCP work? Write down the advantages and disadvantages of DHCP. (12)
   c) What does ICMP stand for? What does it do? Describe the error reporting messages of ICMP. (10)

3. a) "A transport-layer protocol provides logical communication between processes running on different hosts, a network-layer protocol provides logical communication between hosts."- Justify the statement. (08)
   b) What are the essential tools of RDT protocol? Describe the working mechanism of RDT 2.0. How can the performance problem of RDT be solved? (13)
   c) Describe the TCP timers in Brief. (08)
   d) Since UDP is unreliable, why is it used at transport layer? (06)

4. a) Consider the following graph. Assume that link (x,y) cost has been changed from 3 to 55. Show that the poissoned reverse technique can resolve the count to infinity problem in this case. (11)



   b) In theory TCP is fair, but in face it is not. Give an example. (07)
   c) What is the NAT traversal problem for P2P applications, say Skype? (05)
   d) Suppose the values of p and q are 7 and 13 respectively. Find suitable public and private keys for RSA algorithm. Verify the keys for the message "CSE". (12)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the techniques to resolve IPv4 address space problem? Explain in Brief. (10)
   b) What is digital signature? How authentication is provided using digital signature. (10)
   c) What is Diffie Hellman key exchange algorithm? Explain its working principle with an example. (10)
   d) What do firewalls do? Explain shortly. (05)

6. a) "Sockets are the software interface"-Justify the statement. (05)
   b) What are the transport services does an application need? Briefly describe them. (10)
   c) What is the benefit of stateless property of HTTP server? How HTTP servers identify users with stateless property? Explain briefly. (10)
   d) Differentiate GET and Conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cash mechanism? (10)

7. a) Why is it said that FTP sends control information "out of band"? (05)
   b) Suppose Alice, with a web-based email account (such as gmail) sends a message to Bob, who accesses his mail from his mail server using POP3. Discuss how the message gets from Alice's host to Bob's host. Be sure to list the series of application layer protocols that are used to move the message between the two hosts. (10)
   c) Write filtering/forwarding algorithm of a switch. List the actions with corresponding condition(s) that a switch may perform when it receives a frame. (08)
   d) What is service provided by DNS system? Explain load distribution briefly. (06)
   e) If all the links in the Internet were to provide reliable delivery service, would the TCP reliable delivery service be redundant? Why or why not? (06)

8. a) Consider an e-commerce site that wants to keep a purchase record for each of its customer. Describe how this can be done with cookies? (08)
   b) Why are content distribution networks (CDNs) important in internet system? Describe the role of DNS in CDN. (07)
   c) Write down the advantages and disadvantages of polling and Token passing MAC protocol. (09)
   d) Discuss Byte stuffing with an appropriate example. (06)
   e) Why a network interface card is masked as semi-autonomous device? (05)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2015
Department of Computer Science and Engineering
CSE 4105
Computer Networks

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

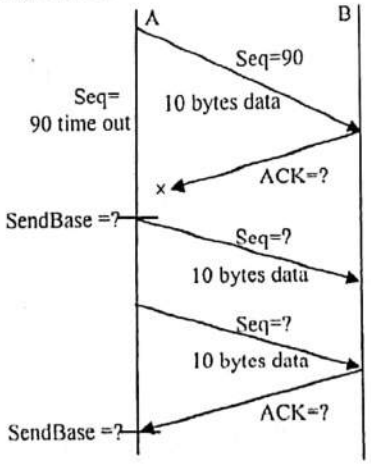
1. a) Define computer network and Internet. Explain Internet on service view points. (09)
   b) Why is it said that packet switching employs statistical multiplexing? Contrast statistical multiplexing with the multiplexing that takes place in TDM and FDM. (09)
   c) "Message segmentation reduces the end to end delay increasing resource usabilities"- Justify the statement. (08)
   d) Explain Internet structure with different tiers. What is the benefit of tierwise ISP connections for internet service? (09)

2. a) Why does TCP socket identification need 4-tuple instead of 2-tuple? What are those tuples? (08)
   b) Why UDP is called a bare-bone protocol in compare of TCP? (06)
   c) In respect of Rdt 2.0, briefly state the objective of the following used techniques: (i) Checksum (ii) ACK packet (iii) NAK packet. (09)
   d) Consider sending a file of 10KB packet with 1 Gbps bandwidth link and 15 msec end-to-end propagation delay. Answer the following questions: (12)
      (i) What is the required transmission time?
      (ii) What is the channel utilization (U)?

      Note that $U = \dfrac{\frac{L}{R}}{RTT + \frac{L}{R}}$

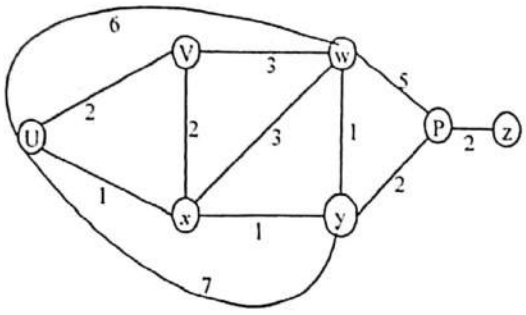
      (iii) For getting the channel utilization 0.5 what will be the transmission time?

3. a) What is the performance problem of Rdt 3.0 (stop-and-wait) protocol? How this problem can be resolved? (07)
   b) Sketch a diagram where slective repeat protocol experiences a dilemma. Give your opinion how this dilemma can be overcome. (10)
   c) Consider TCP's communication scenario between A and B. A and B have initial sequence number 90 and 180 respectively. A is sending a file to B with a packet size of 10 bytes in the following figure. Determine the values of seq#, ACK# and sendBase#. (08)

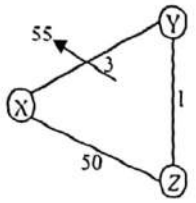


   d) What do you mean by NAT? Explain how NAT operation is performed. (10)

4. a) What is the HOL blocking in the switch fabric of a router? (05)
   b) Consider the following graph. Using Link-state algorithm find the shortest-path tree for router U. (12)



   c) Consider the following graph. Assume that link (x,y) cost has been changed from 3 to 55. Show that the poisoned reverse technique can resolve the count to infinity problem in this case. (11)



   d) Is RIP Intra-AS or Inter-AS routing protocol? Show a simple example of RIP routing table entries. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Briefly explain Nonpersistent HTTP and Persistent HTTP with/without pipeline. Compare TCP connections among the three HTTP methods for a HTML file containing 10 other objects' link. (10)
   b) Differentiate GET and conditional GET commands of HTTP. How does conditional GET command improve performance of web proxy and local cache mechanism? (08)
   c) Does web browsing require UDP service along with TCP? Explain briefly. (05)
   d) "Email demands both push and pull operations"-Justify the statement. (06)
   e) Compare SMTP, HTTP and DNS on the basis of direct transfer and transfer through intermediate hosts. (06)

6. a) Explain common FTP commands and replies. (07)
   b) Explain importance of DNS in service view points. Also explain iterative and recursive DNS queries with appropriate example. (10)
   c) How DNS and CDN together and alone provide better web browsing facilities? (08)
   d) Explain importance of Network Operations Communications Center (NOCC) in content Distribution Network, (05)
   e) "In computer networking switch is transparent to hosts but router is not"-is it true? Justify your answer. (05)

7. a) Why a network interface card is a marked as semi autonomous device? (05)
   b) Explain ideal Multiple Access (MAC) protocol scenario. Compare three different MAC protocols. (09)
   c) Distinguish slotted ALOHA and pure ALOHA protocols. (06)
   d) Write filtering/forwarding algorithm of a switch. List the actions with corresponding condition(s) that a switch may perform when it receives a frame. (09)
   e) Compare routing switches and hubs on operational viewpoints. (06)

8. a) Why does a computer need an MAC address, inspite of having an IP address? (05)
   b) "MAC address of a host never goes outside the network"-is it true? Justify your answer. (05)
   c) Briefly explain various desirable properties of secure communication. (06)
   d) What are shortcomings of symmetric key cryptography? How are those solved in public key cryptography? (07)
   e) What do you mean by digital signature? How does RSA provides digital signature mechanism? (12)

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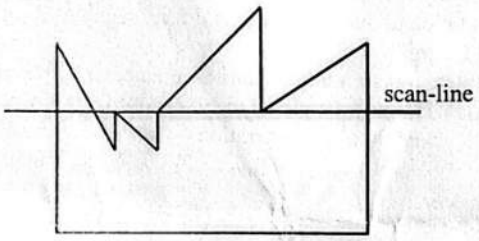
SECTION A

(Answer ANY THREE questions from this section in Script A)

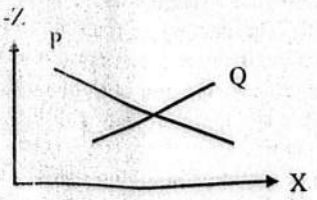
1. a) Suppose, you are designing a CRT monitor for displaying realistic scenes. Choose the appropriate combination of display type (vector/raster) and color production method (beam penetration/shadow mask) for the task and justify your choice. (10) [CO1]
   b) A line segment P_0P_1 is defined by the implicit equation $5x - 8y + 4 = 0$, where the end points are constrained such that $y_{start} = 8$ and $y_{end} = 13$. Using Bresenham's line drawing algorithm, determine the internal points. (12) [CO2]
   c) A graphics system renders two line segments: $L1\ (-1, 7)$ to $(11, 1)$ and $L2\ (5, 4)$ to $(9, 12)$. The visible area is restricted to a square shaped clipping window with $(x_{min}, y_{min}) = (1, 1)$ and $(x_{max}, y_{max}) = (9, 9)$. Using Liang-Barsky line clipping algorithm, determine the portion of the line segments that are visible within the window. (13) [CO3]

2. a) Explain how standard ray tracer be modified to generate soft shadow. (08) [CO3]
   b) Apply the Scan-Line Fill algorithm to demonstrate how the given polygon is filled along the specific scan-line. (10) [CO2]

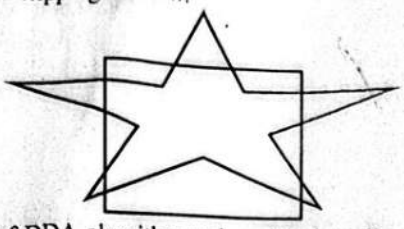


   c) A texture T of size 512×512 is mapped onto a planar surface. You are given a normalized coordinate point on that surface as $P(u, v) = (0.4, 0.6)$. Determine the corresponding texture coordinate $T(x, y)$ to be mapped on P and compute the final color at this point using GL_LINEAR texture filtering method. Given the color of the texels: $T(x, y) = (123, 205, 75)$, $T(x + 1, y) = (200, 150, 100)$, $T(x, y + 1) = (80, 220, 130)$, and $T(x + 1, y + 1) = (160, 90, 240)$. (17) [CO3]

3. a) Explain how the ray tracing method integrates the process of projection, scan conversion and shadow generation. (10) [CO2]
   b) A fractal is generated from an equilateral triangle. In each iteration, every side is replaced using the Koch snowflake construction, and each new edge is transformed into a Cantor edge of the same level. Now, draw the third-generation of the fractal and calculate the dimension of the structure. (12) [CO3]
   c) Given edge-on polygons P and Q in the $(x, -z)$ plane with overlapping (x, y) extents and assuming the view port looks towards the negative z axis. Apply depth sorting algorithm to examine the tests on the polygons and state the final rendering order of them. (13) [CO2]



4. a) Apply Weiler-Atherton polygon clipping algorithm to clip the star shaped polygon using the rectangle as a clipping window. (10) [CO1]



b) State the limitations of DDA algorithm and explain how Bresenham's line drawing algorithm overcomes them? (07) [CO1]

c) Consider a 3D scene where a point light source is located at the center of a sphere of radius 3 with center $C = (-1, 7, -4)$. A light ray emitted from the source intersects the sphere at point p, then strikes a triangle defined by $v_0 = (-5, 1, -5)$, $v_1 = (1, -1, -4)$, $v_2 = (0, 15, -7)$ with approximate surface normal $N = -1.04i + 1.77j + 9.79k$ at point q, and reflects towards the camera at origin $s = (0, 0, 0)$ passing through an image plane at depth 2. Trace the equivalent view ray from camera to the direction $d = (-1, 2, -2)$ and determine the intersection points p and q. (18) [CO3]

SECTION B

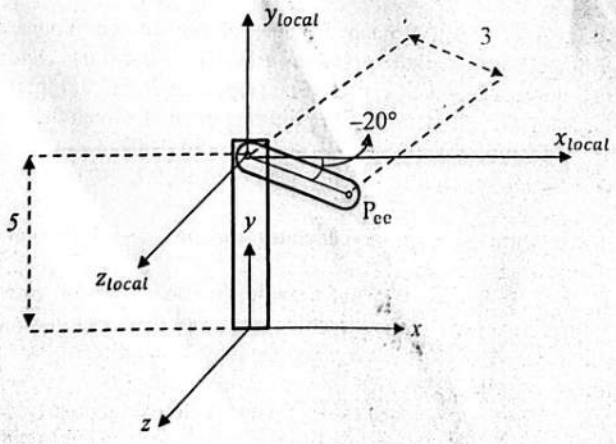
(Answer **ANY THREE** questions from this section in Script B)

5. a) Proof that 3D rotation matrix can be written as $R = \mathbf{I}\cos\theta + \sin\theta\,\mathbf{U} + (1 - \cos\theta)\mathbf{u}\mathbf{u}^T$, where $\mathbf{u}$ is unit axis of rotation, θ = angle of rotation, and $\mathbf{U}$ is skew symmetric matrix satisfying $\mathbf{U}\mathbf{v} = \mathbf{u} \times \mathbf{v}$. Be noted $\mathbf{U}^2 = \mathbf{u}\mathbf{u}^T - \mathbf{I}$. (18) [CO1]

b) Consider the following figure showing a 2 segment robotic arm fixed at origin. Segment 1 has length 5 and placed along y-axis. Segment 2 has length 3, attached to the end of segment 1, and initially it is oriented $-20°$ from its local x-axis. The entire assembly undergoes $+30°$ rotation about global z-axis. Later, segment 2 rotated by $+45°$ about the local y-axis of the joint. (17) [CO3]

Construct the complete transformation matrix sequence to represent the position of the end-effector tip. Also compute the global (x, y, z) coordinates of the tip.

[Hint: The complete operation can be defined as $P_{ee} = T_{global} \cdot T_{L_1} \cdot T_{joint-operation} \cdot T_{init_1} \cdot T_{L_2} \cdot P_{origin}$ where T's are appropriate transformation matrices.]



6. a) A camera in World Coordinate System (WCS) is defined by eye $(7, 8, 9)$, look-at $(1, 1, 1)$, and up $(0, 1, 0)$. The camera is translated (slid) by $(1, -1, 1)$ in its own coordinate system. Determine the new camera origin in WCS after the slide. (13) [CO3]

b) You are given a point $P(2, 4, 5)$, and a projection plane defined by equation $x - 2y + 2z = 3$. Compute isometric projection of the point on the plane. Also, show the isometric projection matrix. (10) [CO1]

c) Derive the general perspective transformation matrix. Verify that it maps (r, t, n) to $(1, 1, -1)$. Symbols have their usual meaning. (12) [CO2]

CamScanner

7. a) State how Ward illumination model estimates anisotropic specular reflection. (07) [CO1]

b) A point $P(1,2,3)$ lies on a surface with normal $\boldsymbol{N} = (0,0,1)$, viewer is at $(1,2,8)$. A monochromatic spot light is positioned at $(4,5,6)$ with light focus vector $\frac{1}{\sqrt{3}}(-1,-1,-1)$ and cutoff angle 30°. Determine whether point P lies inside the spot-light cone. If so, compute the angular attenuation factor with fall off exponent = 2. [Note: Be careful about light vector and focus vector's direction.] (08) [CO2]

c) A triangle in 3D space has vertices and normals: $A(1,1,1), N_A(0,-4,3); B(5,1,1), N_B(0,-3,3); C(3,4,5), N_C(1,-4,2)$. A point $P(3,2,2.33)$ lies inside the triangle. Light and view parameters are: L_{pos} $(6,6,8)$, Eye_{pos} $(6,-2,5)$, $I_L = 1$. You are supposed to render the triangle using Phong shading method, and compute intensity by applying Cook-Torrance illumination model. Compute lerped normal at P using Barycentric interpolation. Then, compute specular intensity at P if surface roughness $m = 0.4$, Fresnel base reflectivity $F_0 = 0.1$. Following equations are given for your convenience, where symbols have their usual meaning. Be sure to clamp negative dot products. (20) [CO3]

$$I_{spec} = I_L \cdot f_r \cdot (N \cdot L), \quad f_r = \frac{DFG}{4(N\cdot L)(N\cdot V)},$$

$$\text{Beckmann Distribution } D = \frac{1}{\pi m^2 (N\cdot H)^4} e,$$

$$\text{Schlick Fresnel } F = F_0 + (1 - F_0)(1 - V \cdot H)^5,$$

$$\text{Cook-Torrance Geometry } G = \min\left(1, \frac{2(N\cdot H)(N\cdot V)}{V\cdot H}, \frac{2(N\cdot H)(N\cdot L)}{V\cdot H}\right).$$

8. a) Neatly write shooting algorithm to compute radiosity of a patch. Why does it converge faster than gathering method? (10) [CO1]

b) A tensor product Bézier surface patch $S(u,v)$ is defined as cubic in u direction and quadratic in v direction. (13) [CO3]

i) Write fully expanded matrix equation to compute $S(u,v)$. Your answer must explicitly show the contents and dimensions of vectors U, V, the blending matrices B_i, B_j and geometry matrix P_{ij}.

ii) Using the components from your matrix expression, write the specific 2D basis function $B_{2,1}(u,v)$ that is used for control point $P_{2,1}$. [1D Bernstein polynomial of degree n at index k is $B_{k,n}(t) = \binom{n}{k} t^k (1-t)^{n-k}$]

c) Consider a quadratic B-spline curve (order $k = 3$) is defined by 4 control points $P_0, \cdots, P_3$ and a non-uniform knot vector $T = \{0, 1, 3, 4, 5, 6, 7\}$. Using the Cox-de Boor recursive formula given below, derive the exact algebraic expression for the basis function $B_{1,3}(u)$. (12) [CO2]

$$B_{k,1}(u) = \begin{cases} 1, & \text{if } u_k \le u \le u_{k+1} \\ 0, & \text{Otherwise} \end{cases}$$

$$B_{k,d}(u) = \frac{u - u_k}{u_{k+d-1} - u_k} B_{k,d-1}(u) + \frac{u_{k+d} - u}{u_{k+d} - u_{k+1}} B_{k+1,d-1}(u), \quad d > 1$$

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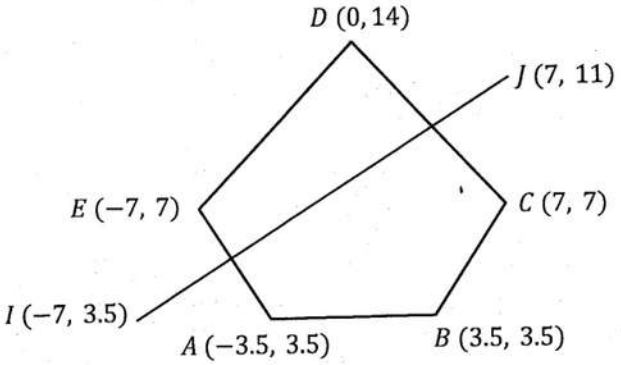
SECTION A

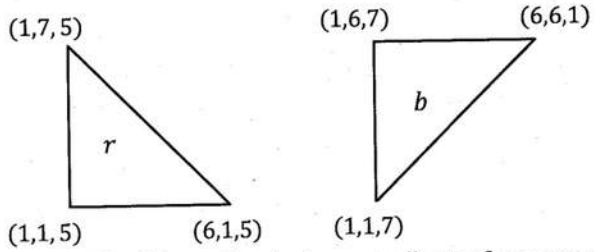
(Answer **ANY THREE** questions from this section in Script A)

1. a) Suppose you are given a task to choose a display system for your project. It needs to be able to draw millions of lines, triangles or characters on the screen. Which display system will you choose? Explain your reasonings. (08) [CO1]
   b) Describe the types of fractals that can be found in the real world. (09) [CO1]
   c) Figure below (not drawn to scale) shows a clipping polygon defined by five vertices. A line IJ is defined by two vertices I and J. Clip the line IJ against the clipping polygon using Cyrus-Beck algorithm and draw the final result after clipping. (18) [CO2]



$$t = \frac{N_i . [P_{E_i} - P_0]}{N_i . D}$$

2. a) What steps are required to scan convert an ellipse using the trigonometric method. (10) [CO2]
   b) The following figure shows two triangles and its vertices. Using Z-buffer, calculate the color of the pixels of color buffer. Show the depth buffer in each step. Assume that the color buffer has 8 (eight) rows and columns. The first triangle is red, denoted by *"r"* and the second one is blue, denoted by *"b"*. Write *"r"* or *"b"* to denote color red and blue, and *"bc"* for background color. (15) [CO2]



   c) Suppose, you are given a task of generating texture coordinates for an actual 3D model of earth. Describe two ways of achieving your goal. (10) [CO1]

3. a) Write an algorithm to determine if a ray intersects a triangle or not. Be sure that it is not the plane containing the triangle but the triangle itself. (12) [CO2]
   b) A line P_0P_1, described by two points $P_0 = (21.2, 12.3)$ and $P_1 = (28.8, 15)$, is given. You have to rasterize the line using Bresenham's line drawing algorithm. Calculate five consecutive points on the line starting from P_0 [excluding P_0]. (15) [CO2]
   c) "On a raster display, diagonal lines feel dimmer than a horizontal or vertical line." Do you agree with the statement? Explain your reasonings. (08) [CO1]

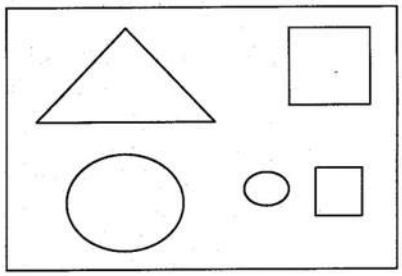
4. a) Describe how mipmapping achieves anti-aliasing. (09) [CO1]

b) Suppose there is a scene consisting of two opaque spheres. A red sphere with center $C_r = (2, 4, -10)$ and radius $R_r = 5$, and a blue sphere with center $C_b = (4, 8, -20)$ and radius $R_b = 3$. We want to calculate the color of a pixel which have direction vector $d = (0.18, 0.36, 0.91)$from the camera origin. Calculate the color of that pixel. For simplicity let's assume there is only white ambient light in the scene. The ray tracing tree shouldn't be deeper than two levels. The background color is black. (20) [CO3]

For your convenience here are some formulas, where the symbols have their usual meanings in this context.

$$t = \frac{-B \pm \sqrt{B^2 - 4C}}{2}; B = 2(s - c).d; C = |s - c|^2 - R^2; r = u - 2(u.n)n$$

c) The following figure shows a 2D slice of a 3D scene. Draw the approximate output of applying (i) KD tree (ii) Octree algorithm to divide the scene into multiple subspaces. (06) [CO2]



SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Derive the scaling matrix for scaling an object by a scale factor 3 along an arbitrary direction given by vector $u = (1, 2)$ rooted at $(4, 3)$. (08) [CO1]

b) Use quaternion basics to show that (13) [CO2]

$$\boldsymbol{qPq}^{-1} = (s^2 - \boldsymbol{v}^2)\boldsymbol{P} + 2s\boldsymbol{v} \times \boldsymbol{P} + 2(\boldsymbol{v}.\boldsymbol{P})\boldsymbol{v}$$

where **P** is a pure quaternion and $\boldsymbol{q} = [s + \boldsymbol{v}]$ is a unit quaternion.

c) The Camera Coordinate System (CCS) is defined by the basis vectors $\boldsymbol{u}(0.782, 0.569, 0.254)$, $\boldsymbol{v}(0.079, 0.927, -0.367)$ and $\boldsymbol{n}(0.334, 0.668, 0.665)$. Currently the camera is at world origin, and you sequentially pitched $(0^\circ \leq \theta^\circ \leq 90^\circ)$ and rolled $(-90^\circ \leq \beta^\circ \leq 0^\circ)$ the camera. (14) [CO3]

If after transform the basis vectors are $\boldsymbol{u}'(0.425, 0.828, -0.366)$, $\boldsymbol{v}'(-0.0044, 0.166, -0.986)$ and $\boldsymbol{n}'(-0.166, 0.955, -0.246)$, find the pitch, θ and roll, β. [Hint: Following equations are given for your convenience.]

$$a \sin x + b \cos x = c$$
$$x = \sin^{-1}(c/\sqrt{a^2 + b^2}) - \tan^{-1}(b/a) + 2n\pi$$
$$or, x = \pi - \sin^{-1}(c/\sqrt{a^2 + b^2}) - \tan^{-1}(b/a) + 2n\pi$$

6. a) Consider the following perspective matrix defined for projection plane containing point $R_0(x_0, y_0, z_0)$, plane normal $N(n_1, n_2, n_3)$, and projection reference point $P_{prp}(a, b, c)$, with $d = d_0 - d_1 = N.R_0 - N.P_{prp}$. (12) [CO3]

$$\begin{bmatrix} d + an_1 & an_2 & an_3 & -ad_0 \\ bn_1 & d + bn_2 & bn_3 & -bd_0 \\ cn_1 & cn_2 & d + cn_3 & -cd_0 \\ n_1 & n_2 & n_3 & -d_1 \end{bmatrix}$$

Show that, for this perspective transform, vanishing point P_v in the direction vector U can be formulated as $P_v = P_{prp} + \left(\frac{d}{K}\right) U$ where $K = N.U$. Show the derivation steps in details.

b) Explain the importance of using Distribution Term, D and Geometry Term, G in their illumination model by Cook-Torrance. How do they estimate those terms? Show with necessary illustration, reasoning and derivation. (15) [CO1]

c) For the given control points (0,0), (0,1), (4,0), (5,0); use Hermite blending functions to generate a cubic curve $Q(t)$. Find the point on the curve at $t = 0.2$. (08) [CO2]

7. a) Let you are drawing a non-uniform cubic B-spline curve $Q(t)$, with control points $(5,6),(10,25),(13,10),(20,30)$ and $(26,15)$ and knot vector $[0,0,0,0,2,4,5,6,7]$. Determine the blending functions for the parameter range $2 \leq t \leq 4$. Also find the point on the curve $Q(t)$ at $t = 3.0$. For your convenience, the recursive Cox-deBoor blending equations are given below, where symbols have their usual meanings. (16) [CO1]

$$B_{k,1}(t) = \begin{cases} 1, & if\ t_k \leq t \leq t_{k+1} \\ 0, & Otherwise \end{cases}$$

$$B_{k,d}(t) = C_k B_{k,d-1}(t) + (1 - C_{k+1}) B_{k+1,d-1}(t)$$

$$C_k = \frac{t - t_k}{t_{k+d-1} - t_k}$$

b) Let we want to define a parametric surface patch $S(u,v)$ using Bezier blending functions. Suppose the patch is quadratic and cubic on the parametric directions u and v respectively. How many control points do you need to supply? Represent the patch in tensor notation, clearly stating each item. Particularly show the Bezier basis matrices for different directions by generating from Bernstein polynomial equation. (11) [CO3]

c) Let a plane P contains the origin and a unit vector **u** is orthogonal to the plane. A reflection across P is a linear transform that maps each point **x** to its mirror image directly across P. Prove that this reflection transform for such a plane can be represented by a 3×3 matrix $\boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^T$. Write out the matrix in component form too. (08) [CO2]
[Hint: Use idea similar to the rotation matrix derivation used by Rodrigues, where $v_{ref} = v_{\perp} - v_{\parallel}$]

8. a) Use Gauss-Seidal iterative approach of solving radiosity equation to find the radiosities of each of the patches having following information. Approximate values up to four decimal places. (20) [CO2]

Patch	1	2	3	4
Emissive Power	0	0	0.6	0.8
Area	0.4	0.6	0.5	0.2
Reflectivity	0.8	0.6	0.3	0.3

		To			
		1	2	3	4
From	1	–	0.5	0.6	0.3
	2	–	–	0.4	0.1
	3	–	–	–	0.2

Form factors

b) Explain why Gouraud surface rendering method produces much better color approximation than constant shading method, especially for curved surfaces represented by polygonal mesh. (07) [CO3]

c) Derive the equations for computing delta form factor of a pixel located at back face of a hemicube. (08) [CO1]

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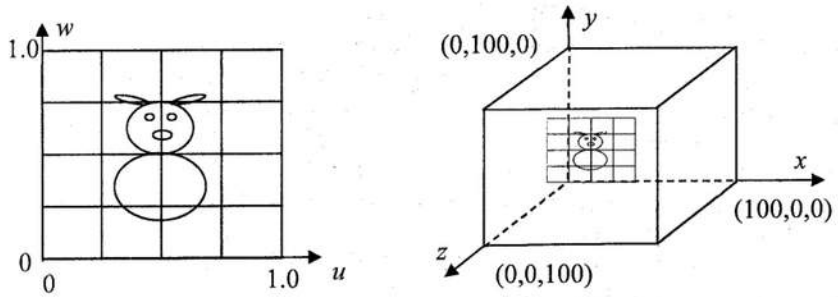
SECTION A

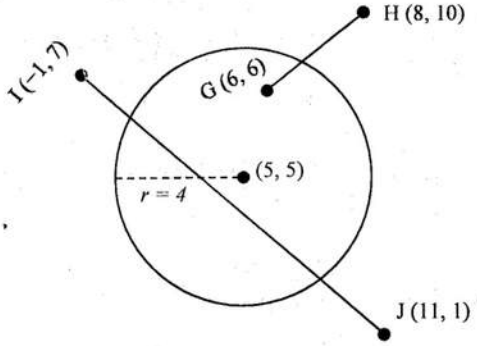
(Answer **ANY THREE** questions from this section in Script A)

1. a) What are the raster and vector graphics? Write the differences between vector and raster graphics. (07) [CO1]

b) Consider the following figure and find the linear functions that map the normalized image onto a square area of 50×50 in the middle of the front face of the cubic object. (12) [CO1]

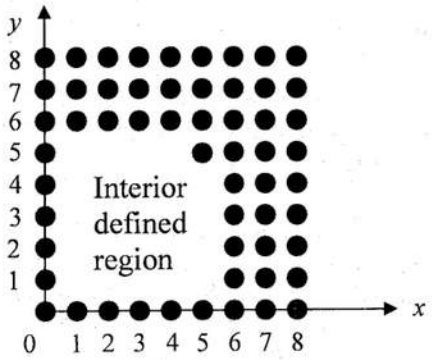


c) Use the Liang-Barsky algorithm to clip the lines in following figure. The clipping window is an axis aligned square that tightly encompasses the given circle. (16) [CO2]



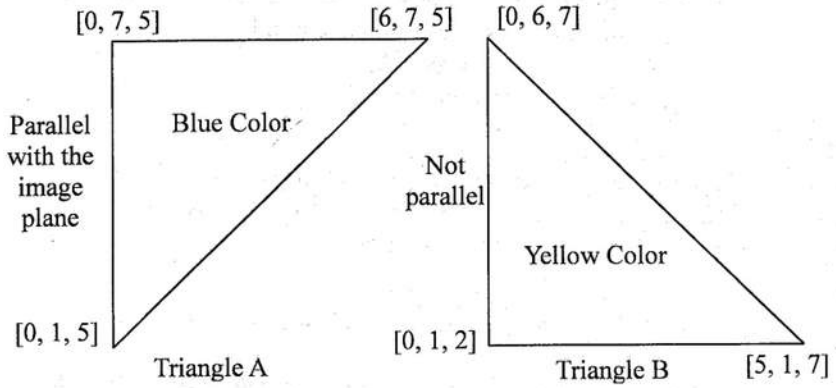
2. a) Indicate which raster locations would be chosen by Bresenham's algorithm when scan converting a line from point $(1, 9)$ to $(6, 2)$. (12) [CO1]

b) How would a flood-fill algorithm fill the region shown in following figure, using the 8-connected definition for region pixels? (13) [CO3]



c) When eight-way symmetry is used to obtain a full circle from pixel coordinates generated for the 0° to 45° or the 90° to 45° octant, certain pixels are set or plotted twice. This phenomenon is sometimes referred to as overstrike. Identify where overstrike occurs. (10) [CO1]

3. a) Using a 8×8 display, show how the z-buffer algorithm would determine the color of each pixel for the given triangles A (Blue Color) and B (Yellow Color) in following figure. Write 'B' for blue color and 'Y' for yellow color in the color buffer. (15) [CO2]



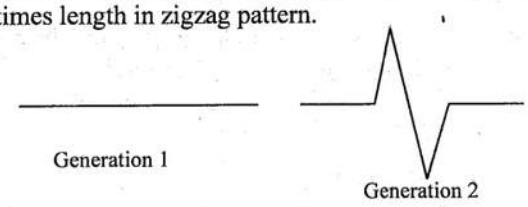
b) Explain Sutherland-Hodgeman polygon clipping algorithm. Mention it's limitations. (12) [CO3]

c) Briefly discuss how the roughness of a surface can be simulated using texture mapping. (08) [CO1]

4. a) Let, $Q(t) = (1+t^2)p_1 + (5t - 5t^2)p_2 + t^2p_3$, and $R(t) = (1+t^2)^2p_3 + (5t - 5t^2)p_4 + t^2p_5$ be the parametric equations of two curves, where t is the parameter and P_1, P_2, P_3, P_4, P_5 are the points on the curves. Determine whether $Q(t)$ and $R(t)$ are C^0, C^1, and G^2 continuous. (07) [CO1]

b) A Bezier curve is to be drawn given the control points $P_1(0,0)$, $P_2(3,3)$, $P_3(8,1)$, $P_4(10,2)$. Calculate the coordinates of the points on the curve corresponding to the parameter $t = 0.2, 0.4, 0.5, 0.6, 0.8$. Draw a rough sketch of the curve and show coordinates of the various points on it. (16) [CO3]

c) Two generation of a fractal is shown below, where middle one third of original curve is replaced with 4-times length in zigzag pattern. (12) [CO3]



i) Draw the third generation,
ii) Determine length of n-th generation of the fractal,
iii) Determine the dimension of the fractal.

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Find the composite transformation matrix (affine matrix) that transforms the triangle with vertices $A(1,2)$, $B(5,2)$, and $C(3,5)$ into vertices $A'(-1,-2)$, $B'(-4,-5)$, $C'(2,-2)$ respectively. (14) [CO3]

b) Determine the reflection of a 2D point $P(5,9)$ about a line $L(t) = (2,11) + (3,9)t$. Let you are given with the following reflection transformation matrix: (12) [CO2]

$$M_L = \frac{1}{m^2+1}\begin{bmatrix} 1-m^2 & 2m & -2bm \\ 2m & m^2-1 & 2b \\ 0 & 0 & m^2+1 \end{bmatrix}$$

c) Show that the path of a ray transmitted through a surface is $\mathbf{t} = \mathbf{z} - (\sqrt{1-|\mathbf{z}|^2})\mathbf{n}$, where $\mathbf{z} = \frac{\eta_1}{\eta_2}(\mathbf{u} - (\mathbf{u}.\mathbf{n}))\mathbf{n}$, $\mathbf{u}$ is an incoming ray, $\mathbf{n}$ is the surface normal, and η_1, η_2 are the refractive indices of the media. (09) [CO3]

6. a) The following matrix is a general parallel projection matrix with direction of projection (DOP) onto an arbitrary plane with plane normal N, passing through P_0. Now, determine the parallel projection of the point $(2,3,4)$ if DOP is $(1,1,1)$ and projection plane is $2x + y + z = 8$. Symbols have their usual meanings in the given matrix. (12) [CO2]

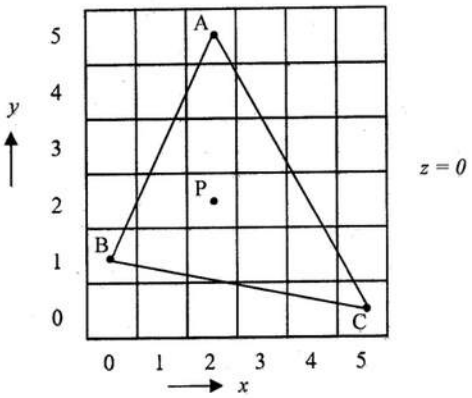
$$Par_{V,N,P_0} = \begin{bmatrix} d_1 - an_1 & -an_2 & -an_3 & ad_0 \\ -bn_1 & d_1 - bn_2 & -bn_3 & bd_0 \\ -cn_1 & -cn_2 & d_1 - cn_3 & cd_0 \\ 0 & 0 & 0 & d_1 \end{bmatrix}$$

b) Suppose you want to align vector $\bar{p} = (0, 1, 1)$ with the vector $\bar{q} = (1, 0, 1)$ by rotating around the axis $(-1, -1, 1)$. Consider the given quaternion-based rotation matrix $M_R(\theta)$, where symbols have their usual meanings. What will be the angle of rotation? (12) [CO3]

$$M_R(\theta) = \begin{bmatrix} 1-2b^2-2c^2 & 2ab-2sc & 2ac+2sb \\ 2ab+2sc & 1-2a^2-2c^2 & 2bc-2sa \\ 2ac-2sb & 2bc+2sa & 1-2a^2-2b^2 \end{bmatrix}$$

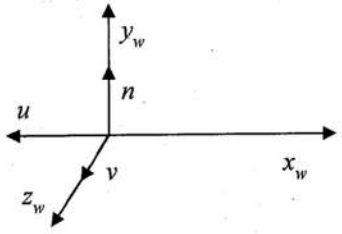
c) Derive a transformation matrix to convert a frustum shape eye-space view volume in a canonical view volume in normalized device coordinate space. (11) [CO1]

7. a) Consider a triangular fragment ABC as shown below. Let, the vertex normals are $N_A = (0, 0, 1)$, $N_B = (0, 0.2, 1)$, and $N_C = (0, -0.2, 1)$. If the light source is at $(7, 3, 4)$, viewer is at $(-6, 3, 3)$ with light color $(0.9, 0.9, 1)$, surface materials reflectivity $(0.5, 0.6, 0.4)$ and you are rendering the surface with Phong shading then determine the diffuse color at point P. (14) [CO1]



b) How can you generate soft shadows using ray tracing? Explain. (06) [CO2]

c) The current position of a camera is at $(10, 20, -15)$, looking at $(35, 40, 20)$, and head up is along $(3i + 4j - 5k)$. Find the transformation matrix that would align the camera looking direction along with positive y-axis, and the direction that points to the right of the camera along with the negative x-axis as shown in figure below. (15) [CO1]



8. a) Let a ray is defined by $\mathbf{s} = (0, 0, 0)$, $\mathbf{d} = (-1/\sqrt{14}, 3/\sqrt{14}, -2/\sqrt{14})$. (15) [CO2]

i) Determine if the ray intersects the triangle defined by the points $(-5, 1, -5)$, $(1, -1, -4)$, and $(0, 18, -7)$,

ii) Also find the transmitted ray considering the triangle as transparent surface with $\eta = 1.5$.

b) Consider following two tables showing information regarding four patches and their form factors. Determine radiosity of each patch. (15) [CO1]

Patch No	1	2	3	4
Reflectivity, ρ_i	0.2	0.5	0.6	0.4
Emissive intensity, E_i	0.8	0	0	0
Patch area	0.2	0.6	0.5	0.7

	1	2	3	4
1	1	0.6	0.6	0.3
2	–	1	0.3	0.1
3	–	–	1	0.1
4	–	–	–	1

Form Factor, F_{ij}

Approximate values up to three decimal places.

c) Given that a hemicube is defined at origin with the z axis vertical. What is the delta form factor of the hemicube pixel on the right face $x = 1$, if $(y_{min}, y_{max}, z_{min}, z_{max}) = (0.55, 0.65, 0.35, 0.45)$. (05) [CO3]

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

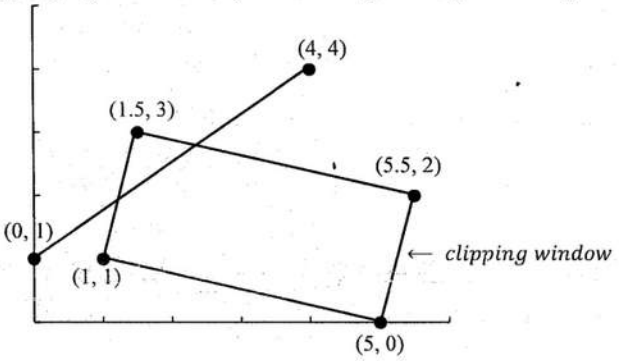
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Briefly explain the different functionalities of 3D computer graphics pipeline. (10)
   b) Show that two successive reflections about any two lines in the xy plane that go through the origin is equivalent to a rotation about the origin. (12)
   c) Use quaternion mathematics to derive a 3D rotation matrix about a unit vector $\boldsymbol{u}$. (13)

2. a) Derive a perspective projection matrix if the center of projection is at (a, b, c), the projection plane normal $N = n_1 i + n_2 j + n_3 k$ and reference point $R_0(x_0, y_0, z_0)$. (13)
   b) Determine the oblique projection of a VCS point $P(2, 2, 5)$ onto a view plane with $N = k$ and $R_0(1, 0, 1)$. Use azimuth = 30° and elevation = 45°. (10)
   c) Consider the following figure showing a rectangular clipping window and a line segment. Apply Cyrus-Beck line clipping algorithm to clip the line segment against the given window. Find intersection points. (12)

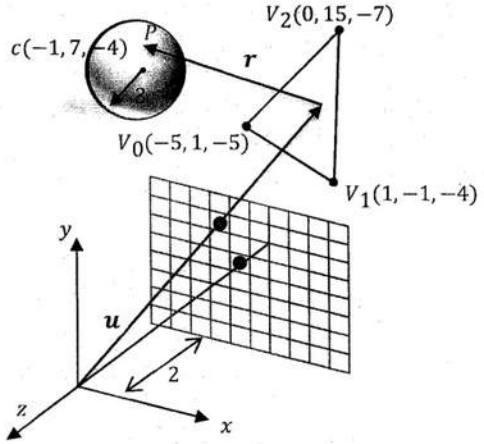


3. a) There are two monochromatic light sources L^0 and L^1 in a scene. The light intensities and corresponding material's reflectivity at vertex V are given in the following table. (12)

	Ambient	Diffuse	Specular
L^0	0.1	0.9	0.9
L^1	0.2	0.7	0.5
V	0.2	0.6	1.0

Furthermore, the shininess at $V = 3.0$, vertex normal at V is $[0 \quad 1 \quad 0]^T$, emissive light at V is 0.05. The position vectors of L^0, L^1, V and eye are, respectively, $[0 \quad 5 \quad 5]^T$, $[5 \quad 5 \quad 5]^T$, $[0 \quad 0 \quad 5]^T$ and $[0 \quad 0 \quad 0]^T$. Use Blinn-Phong illumination model to approximate the total intensity at V.

   b) Consider, the information given in the following figure. Determine whether the reflected ray $\boldsymbol{r}$ intersects the sphere at P. If so, find the P. Here, it is given that the approximated triangular surface normal $N = -1.04i + 1.77j + 9.79k$. (12)



   c) Define and explain the basic radiosity equation. State a method to solve the equation. (11)

4. a) Find Camera Coordinate System (CCS) value for a world point $P(1, 2, 3)$ if the camera is at $(3, 3, 7)$ focusing towards $(1, 1, 1)$ with up vector $U = i + j$. (12)
   b) Write down the 4×4 rotation matrix M that transforms the orthonormal 3D vectors $\boldsymbol{u}(x_u, y_u, z_u)$, $\boldsymbol{v}(x_v, y_v, z_v)$, and $\boldsymbol{w}(x_w, y_w, z_w)$ to orthonormal 3D vectors $\boldsymbol{a}(x_a, y_a, z_a)$, $\boldsymbol{b}(x_b, y_b, z_b)$, and $\boldsymbol{c}(x_c, y_c, z_c)$ such that $M\boldsymbol{u} = \boldsymbol{a}$, $M\boldsymbol{v} = \boldsymbol{b}$, and $M\boldsymbol{w} = \boldsymbol{c}$. (16)
   c) Define z-fighting problem. Mention some strategies to overcome the issue. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define horizontal and vertical retrace. (04)
   b) If a circle's origin is centered at point (5, 5) and radius is equivalent to 5, then rasterize the circle using midpoint circle drawing algorithm. (14)
   c) If $(x_0, y_0) = (5, 15)$ and $(x_{end}, y_{end}) = (11, 9)$, use Bresenham's algorithm to compute pixels for line segment. (09)
   d) What are the major side effects of scan conversion? Explain. (08)

6. a) Write down some limitations of z-Buffer algorithm. Explain how A-Buffer method overcomes the drawback of z-Buffer. (08)
   b) Explain the basic idea of Painter's algorithm. How does the algorithm deal with the situation when the polygons are such that their x, y and z-extents overlap? (10)
   c) Consider two triangles, ΔABC defined by the points $A(1, 7, 5)$, $B(1, 1, 5)$, $C(6, 1, 5)$ and ΔDEF defined by the points $D(6, 5, 2)$, $E(2, 0, 3)$, $F(7, 0, 7)$. Apply z-Buffer algorithm to the triangle surface using a 8×8 depth buffer. Show z-Buffer contents for all the pixels. Also show the frame buffer content if the surface colors of ΔABC and ΔDEF are RED and GREEN respectively. (17)

7. a) Equation of two curves Q and R are given as follows: (09)

$$Q(t) = (1 + t^2)P_1 + (5t - 5t^2)P_2 + t^2P_3$$
$$R(t) = (1 + t^2)P_3 + (5t - 5t^2)P_4 + t^2P_5$$

   For both the curves $0 \leq t \leq 1$. Are both the curves C^0 continuous at $Q(1)$ and $R(0)$? Are both the curves C^1 and G^1 continuous at $Q(0)$ and $R(0)$? Why or why not? Explain your answer.
   b) Show that a cubic Bézier curve can be presented as following matrix formulation; symbols have their usual meanings. (14)

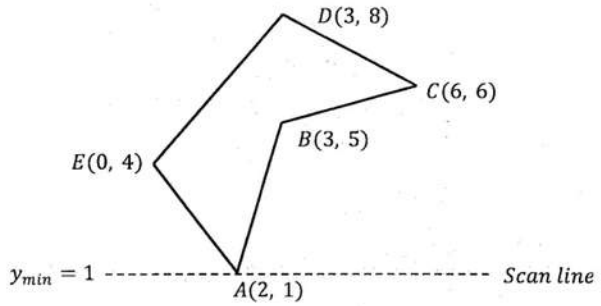
$$P(t) = [t^3 \quad t^2 \quad t \quad 1] \begin{bmatrix} -1 & 3 & -3 & 1 \\ 3 & -6 & 3 & 0 \\ -3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} P_1 \\ P_{C_1} \\ P_{C_2} \\ P_2 \end{bmatrix}$$

   Also draw a rough sketch of the curve if $P_1(0, 0)$, $P_{C_1}(0, 3)$, $P_{C_2}(2, 0)$ and $P_2(4, 0)$.
   c) An open uniform B-spline curve $Q(u)$ of order $K = 3$ is constructed with the control points $P_0(0, 0, 0)$, $P_1(1, 1, 0)$, $P_2(2, 0, 0)$, $P_3(3, 1, 0)$ and $P_4(4, 0, 0)$. Determine the knot vector and parametric range for each segment of the curve. Also determine the blending functions for the last segment of the curve. (12)

8. a) How does the texture magnification effect and texture minification effect occur? How to determine the appropriate mipmap level in mipmapping? (05)
   b) How does a CRT monitor display color pictures? Briefly explain. (07)
   c) Define fractal. Generate a fractal image using the following language. (12)

$$AA[A[B]AA(B)]AAAA(A[B]AA(B))$$

   where Alphabet: $\{A, B, [,], (,)\}$,
   Rule: $A \Rightarrow AA$, $B \Rightarrow A[B]AA(B)$, $[] \Rightarrow left\ brance$, $() \Rightarrow right\ brance$.
   d) Consider the scan line Fill algorithm. Determine the content of active edge table to fill the polygon. What are the global edge table for the polygon? (11)



KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2020
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 1.5 hours FULL MARKS: 120

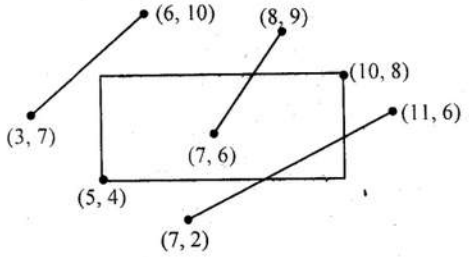
N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) Find mirror reflection of a point $P(3, 3, 3)$ about a plane defined by the vertices (1, 0, 0), (0, 1, 0) and (0, 0, 1). (14)
   b) Let a View-Coordinate System (VCS) point $P(1, 1, 7)$. Find the Normalized Device Coordinate System (NDCS) value of P, if the view volume is defined by the following parameters and projection reference point is at VCS origin: (16)
   $left = -1$, $right = 1.2$, $top = 1$, $bottom = -0.8$, $near = 1$, $far = 8$.

2. a) Using Liang-barsky method clip the lines against the clipping rectangle shown in figure below. (16)



   b) Find the parallel projection of a point $P(5, 7, 8)$ with DOP, $\bar{v} = I + 2J + 2K$ onto a plane with plane normal $N = K$, and $P_0(0, 0, 1)$. (14)

3. a) Let a plane defined by the points (0, 0, 0), (0, 0, 2) and (5, 1, 1) is illuminated with a directional white light with direction vector (3, –3, 1). Find the diffuse color for all vertices and specular color for the last vertex by using Phong illumination model. Consider following parameters: (15)
   diffuse reflectivity = (0.4, 0.6, 0.1), *specular reflectivity* = (0.8, 0.9, 0.9), *shininess* = 8, *viewer position* = (6, 2, 3), *vertex normal* = *surface normal*, *no ambient light* and *no attenuation*.
   b) Determine if a ray with $s = I + J + 2K$ and $d = -2K$ intersects the triangle defined by the points (1, 0, 1), (0, 2, 1), (0, 1, 2). Use Badouel's algorithm to verify. (15)

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Define horizontal and vertical retrace. (06)
   b) How does a CRT monitor display color pictures? Briefly explain. (09)
   c) If a circle's origin is centered at point (10, 10) and radius is equivalent to 8, then rasterize the circle using Bresenham's circle drawing algorithm. (15)

5. a) Equation of two bezier curves Q and R are given as follows: (08)

$$Q(t) = (1 + t^2)P_1 + (5t^2 - 5t)P_2 + t^2P_3$$

$$R(t) = t^2P_1 + (5t^2 - 5t)P_2 + (1 + t^2)P_3$$

For both curves $0 \le t \le 1$.

Are both curves C^0 continuous at $Q(0)$ and $R(1)$. Are they both C^1 and G^1 continuous at $Q(0)$ and $R(0)$? Why or why not? Explain your answer.

   b) Find the equation of a Hermite cubic spline with the two end points as (1, 1) & (6, 5) and corresponding tangent vectors as (0, 4) & (4, 0). (13)
   c) How is Koch curves generated? Explain with suitable example(s)? (09)

6. a) How is painting priority of a surface determined by painter's algorithm? (08)
   b) Explain z-buffer algorithm in detail for visible surface detection. What are the advantages and disadvantages of z-buffer algorithm? (13)
   c) Define texture minification. Why is mipmapping used in computer graphics? Briefly explain. (09)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2019
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

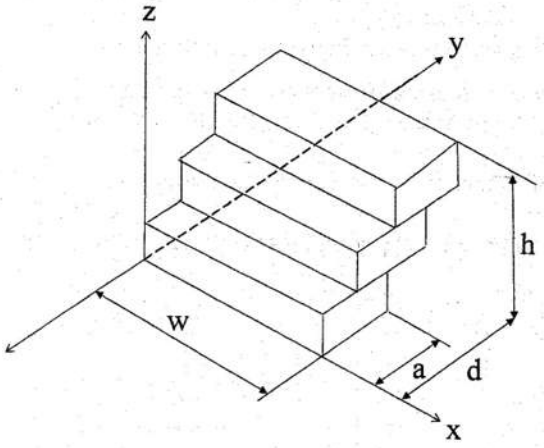
1. a) Define and mention the importance of following transformation in a Computer graphics system. (10)
   i) Modelling transform.
   ii) Viewing transform.

   b) In 2D, it is possible to rotate data using a sequence of a shear in x, followed by a shear in y, and then a non-uniform scaling transform. Rather than using the rotation matrix, you are instructed to use the above mentioned method to perform a θ° rotation. In that case- (15)
   i) What should the X and Y scale factor be?
   ii) Determine the X and Y shear terms and their relationship. As a reminder, the homogeneous shear matrices are given below.

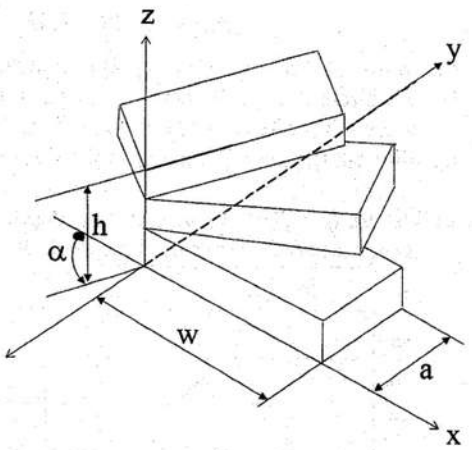
$$shear(x) = \begin{bmatrix} 1 & Sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad shear(y) = \begin{bmatrix} 1 & 0 & 0 \\ Sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

   c) Let a triangle is defined by the points A (1, 0, 0), B (0, 2, 0) and C (0, 1, 1). Determine if a ray with $\mathbf{s} = i + j + 3k$ and $\mathbf{d} = -2k$ intersects the triangle or not? First find the intersection of ray and triangular plane and then use Badouel's algorithm to verify that the intersection lies within the triangle edges. (10)

2. a) Suppose you have to draw a stair having N steps, each step has width w and depth a; the total stair height is h; for a straight forward stair, the total depth is d. See the left figure below, where a stair with N = 3 is shown. Assume that a function *Drawstep(M)* is available to draw a step, subject to a homogeneous transformation M. A step in local coordinates is a cube $-1 \le x,y,z \le 1$, centered around the origin. Furthermore, suppose that a standard set of basic modeling transformation functions for translation, rotation and scaling are available. (11)
   i) Define a matrix M_i using basic transformation functions along with their parameters to draw step i, i =1, …, N for the straight forward stair.
   ii) Consider a spiral stair (see the right Figure below), where the total rotation is α. Define the matrix M_i again for drawing a step i, i =1, …, N for this spiral stair.



Straight Forward Stair



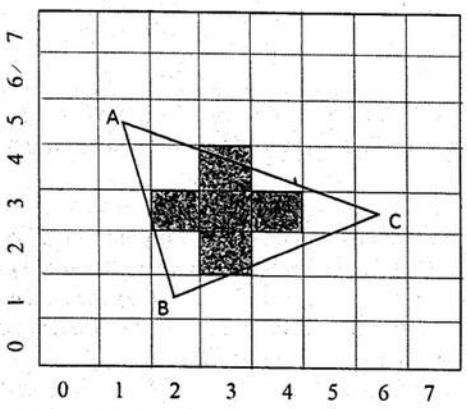
Spiral Stair

   b) Find mirror reflection of a point P (2, 2, 1) about a plane defined by the points (1, 0, 1), (1, 0, 0), and (0, 1, 0). (12)

c) Derive a transformation matrix that transform a world coordinate point to camera coordinate. Be sure to explain how the rotation matrix involved in above transformation is derived. Also consider the camera is defined by the three parameters *position, look-at* and *up-vector.*

3. a) Compare between perspective and axonometric projection. (05)
   b) Derive a projective transformation matrix that perspectively project any point in view-coordinate system (VCS) onto a projection plane parallel to xy-plane and placed at distance –d. Consider the center-of-projection (COP) is at C (x_c, y_c, z_c). (12)
   c) A triangular plane defined by the points (0, 2, 0), (3, 3, 0) and (1, 0, 1) is illuminated by a monochromatic point light source with light intensity 0.9 and positioned at (4, 6, 2). Find the vertex intensities using Phong illumination model if the viewer is at (1, 1, 5), diffuse and specular reflectivity co-efficient of the surface are 0.6 and 0.8 respectively. Also, consider the shininess factor = 8, no ambient light, radial attenuation constraints $a_0 = 0$, $a_1 = 1$, $a_2 = 0$, and vertex normal equals to the surface normal. (15)
   d) State the necessity of near clipping plane while defining a view volume. (03)

4. a) Explain how the Ray tracing method integrates the process of projection, scan-conversion and shadow generation. (10)
   b) Let a triangular face is rasterized to pixel coordinates A (1, 5), B (2,1), C (6,3). Say, RGB vertex intensities are I_A (0.9, 0.1, 0.2), I_B (0.1, 0.8, 0.2) and I_c (0.2, 0.3, 0.7). Use Gauraud shading method to render the triangular face showing the interval pixel intensities of the highlighted pixels shown in figure below. (14)



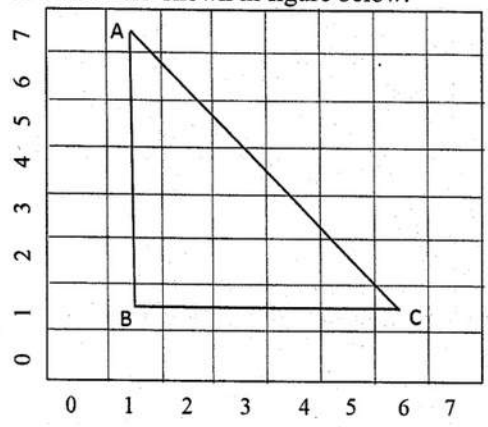
   c) Formulate a transformation matrix that converts an asymmetric parallelepiped view volume to a canonical view volume. Also mention, why should we do the conversion? (11)

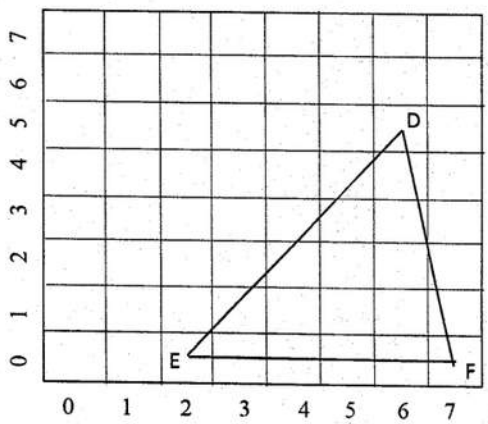
SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Digitize a line (3, 12) to (6, 2) on a raster screen using Bresenham's straight line algorithm. (10)
   b) If a circle's origin is centered at point (30, 40) and its radius is equivalent to 8, then rasterize the circle with midpoint circle algorithm. (10)
   c) Find out the decision parameter values of a quadrant of an ellipse using midpoint ellipse algorithm. (15)

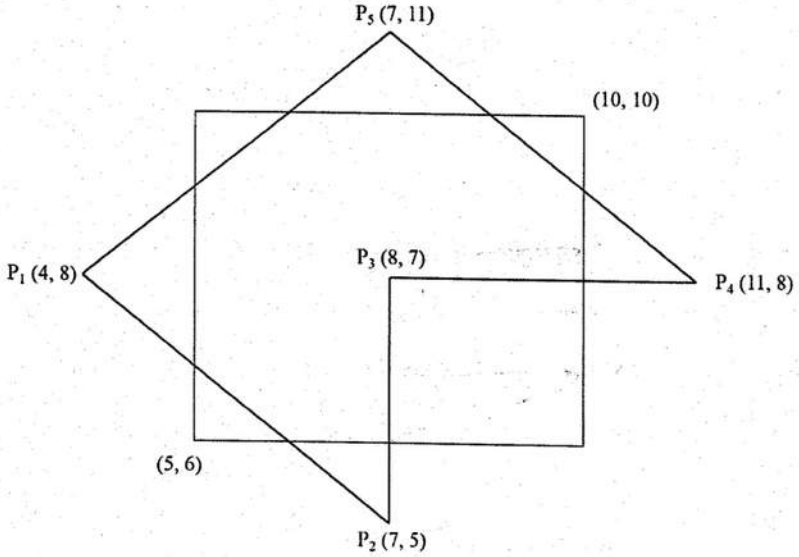
6. a) Use an 8x8 Z-buffer to store the depth values after applying Z-buffer algorithm onto the triangles ABC and DEF shown in figure below. (17)





Here the z-coordinates of the points A, B, C, D, E and F are 5, 5, 5, 2, 3 and 8 respectively. Find the internal Z-values of the triangle before applying the algorithm and show Z-buffer contents after each step. Also show the frame buffer content if the triangle surface of ABC and DEF are RED and GREEN color respectively.

b) Consider the polygon defined by the points P_1, P_2, P_3, P_4 and P_5 as shown in figure below. Use Sutherland-Hodgeman algorithm to clip polygon against the rectangular window defined by the point $(x_{min}, y_{min}) = (5, 6)$ and $(x_{max}, y_{max}) = (10,10)$. Compute the final points list. (13)



c) Define fractal. Draw the first three iterations of Koch snowflake fractal. Show each iteration separately. (05)

7. a) Write about the differences between Object-space method and Image-space method. (08)
   b) How painting priority of a surface is determined by painter's algorithm? (10)
   c) What are the major mapping methods? Give description of them. (07)
   d) Give description of Linear Texture and Cylinder Texture mapping in 2d. (10)

8. a) What is aliasing problem? How one can solve it? (10)
   b) Give an example of special case in scan-line polygon filling algorithm. (05)
   c) What do you know about geometric continuity and parametric continuity? Explain C^0, C^1, C^2 (10)
   d) Calculate Hermite Blending Functions from cubic Hermite Spline Curve having four control points. (10)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2018
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is computer graphics deal with? Mention some prominent applications of it. (06)
   b) Explain the geometry stage of graphics rendering pipeline with necessary illustration. (12)
   c) Find a transformation matrix that aligns line L_1, with line L_2 given by the following equations: (17)

$$L_1 = \begin{cases} \bar{v} = 2I - 3J + 4K \\ P_0 = (1,1,0) \end{cases}$$

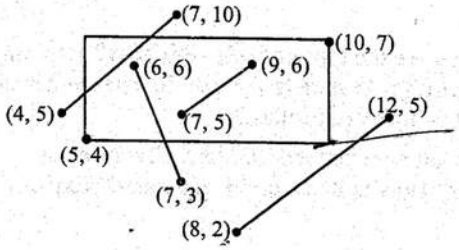
$$L_2 = \begin{cases} x = 1 + 3t \\ y = 4t \quad 0 \le t \le 1 \\ z = 2 - 5t \end{cases}$$

2. a) Derive a transformation matrix to convert an asymmetric view frustum to a Canonical View Volume (CVV). (15)
   b) What is projection? Find a parallel projection of the point (7, 8, 9) with DOP, $\bar{v} = I + 2J + K$ onto a projection plane parallel to xy plane and cuts z-axis at a distance of 1. (10)
   c) What are the parameter needed to define a View-Coordinate System (VCS)? How can a world coordinate point be converted to VCS point? (10)

3. a) Let a VCS point $p(3, 2, 6)$, projection reference point is at VCS origin, and a symmetric view volume is defined by the following parameters: (14)

$$left = right = 1,\ bottom = top = 0.8,\ near = 1,\ far = 7$$

   Find the Normalized Device Coordinate System (NDCS) value of the point p.
   b) Clip the lines presented in the following figure using Cohen-Sutherland algorithm. Also mention how CVV helps 3D clipping. (12)



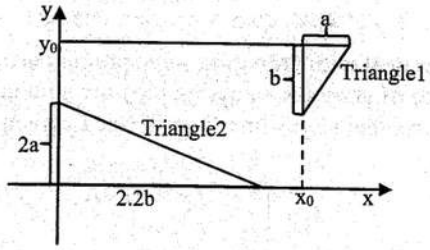
   c) Show that the path of a ray transmitted through a surface is $t = z - \left(\sqrt{1 - |z|^2}\right).n$, where $z = \frac{\eta_1}{\eta_2}(u - (u.n))n$, u is an incoming ray, n is the surface normal, and η_1, η_2 are the refractive indices of the media. (09)

4. a) "Phong shading is more realistic than Gouraud shading, but expensive"– Justify the statement. (12)
   b) Let a surface is illuminated by an RGB light source with light intensity (0.9, 1, 0.9), positioned at (5, 4, 2). A vertex (4, 1, 3), whose surface normal is (0, 2, –1), is viewed from (3, 2, 2). Find the intensity that the viewer will see at the vertex, if the surface diffuse and specular reflectively are (0.4, 0.3, 0.6) and (0.8, 0.8, 0.8) with shininess = 10. Consider there is no ambient light, and no attenuation of light. (15)
   c) Explain a method to find the intersection of an incoming ray with a sphere. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the raster and vector graphics? Write the differences between vector and raster graphics. (07)
   b) Convert these homogeneous points to Cartesian: (0, 1, 3, 2) and (2, 1, 3, 4). (07)
   c) Transform Triangle1 to Triange2 in the following figure by a sequence of basic transformations. Write the transformation matrices in 2D Homogeneous Coordinates. Compute the final transformation matrix. Note that the size of the triangle has also been changed; see the following figure for the dimensions. (14)



   d) Explain the difference between "Object Coordinates", "World Coordinates", and "Camera Coordinates". (07)

6. a) What are the three major efficiency ingredients in Bresenham's line drawing algorithm? (07)
   b) Consider the line from (0, 0) to (–8, –4), use general Bresenham's line algorithm to rasterize this line. Evaluate and tabulate all the steps involved. (12)
   c) Write down the difference between DDA line drawing algorithm and Bresenham's line drawing algorithm. (06)
   d) Explain the steps in Bresenham's circle drawing algorithm. (10)

7. a) How many real values are necessary to describe the shape of a cubic rational Beizer curve in three dimensional space? Provide a short justification for you answer. (06)
   b) Prove that $R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2)$, where $R()$ is a rotation function and θ_1, θ_2 are angle of rotation. (08)
   c) Define Hermite Surface. Find the geometry vector G_{Hx} for Hermite Surface. Also find the first of the vecrtor. (10)
   d) Construct the Beizer,curve of order 3 with 4 polygon vertices A(0, 1), B(3, 3), C (4, 5) and D (7, 5) (11)

8. a) What is fractals? What are the types of self-similarity? Explain with example(s). (08)
   b) Explain z-Buffer algorithm in details for visible surface detection. What are the advantages and disadvantages of z-Buffer algorithm? (13)
   c) What is continuity? Compare between G^1 and C^1 continuity. (06)
   d) What is Koch Curves? How is Koch curve generated? Explain with suitable example(s)? (08)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2017
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

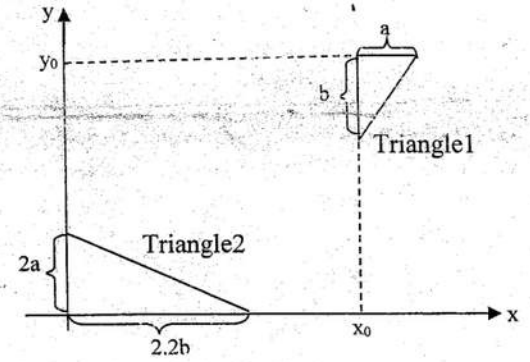
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

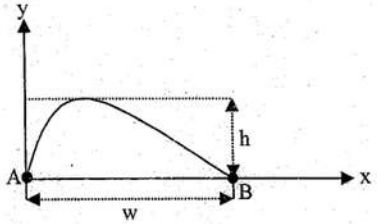
1. a) Explain the differences between "object coordinates", "world coordinates", and "camera coordinates". (08)
   b) When can you convert a quadratic Beizer curve to a cubic Hermite curve? (07)
   c) In Breshenham's algorithm, we saw the efficiency of incremental evaluation. For the function, $f(x) = y = x^2 - 3$, compute the incremental evaluation function. (10)
   d) If $(x_0, y_0) = (5,4)$ and $(x_{end}, y_{end}) = (8,10)$, use Breshenham's algorithm to compute pixels for the line segment. (10)

2. a) Prove that $R(\theta_1).R(\theta_2) = R(\theta_1 + \theta_2)$ where $R()$ is a rotation function and θ_1, θ_2 are angle of rotation. (07)
   b) Transform Triangle1 to Triangle2 in the following figure by a sequence of basic transformations. Write the transformation matrices in 2D homogeneous coordinates. Compute the final transformation matrix. Note that size of the triangle has also been changed; see the figure for the dimensions. (14)



   c) Reflect the diamond-shaped polygon whose vertices $A(-1,0), B(0,-2), C(1,0)$ and $D(0,2)$ about the line $y = x + 2$. (14)

3. a) We want to draw a cubic Beizer curve starting from $A = (0,0)$ to the point $B = (w,0)$. At A the curve is tangent with y-axis and at B the curve is tangent with x-axis. (see figure below). Draw control points and give the possible positions of them (ignoring h), use new parameters where necessary. (10)



   b) Define fractal. What are the classifications of fractal objects? How is fractal generated? (09)
   c) Find fractal dimension of Koch curve. (05)
   d) Explain depth-buffer algorithm in details for visible surface detection. What are the advantages and disadvantages of z-buffer algorithm? (11)

4. a) How is painting priority of a surface determined by painter's algorithm? (09)

b) A Beizer curve is to be drawn given the control points $P_1(40,40), P_2(10,40), P_3(60,60), P_4(60,0)$. Calculate the coordinates of the points on the curve corresponding to the parameter $t = 0.2, 0.4, 0.6$. Draw a rough sketch of the curve and show coordinates of the various points on it. (14)

c) What is the advantage of using curved surfaces for rendering? How to display it efficiently in practice? (07)

d) What are the drawbacks of Digital Differential Analyzer (DDA) algorithm? (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the three stages of computer graphics pipe-line? Briefly explain each of them. (10)

b) Let a line L, defined in parametric form below. Determine a rotation matrix about L with rotation angle $= 30°$. (14)

$$L: \begin{cases} x = 2 + 3t \\ y = 4t \\ z = 3 + 4t \end{cases} \quad 0 \le t \le 1$$

c) When performing Cohen-Sutherland line clipping, how do we use outcodes to check for trivial rejection (what exact operation and comparison would you use?) (06)

d) What is GPU? Can it be used in Phong shading? Why? (05)

6. a) Explain the importance of projection in computer graphics. Find a parallel projection matrix with DOP $V = aI + bJ + cK$ and xy as projection plane. (10)

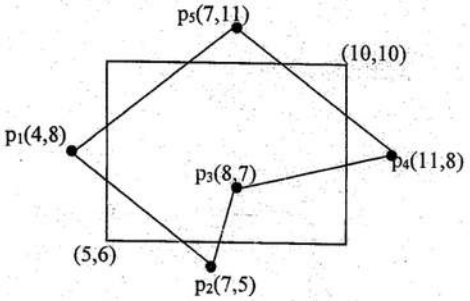
b) Consider a point $P = (2,3,4)$ in eye space. Let a symmetric projection window is defined with aspect ratio $= 1.25$, fovy $= 60°$, near and far plane of the view volume is at a distance 1 and 5 respectively. Find the perspective transformation of the point P in normalized device coordinate space. (12)

c) What is the importance of maintaining pseudo depth in perspective transformation? (05)

d) Describe the anomalies of perspective projection. (08)

7. a) What is view volume? Describe how a parallelepiped view volume can be converted to canonical view volume. (10)

b) Consider the polygon defined by the points p_1, p_2, p_3, p_4 and p_5 as shown in figure below. Use Sutherland-Hodgeman algorithm to clip polygon against the rectangular window defined by the point $(x_{min}, y_{min}) = (5,6)$ and $(x_{max}, y_{max}) = (10,10)$. Compute the final points list. (15)



c) In ray tracing a large computational cost is associated with determining ray-object intersections. Suggest a method that may reduce this cost. (10)

8. a) The normal at a vertex $(0,1,2)$ on a surface is $(0,4,5)$. The light source is at $(0,1,4)$. The diffuse color of the light is $(0.9,0,0.2)$. The diffuse reflectivity of the surface are $(0.4,1,0)$. What will the RGB color of the vertex be? Assume there is no specular, emmisive or ambient component or light attenuation. (10)

b) Explain how light intensity attenuation can be modeled. (07)

c) Let S, be a sphere of radius 10 centered at $(10,-2,-5)$. Determine if a ray $S = 2J + 5K$ and $d = I - 2K$ intersects the sphere. (10)

d) Compare the working principle of Phong and Gouraud shading. (08)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2016
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

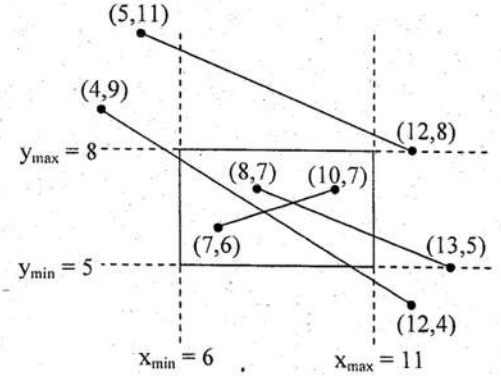
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What does computer graphics deal with? We have cameras to shoot photographs or videos, then what are the driving factors of computer graphics? (07)
   b) Let a unit square defined by $0 \le x \le 1$, and $0 \le y \le 1$ in object coordinate system. New positions X′ are derived from the local positions X by X′ = MX, where M is a transformation matrix. Give a matrix M, expressed in terms of basic modeling transformations, to obtain a rectangle with width = 2 and height = 3, centered around the origin of world coordinate system, and rotated over an angle of 30°. (13)
   c) Why are matrix representations used to describe point transformations in computer graphics? Derive the basic rotation matrix for a rotational transformation about z-axis. (10)
   d) Explain the properties of rigid body transformation and affine transformation. (05)

2. a) Find the mirror reflection of the point (3, 1, 6) with respect to the plane defined by the points (0, 1, 0), (0, 0, 1), and (1, 0, 0). (14)
   b) Explain the working principle of 2D Sutherland-Hodgeman polygon clipping algorithm with necessary illustrations and equations. How can it be extended to 3D? (13)
   c) Briefly explain the taxonomy of planar geometric projection. (08)

3. a) Let the view coordinate system (VCS) is defined by the parameter *eye* $(0, 0, 1)_{WC}$, *look-at* $(0, 0, 0)_{WC}$, *head-up* $\bar{V} = j$. If the projection plane is parallel to the xy plane of the VCS and cuts the negative z-axis of VCS at a distance $d = 1$, find the perspective projection of the point $(1, 1, -1)_{WC}$ on the projection plane considering the VCS origin as a center of projection. (14)
   b) Use Liang-Barsky line clipping algorithm to clip the lines in the following figure. (14)



   c) Briefly explain principal vanishing points for standard perspective projection. Use necessary diagram. (07)

4. a) Let a triangular facet, tessellated part of a large curved surface, is defined by the vertices A (1, 4, 0), B (5, 6, 0), C (4, 0, 0) and their vertex normal are $N_A = -i + j + 3k$, $N_B = j + 3k$, $N_C = i + 4k$ respectively. If the triangle is illuminated by a monochromatic point light source positioned at (–2, 3, 2), and the diffuse reflectivity of the surface is 0.7 then find the diffuse reflection intensity of point P (3, 3, 0) that a viewer will see from position (6, 3, 1). Assume any missing values (if necessary). (15)
   b) Compare among flat shading, Gouraud shading, and Phong shading techniques. (09)
   c) Show that the path of a refracted ray is $t = z - \left(\sqrt{1-|z|^2}\right)n$, where $z = \frac{\eta_1}{\eta_2}(u - (u.n))n$, u is incoming ray, n is surface normal, and η_1, η_2 are the refractive indices of the media. (07)
   d) What are the trade-offs of ray tracing method? (04)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Indicate which raster locations would be chosen by Bresenham's algorithm when scan-converting a line from pixel coordinate (1, 1) to pixel coordinate (8, 5). (12)
   b) What are the major side effects of scan conversion? (08)
   c) Using midpoint algorithm, draw an ellipse with center (2, 2), major axis = 12, and minor axis = 10. (15)

6. a) How do you convert RGB color space into HSI color space? Explain briefly with appropriate figure. (12)
   b) What are the possible representations of a curve? Why is the parametric curve of degree 3 (cubic) preferable? (10)
   c) Define chromatic light and achromatic light. What are the properties of chromatic light? Explain briefly. (07)
   d) What is a raster display? Explain the necessity of scan conversion. (06)

7. a) Define fractal. Generate a fractal image using the following language (12)

   AA[A[B]AA(B)]AAAA(A[B]AA(B))

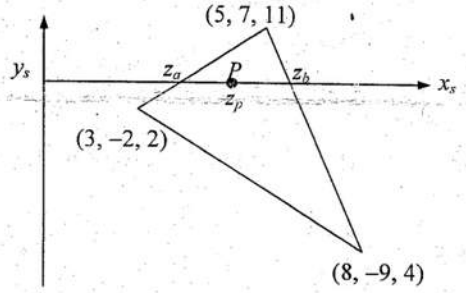
   where Alphabet {A, B, [,], (,)}

   Rule A ⇒ AA

   B ⇒ A[B]AA(B)

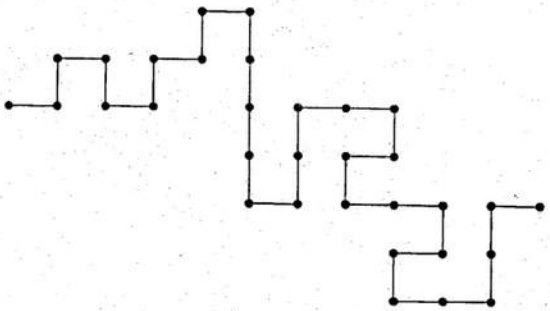
   [] ⇒ left branch

   () ⇒ right branch
   b) What are the difficulties encountered in implementing the painter's algorithm? When does *z*-fighting problem arise? (08)
   c) How does the *z*-buffer algorithm determine which surfaces are hidden and hence compute the z-value at point P of the following figure. (10)



   d) How can you generate the terrain in computer graphics? Explain. (05)

8. a) What do you know about geometric continuity and parametric continuity? Explain G^0, G^1, and G^2 geometric continuity. (10)
   b) Define Hermite surface. Find the geometry vector G_{H_X} for Hermite surface. Also find twist of the vector. (11)
   c) Derive the equation for calculating fractal dimension and hence find the dimension of the following fractals. (07)



   d) What is texturing? Compare between bump mapping and frame mapping techniques. (07)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2015
Department of Computer Science and Engineering
CSE 4207
Computer Graphics

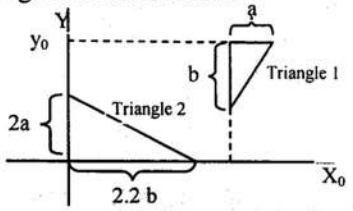
TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What are the raster and vector graphics? Write the differences between vector and raster graphics. (07)
   b) We wish to scale a triangle A(0,0), B(1,1), C(5,2) to twice its size. While keeping vertex C fixed. This scaling can be expressed as a composite transformation. Describe and write the matrices for the component transformations. (10)
   c) Transform Triangle1 to Triangle2 in the following figure by a sequence of basic transformations. Write the transformation matrices in 2D Homogeneous coordinates. Compute the final transformation matrix. Note that the size of triangle as also been changed; see the figure for dimensions. (13)

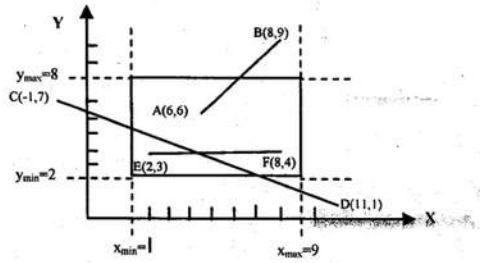


   d) What are the drawbacks of DDA algorithm over Bresenham's line drawing algorithm. (05)

2. a) Scan convert the line segment from point (5,8) to point (9,12). (10)
   b) Why is the point to point line error always $\leq 1/2$ for the mid-point scan conversion algorithm? (10)
   c) Prove that $R(\theta_1).R(\dot{\theta}_2)=R(\theta_1+\theta_2)$ Where R( ) is a rotation function and θ_1, θ_2 are angle of rotation. (08)
   d) Convert these homogeneous points to Cartesian (0,1,3,2),(2,1,3,4),(2,4,3,5). (07)

3. a) When performing Cohen-Sutherland line clipping, how do we use outcodes to check for trivial rejection (what exact operation and comparison would you use?). (06)
   b) What are the new coordinates of the diamond shaped polygon whose vertices are A(-1,0), B(0,-2), C(1,0) and D(0,2). About the horizontal line y=2 after the reflection. (11)
   c) Derive the perspective projection matrix. In a perspective projection matrices with $z_0=6$, scene point (-20,8,15) projects to what (x,y) point on the virtual image plane? To what (x,y) point on the real image plane? (11)
   d) Describe the traditional graphics pipeline and explain what is done in each stage. (07)

4. a) Use the Liang-Bersky algorithm to clip the lines in following figure (12)



b) What are the steps to calculate a transformation matrix to find the mirror reflection of an object with respect to an arbitrary plane whose normal is N and a point P is on the plane. (10)

c) Derive a 3D transformation matrix to rotate an object by θ degree with respect to z-axis. (09)

d) State the name of different types of parallel and perspective projections. (04)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Why do we need illumination models? Derive the Phong specular reflection model and combine diffuse and specular reflection. (13)

b) What is error diffusion? Write the algorithm for error diffusion image representation. (08)

c) Describe the strength and weakness of Ray tracing, using the theory of "The Rendering Equation". (06)

d) Compare between Phong surface rendering method and Gouraud surface rendering method. Which is better and why? (08)

6. a) Consider a triangle which has three vertices as $P_1(1,2)$, $P_2(5,0)$ and $P_3(3,4)$. A scan line S go through it and intersection points are $P_4(3,1)$ and $P_5(4,1)$. Inside the triangle there is a point P(3.5,1) on scan line. Normals for three vertices are $N_1(0,1,0)$,$N_2(0,0,1)$ and $N_3(0,0.7,0.7)$. We also have the following values: L=(0,0.7,0.7), v=(0,0,1) ,K_a=0.1,I_a=0.5,K_d=0.5, K_s=0.2, n_s=1,I_L=0.6, where symbols denote usual meanings. Find Intensity I_4, I_5, and I_p on scan line with Gouraud shading. (22)

b) Prove that reflected ray path, (13)

$T=\frac{n_i}{n_r}u-(\cos\theta_r-\frac{n_i}{n_r}Cos\theta_i)n$, where u is incoming ray and n is normal.

7. a) Let S_1, be a sphere of radius 8 centered at (2,4,1) and S_2 a sphere of radius 10 centered at (10,-2,-5). Determine if a ray with s=2J+5K and d=I-2K intersects the spheres and draw necessary diagram to show the intersections. (14)

b) What do you mean by depth buffer method and z-fighting? (06)

c) Define Hermite Surface. Find geometry vector G_{HX} for Hermite surface. Also find twist of the vector. (10)

d) What is terrain in computer graphics? How do you generate terrain? (05)

8. a) What is particle system? How particles are generated? (07)

b) Define fractal. What are the classifications of fractal objects? How is fractal generated? (09)

c) What do you know about geometric continuity? Explain G^0, G^1 and G^2 geometric continuity? (08)

d) Explain briefly about natural cubic splines and cubic Bezier curves. (11)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) "Asymmetric cryptosystems are based on clever application of number theory concepts" – individually establish it for RSA, ElGamal and Paillier cryptosystems about different keys. (12) [CO2]
   b) Establish usages of both block and stream ciphers in cases of two asymmetric and one symmetric cryptosystems. If not possible to deploy, also establish it. (12) [CO3]
   c) Discuss properties of digital signatures. Analyze them about ElGamal digital signature. (11) [CO2]

2. a) Discuss characteristics of cryptographic hash functions. Digital signature, asymmetric cryptosystem and hash function need to combine concurrently – why and how? (13) [CO1]
   b) Define and discuss characteristics of 'Zero Knowledge Proof'. Using a block diagram depict the symmetric cryptosystem along with the ways of the key distribution. (14) [CO2]
   c) Derive a bonding among order, prime, and primitive element considering the prime 13 and order 2. (08) [CO4]

3. a) Explain how triple DES performs data encryption. Using a block diagram illustrate the major steps of each cycle of AES algorithm. (11) [CO1]
   b) Discuss the following attacks: (i) Ciphertext-only, (ii) Chosen-plaintext, (iii) DoS, and (iv) DDoS. How Known-plaintext-attack can be mounted on RSA cipher? Using the required parameters, derive the mechanism. (15) [CO4]
   c) Define Firewalls. Discuss major types of Firewalls. Discuss various network threats elaborately. (09) [CO2]

4. a) Describing required conditions and parameters, discuss how RSA and ElGamal cipher can be used to develop product cipher. Derive re-randomization property of ElGamal cipher. (15) [CO3]
   b) 'Confusion' and 'Diffusion' are the ultimate goal of every cryptosystem – why? Discuss Shannon characteristics of good cipher. (09) [CO1]
   c) Establish a comparison between AES and DES. Show that the encryption and decryption operations of Caesar cipher work according to substitution cipher. (11) [CO2]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define the Discrete Logarithm Problem (DLP) and explain why it is considered computationally difficult to break in cryptography. (09) [CO1]
   b) In the Diffie-Hellman Key Exchange (DHKE) protocol, each participant selects a private key 'a' from the range $\{2, \dots, P\text{-}2\}$ where P is a prime number. Explain why the values $a = 1$ and $a = P - 1$ are excluded from this range. (11) [CO2]
   c) Explain scalar multiplication operation on Elliptic Curve (EC). Given an EC (15) [CO3]
   $$E: y^2 = x^3 + 3x + 5 \bmod 17$$ and point $P = (1,3), Q = (4,8)$,
   compute $2P = P + P = (x_3, y_3)$ and $P + Q = (x_4, y_4)$. Consider the equations:
   $$x_3 = S^2 - x_1 - x_2 \bmod P$$
   $$y_3 = S(x_1 - x_3) - y_1 \bmod P$$
   $$S = \begin{cases} \dfrac{3x_1^2 + a}{2y_1} \bmod P \\ \dfrac{y_2 - y_1}{x_2 - x_1} \bmod P \end{cases}$$

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6. a) Explain Post-Quantum Cryptography (PQC) and discuss its impact on symmetric and asymmetric cryptographic algorithms. (07)
   b) Explain the DHKE protocol and its key setup. Describe the EC-DHKE protocol and compare it with standard DHKE in terms of security and key size. (13) [C
   c) Define Transport Layer Security (TLS) and its purpose. Describe the TLS 1.2 handshake process. Compare TLS 1.2 and TLS 1.3 in terms of handshake efficiency and latency. Is O-RTT handshake supported in TLS 1.3, and under what conditions? (15) [CO4

7. a) Explain the N^2 key distribution problem in symmetric key systems. (05) [CO1]
   b) Define Elliptic Curve Discrete Logarithm Problem (ECDLP) and explain why computing it is considered difficult. (05) [CO2]
   c) Explain Elliptic Curve Cryptography (ECC) by describing its key generation, encryption and decryption processes. (10) [CO3]
   d) Illustrate the following web attacks with diagrams showing attacker-victim-server interactions, and explain effective mitigation techniques for each: (15) [CO4]
      i) Cross-site Request Forgery (CSRF).
      ii) Server-side Request Forgery (SSRF).
      iii) Cross-site Scripting (XSS).

8. a) Various malicious codes have different characteristics – discuss them briefly. Explain characteristics of good viruses. (05) [CO2]
   b) Define the Point at Infinity in ECC and explain the conditions under which it is reached. (06) [CO2]
   c) Explain homomorphic encryption and its types. Derive the equations of homomorphic operations in the Paillier cryptosystem and illustrate each with an example. (12) [CO3]
   d) Illustrate a Man-in-the-Middle (MITM) attack on the DHKE protocol and discuss methods to prevent such attacks in public key cryptosystems. (12) [CO3]

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2023
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define and discuss goals of computer security. Using a block diagram show relation and explain terminologies of any cryptosystem. (08) [CO1]
   b) Discuss Shannon characteristics to describe characteristics of good cipher. "Ciphertext size should not be larger than plaintext" – show how RSA and ElGamal cryptosystems retain this criterion. (14) [CO2]
   c) Define and discuss characteristics of zero-knowledge proof. How and why digital signatures with public key cryptography is required to combine? Explain the procedure. (13) [CO2]

2. a) Using block diagrams show that asymmetric cryptosystem is alluring than symmetric cryptosystem. Illustrate the major steps of each cycle of AES algorithm using an appropriate diagram. (13) [CO1]
   b) Discuss characteristics of cryptographic hash function. Asymmetric cryptosystem and hash technique need to combine concurrently – why and how? (13) [CO1]
   c) Various malicious codes have different characteristics – discuss them briefly. Explain characteristics of good viruses. (09) [CO2]

3. a) Discuss properties retained by digital signatures (DS). Derive the steps of ElGamal digital signature. Show that it is capable to satisfy properties of DS. (13) [CO3]
   b) Define the term cryptanalysis. Explain three terms allied with attack on cryptanalysis. (10) [CO1]
   c) Explain the working procedure of Double DES and Triple DES. Establish a comparative analysis between AES and DES based on various features of design. (12) [CO2]

4. a) ElGamal cipher can be developed to form 'product cipher' and 're-randomized cipher'. How? Show step by step. (14) [CO4]
   b) Derive encryption-decryption procedure of Caesar cipher and Vernam cipher. For RSA algorithm from $ed \equiv 1 \; mod \; \varphi(n)$, establish that many values may exist for d for a single value of e, where symbols have their usual meanings. (14) [CO3]
   c) Derive a bonding among order, prime and primitive element considering the prime 13 and order 2. (07) [CO4]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are covert channels? Differentiate between error and fault. (08) [CO2]
   b) What is 'Salt' in rainbow table attack? How does viruses gain control in file directory? (10) [CO1]
   c) What is segmentation fault in buffer overflow attack? Explain with figure. Also, discuss the countermeasures of buffer overflow attack. (10) [CO2]
   d) How webcams are used in Distributed Denial of Service (DDOS) attack? (07) [CO3]

6. a) Explain how paging removes the drawbacks of segmentation. (08) [CO3]
   b) Show how integer overflow attack compromises a program's reliability and security. (15) [CO4]
   c) How does a Demilitarized Zone (DMZ) network work? "Paillier cryptosystem permits addition of two ciphertexts, or one ciphertext and one plaintext or multiplication of one ciphertext and one plaintext" – justify it. (12) [CO4]

7. a) "Format string vulnerability is responsible to crash a program" – explain it. (05) [CO2]
   b) Explain relocation with example. What are the advantages of segmentation in operating system? (10) [CO2]
   c) Write short notes on the followings: (12) [CO1]
      (i) Trojan horse, (ii) Boot sector viruses, (iii) Dictionary attack.
   d) What is macro? Discuss how does document virus attack work in excel file using Dynamic Data Exchange (DDE). (08) [CO4]

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8. a) What is a virus signature? How does antivirus work on a system? (10) [CO2]
   b) Differentiate between Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS). (10) [CO2]
   c) What is light weight cryptography? Give an example and working procedure of it. (10) [CO4]
   d) Describe Stateful Inspection firewalls. (05) [CO1]

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2022
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Suppose to withdraw money from your Bank account when you put a signature over your Bank cheque, what are the properties essential to retain concerning it? Explain them. Now, analyze that RSA algorithm is capable to fulfill those properties or not. Establish your intuition relating the specific portion of RSA algorithm. (13)
   b) Differentiate between 'Cryptography' and 'Cryptanalysis'. Justify them, i.e., how they are associated with Caesar cipher technique. (10)
   c) List the trends in case of any good cipher. Analyze relevancy of 'Known-Plaintext-Attack' with RSA encryption algorithm. (12)

2. a) Discuss major goals (i.e., CIA) of security. Using a diagram state and explain ingredients of a cryptosystem. (09)
   b) Explain properties of cryptographic hash function. How and why data encryption and hash techniques need to perform concurrently? Propose an outcome from your thinking. (11)
   c) Discuss pros and cons of 'Stream cipher' and 'block cipher'. Assess the working procedures of Caesar cipher, RSA cipher, Vernam cipher etc., with block and Stream ciphers. (15)

3. a) Relating order, prime and primitive element, demonstrate a bonding. Also, judge your bonding for a prime 13, and order 2. (08)
   b) Explain how triple DES performs data encryption. Using a block diagram illustrate the major steps of each cycle of AES algorithm. (11)
   c) Discuss characteristics maintained by "Good Viruses". (06)
   d) Considering two messages m_1 and m_2, develop a model of ElGamal product cipher. Also demonstrate the decryption technique. Step wise write the whole procedure. (10)

4. a) Design a procedure to develop data encryption and digital signature for RSA algorithm together. (08)
   b) "ElGamal digital signature is based on discrete logarithm"—how discrete logarithm manages key generation, signing, verification, etc.? Explain. (11)
   c) How ElGamal cipher can be modified to re-randomize itself? Explain. (07)
   d) Discuss various aspects that make computer networks vulnerable. (09)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain relocation of address with example. What are the complementary goals to protect object of OS? (09)
   b) Discuss how ECC works based on the basic of trapdoor function. (10)
   c) Explain the types of Salami attack. What are the impacts and preventions of DHCP spoofing attack? (10)
   d) Differentiate among black hat, white hat and grey hat hackers. (06)

6. a) What is polymorphic virus? How does an Email impersonation attack work? (08)
   b) What are the considerations before committing to a hybrid firewall solution? Let, a businessman works with a public website, accessible from the internet, which put the entire internal network at high risk. Explain which network can help him to conduct his business securely and how? (15)
   c) Explain with figure how paged segmentation removes the problems associated with segmentation and paging. (12)

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7. a) Discuss the problems created by format string attack and countermeasures for them. (15)
   b) Demonstrate the ways to keep devices safe from wiretapping. (12)
   c) Which one is preferable, RSA or, ECC? Explain why? (08)

8. a) Suppose an attacker may introduce extra code, sending new instructions to the application to gain access to IT systems. If attackers know the memory layout of a program, they can intentionally overwrite the executable code with their own code. What kind of attack is this? What are the countermeasures for it? (15)
   b) Explain how documents virus works in Microsoft's excel Microsoft's word file using DDE protocol. (10)
   c) What are the responsibilities of firewall administrators? What are the challenges of intrusion detection system? (10)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Explain the terms: 'Security services', 'Cryptosystem', 'Cryptoanalysis' along with the major goals of security. Also explain a conventional cryptosystem by using a proper diagram. (13)
   b) Specify Shannon characteristics to explain good ciphers along with 'confusion' and 'diffusion'. Using a block diagram depict the symmetric cryptosystem along with the ways of the key distribution. (14)
   c) Discuss the benefits of an asymmetric cryptosystem. For RSA algorithm from $ed \equiv 1 \; mod \; \varphi(n)$ show that for a single value of e, there may exist many values of d, where symbols have their usual meanings. (08)

2. a) Define chosen-plaintext attack. Show how RSA algorithm suffers with this attack. (08)
   b) Explain the working procedure of Caesar cipher. How it is an example of substitution ciphers. (07)
   c) Discuss the properties of digital signatures. RSA algorithm can be used to implement product cipher. How? Show it? (10)
   d) Discuss the techniques, pros and cons of block and stream ciphers. What are the trends of maintaining security? Explain them. (10)

3. a) Explain the working procedure of ElGamal encryption algorithm. Why it is said to be a probabilistic encryption algorithm? Explain. (13)
   b) Explain ElGamal digital signature algorithm in detail. Also give a working example of it. (13)
   c) Discuss the major characteristics of cryptographic hash function? Why it is one-way? (09)

4. a) What is firewalls? Discuss the characteristics of various malicious code. (09)
   b) Explain AES algorithm with a block diagram. Establish a comparative study between AES and DES. (14)
   c) Discuss the characteristics of good viruses. (05)
   d) How triple DES is better than DES algorithm? Explain primitive element of Elgamal cryptosystem. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is segmentation fault? Explain Buffer Overflow attack with proper code and stack diagram. (15)
   b) Show how Integer Overflow attack compromises a program's reliability and security. (15)
   c) Explain document virus using Microsoft's Dynamic Data Exchange (DDE) protocol. (05)

6. a) What are the problems associated with Segmentation and Paging? Explain how Paged Segmentation removes them. (15)
   b) Explain some countermeasures that are used against Buffer Overflow attack. (15)
   c) What is format string? (05)

7. a) Explain attacks on format string vulnerability. (15)
   b) Write short notes on the followings: (12)
      (i) DHCP Spoofing Attack. (ii) Rainbow Table Attack.
      (iii) Polymorphic Viruses. (iv) Dictionary Attack.
   c) What are covert channels? Differentiate between errors and faults. (08)

8. a) How does viruses attach with original programs? (10)
   b) Explain the working principle of Elliptic Curve Cryptography (ECC). What are the reasons to prefer ECC over RSA? (10)
   c) What is salt? How does it help against Rainbow Table attack and make password hashing more secure? (10)
   d) Explain SQL injection and web bug. (05)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer ANY THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer ANY THREE questions from this section in Script A)

1. a) Define the terms: 'confusion', 'diffusion', 'cryptography' and 'cryptanalysis'. Discuss Shannon characteristics of good cipher. (12)
   b) 'RSA is a commutative like cryptosystem' – how? Explain deliberately. (08)
   c) Define cryptosystem. Discuss about the ingredients of a cryptosystem. (08)
   d) Define 'known-plaintext' attack. How this attack can mount over RSA? (07)

2. a) Discuss characteristics of digital signature. Explain ElGamal digital signature algorithm with an example. (12)
   b) Digital signature and public key cryptography can be combined – why and how? Explain. (08)
   c) Using an example explain ElGamal cryptosystem. How does it relate with discrete logarithm problem? (10)
   d) Discuss threats to e-mail. (05)

3. a) Discuss 'stream cipher' and 'block cipher'. Explain a specific key exchange protocol to distribute a key. (13)
   b) Explain AES algorithm using a block diagram. Make a comparison between DES and AES. (13)
   c) Define sensitive data. Discuss the factors that can make data sensitive. (09)

4. a) What is meant by homomorphic encryption? Show that RSA and ElGamal cryptosystems are homomorphic. (10)
   b) What are the parameters that make a computer network vulnerable? Discuss them briefly. (10)
   c) Discuss the dimensions of reliability and integrity of database. What are the requirements for database security? Explain them. (10)
   d) Explain the mechanism of triple DES algorithm. (05)

SECTION B

(Answer ANY THREE questions from this section in Script B)

5. a) What is salt and how does it make password hashing more secure? Also explain rainbow table attack and how salt can help against rainbow table attack. (15)
   b) How does the attacker know what algorithm and salt to use in a dictionary attack? (06)
   c) Explain Buffer Overflow attack with proper code and stack diagram. Also discuss about the countermeasures that are used against Buffer Overflow attack. (14)

6. a) Explain how Format String attack works with appropriate activation record diagrams. Also explain different uses of Format String attack with proper example(s) and stack diagrams. (15)
   b) Demonstrate Integer Overflow attack using suitable example(s). (07)
   c) Explain SQL Injection attack with practical example. Also discuss about the prevention techniques used against SQL Injection attack. (13)

7. a) Explain Persistent, Reflected and DOM-based Cross-Site Scripting (XSS) attacks with example(s). (15)
   b) Explain the methods of preventing XSS attacks. (10)
   c) What are web bugs? Explain how attackers bypass CAPTCHA protection. (10)

8. a) Explain how combining Paging with Segmentation provides better performance for protection in general purpose operating system compared to using them separately. (09)
   b) What is a Covert Channel? How to create Covert Channels? (06)
   c) Briefly explain the following attacks: (i) Ransomware, (ii) Salami Attack, (iii) Tapjacking, (iv) Document virus using Microsoft's Dynamic Data Exchange. (16)
   d) What is Access Control Directory? (04)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Through Shannon characteristics, discuss the characteristics of good cipher. Define the terms: 'confusion' and 'diffusion'. (10)
   b) Define known-plaintext attack. How it can be associated with RSA cryptosystem? (10)
   c) On same modulus using RSA, how multiple parties can encrypt data? (09)
   d) Show that "RSA cryptosystem is homomorphic". (06)

2. a) Discuss properties of digital signature. Explain ElGamal digital signature algorithm with an example. (12)
   b) Explain the working procedure of ElGamal cryptosystem specifying that it is based on discrete logarithm problem. (10)
   c) Make a comparison between the mechanism of AES and DES. Describe AES algorithm using a block diagram. (13)

3. a) Define cryptographic hash function. Discuss properties of it. (07)
   b) How and why you combine digital signature and public key cryptography? Explain. (08)
   c) Discuss the parameters that make a computer network vulnerable. (10)
   d) Define sensitive data. Discuss the factors that can make data sensitive. (10)

4. a) How documents can be signed using public key cryptography and one way hash function? Discuss. (07)
   b) Discuss 'stream cipher' and 'block cipher'. Explain a specific key exchange protocol to distribute a key. (13)
   c) How 'primitive element' relates with ElGamal? (08)
   d) What are the requirements of database security? Explain them. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Give an intuitive explanation of the working principle of Elliptic Curve Cryptography (ECC). Also explain how encryption and decryption work in ECC. (13)
   b) Explain ECC based digital signature with mathematical proof. (10)
   c) What are the reasons to prefer ECC over RSA? (05)
   d) What is tapjacking? Explain the prevention mechanism used against tapjacking in context of Android operating system. (07)

6. a) Write short notes on following attacks: (16)
      i. Page-in-the-Middle.
      ii. Program Download Substitution.
      iii. SQL Injection.
      iv. Man-in-the-Browser.
   b) Explain how attackers use User-in-the-Middle attack to bypass CAPTCHA protection. (06)
   c) What are web bugs? Explain how websites can be protected against unwanted change both from administrator's and users' perspective. (13)

7. a) With appropriate code and stack diagram explain how Buffer Overflow can be exploited to execute attacker's provided code. Also explain some common counter measures that are used against Buffer Overflow attack. (18)
   b) Explain the uses of Format String attack with appropriate example(s) and stack diagrams. (11)
   c) Explain Integer Overflow attack using suitable example(s). (06)

8. a) Write short notes on the followings: (14)
      i. Salami Attack.
      ii. Covert Channel.
      iii. Document Virus using Microsoft's Dynamic Data Exchange (DDE).
   b) Explain the problems associated with Access Control Directory. (06)
   c) Explain the problems associated with Segmentation and Paging. Also discuss how combining Paging with Segmentation provides better performance for protection in general purpose operating system. (15)

CR CSE 14

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2017
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define cryptosystem. Discuss about the ingredients of a cryptosystem. (08)
   b) Through Shannon characteristics, discuss the characteristics of good cipher. Compare between block and stream ciphers. (10)
   c) Define chosen-plaintext attack. Discuss a specific key exchange protocol to distribute a key. (10)
   d) Define cryptographic hash function. Discuss some properties of it. (07)

2. a) Explain the mechanism of triple DES encryption. Using a block diagram explain AES algorithm. (13)
   b) Discuss properties of digital signature. Explain Elgamal digital signature algorithm with an example. (12)
   c) A computer network can become vulnerable. How? Explain the reasons. (10)

3. a) In RSA, how $ed = 1$? On same modulus using RSA, how multiple parties can encrypt data? (10)
   b) Define sensitive data. Discuss the factors that can make data sensitive. (10)
   c) Explain why and how digital signature and public key cryptography can be combined? (08)
   d) Explain the steps of signing documents with public key cryptography and one way hash functions. (07)

4. a) Discuss the dimension of reliability and integrity of database. What are the requirements of database security? Explain them. (10)
   b) Define 'primitive element' with example. How it is associated with Elgamal cryptosystem? (10)
   c) Define 'confusion' and 'diffusion'. Give an overview of DES encryption technique. (08)
   d) How is trade secret protection vanished by 'reverse engineering'? Explain. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Consider the code snippet shown in below: (07)

```
int main()
{
  int pass = 0;
  char buff[3];
  printf("\n Enter the password:\n");
  gets(buff);
  if(strcmp(buff,"cse"))
  {
   printf("\n Wrong password\n");
  }
  else
  {
   printf("\n Correct password\n");
   pass = 1;
  }
if(pass)
 {
  /*Give admin priviledge to the  user*/
  printf("\n Admin priviledge given to the user\n");
  }
  return 0;
}
```

Explain how buffer overflow attack can be utilized to get admin privilege without even knowing the actual password.

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b) With appropriate stack layout explain how buffer overflow attack can be used to modify return address of a function. Design an input payload for the code snnipet shown in the following to transfer the execution to secret function. (15)

```
void secretFunction()
{
printf("Can't be called");
}
void test()
{
char buff[10];
gets(buff);
}
int main()
{
test();
return 0;
}
```

```
Dump of assembler code for function
secretFunction:
0x004014d0<+0>: push %ebp
0x004014d1<+1>: mov %esp, %ebp
.
.
.
```

c) Explain how buffer overflow can be exploited to execute attacker's provided code. (06)
d) Explain integer overflow attack with an example. (07)

6. a) Discuss some prevention mechanisms that are used against buffer overflow attack. (12)
b) Explain why the code snnipet shown in the below is vulnerable to format string attack. (06)

```
int main (int argc, char *argv[])
{
char *secret = "test";
char *secret2 = "test2";
printf(argv[1]);
return 0;
}
```

c) Explain how format string vulnerability can be utilized for: (i) Crashing a program (ii) Viewing the stack (iii) Viewing memory at any location (iv) Writing an integer to nearly any location in process memory. (17)

7. a) Suppose, you want to perform brute force attack on a system which allows password length between 5 to 8 characters limited to character set:{'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', '1', '2', '3', '4'}. Now, write a pseudocode to generate all possible passwords for your brute force attack. (12)
b) Explain how the tapjacking attack works. (05)
c) What are covert channels? Explain how covert channels can be created. (12)
d) Write short notes on: (i) Trapdoors (ii) Keystroke logging (06)

8. a) Explain how Dynamic Data Exchange (DDE) and macro in Microsoft Office can be exploited to run malicious code. (05)
b) Explain the working principle of ransomware. (05)
c) Explain the terms: i) denial-of-service (DoS) attack and ii) distributed DoS attack. (08)
d) Discuss the threats to E-mail. Also discuss how to defend against it. (07)
e) What is meant by malleability? Show that both RSA and Elgamal are malleable cryptosystem. (10)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2016
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define the terms: cryptography, cryptanalysis. Discuss Shannon characteristics of good cipher. (10)
   b) Define known-plaintext attack. How can it be applied for RSA cryptosystem? (10)
   c) Explain with example: a) substitution ciphers, b) transposition ciphers. Discuss a specific key exchange protocol to distribute a key. (10)
   d) How is trade secret protection vanished by 'reverse engineering'? Explain. (05)

2. a) What are the characteristics of digital signature? Explain ElGamal digital signature algorithm with an example. (12)
   b) Define cryptographic hash function. Discuss some properties of it. (08)
   c) Explain how $ed = 1$ in RSA encryption algorithm? [Where d and e have their usual meanings]. (07)
   d) Digital signature and public key cryptography can be combined. Why and how? Explain. (08)

3. a) Discuss the steps to sign documents with public-key cryptography and one way hash functions. (07)
   b) What are the parameters that make a computer network vulnerable? Discuss them briefly. (10)
   c) What are the limitations of RSA cryptosystem? Make a comparison between RSA and ElGamal cryptosystems. (10)
   d) Explain the terms: (i) denial-of-service (DOS) attack and (ii) distributed denial-of-service (DDOS) attack. (08)

4. a) Discuss the threats to E-mail. Also discuss how to defend against it. (07)
   b) Define sensitive data. Discuss the factors that can make data sensitive. (10)
   c) Discuss the dimensions of reliability and integrity of database. What are the requirements for database security? Explain them. (10)
   d) What is meant by homomorphic encryption? Explain with examples. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) If two users share access to a segment, they must do so by the same name. Must their protection rights to it be the same? Why or why not? (05)
   b) Why is Kerberos a credible security solution? Describe, how is authentication and access authorization implemented in Kerberos. (10)
   c) Make short notes over (i) Loose-lipped systems (ii) Access control matrix (08)
   d) Briefly explain the operational modes of Intrusion detection system. (12)

6. a) What do you mean by Wired Equivalent Privacy (WEP)? Briefly explain all the possible attacks on wifi protected access (WPA & WPA2) networks. (09)
   b) Draw and explain each of the constructing components of wireless data frame. (09)
   c) Suppose Alice is sending an Email to Bob, but Alice wants her Email to be anonymous. How can Alice achieve this? What are the threats about which Alice is afraid of? (12)
   d) Describe how RSA SecurID suite works. (05)

7. a) Suppose, while browsing web, your machine got attacked by a virus. What are the best places for that virus to make itself hidden and gain control over your system? Describe the situation before and after this infection. (09)
   b) What are the error detecting codes that can be used to detect when digital objects are surreptitiously altered? (10)

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c) Suppose, an attacker wants to attack a target server which has higher bandwidth than the attacker. Explain an attack with necessary figure by which the attacker can overflow the target server's maximum connection capacity. (10)

d) Write down a short note on Smurf Attack. (06)

8. a) What do you mean by security through obscurity and challenge-response system? Briefly explain. (10)

b) How would you distinguish a risk from other project events? Discuss the strategies for dealing with risks. (10)

c) What type of attacks "Keystroke loggers" is? Briefly explain. (08)

d) Suppose, a system is being accessed by different users with different privileges. What are the principles that you as an administrator would adopt while giving access to the users. (07)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2013
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define the terms: 'Ciphertext-only attack', 'Known-plaintext attack' and 'Chosen-plaintext attack'. Discuss Shannon characteristics of good cipher. (12)
   b) Explain signing and verification mechanisms of ElGamal Digital Signature scheme with an example. (13)
   c) Discuss how can a computer network become vulnerable? Briefly explain the reasons. (10)

2. a) Discuss the terms: 'Stream ciphers' and 'Block ciphers'. Discuss a specific key exchange protocol to distribute a key. (12)
   b) Discuss the characteristics of relation between the two keys of public key cryptography. Explain how RSA keys are generated employing extended Euclid's gcd algorithm. (13)
   c) Define sensitive data. Discuss the factors that can make data sensitive. (10)

3. a) Discuss the basic protocol of signing documents with public-key cryptography and one-way hash function. (08)
   b) ElGamal encryption mechanism is non-deterministic, how? Explain with a suitable example. (12)
   c) Explain the steps of AEs algorithm with a block diagram. (08)
   d) Discuss the threats to E-mail. Also discuss how to defend against it. (07)

4. a) Define cryptographic hash function. Discuss the properties of it. (08)
   b) Explain the mechanism of triple DES encryption. (07)
   c) Explain the homomorphic encryption mechanism with possible merits and demerits. (10)
   d) Discuss the dimensions of reliability and integrity of database. What are the requirements for database security? Explain them. (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is information security? "Security measures must balance between security and productivity"- justify. (12)
   b) Define risk and risk analysis. Discuss the steps of risk analysis. (11)
   c) Write notes on- i) Confidentiality, ii) Integrity, and iii) Availability (12)

6. a) Differentiate between cryptography and cryptanalysis. Encrypt the following message using column transposition method with five column and Vernam method. (15)
   "CAESAR WAS KILLED BY BRUTUS"
   b) What is the basic theme of OS security? What is fence register? How does fence register provide extra benefit in relocation method? Explain. (12)
   c) What are the differences between paging and segmentation? (08)

7. a) What are malicious error and non-malicious error? Discuss six categories of non-malicious errors. (10)
   b) Explain risk exposure and risk leverage with example. (10)
   c) Discuss exhaustive attack and one time password with an example. Differentiate between copyright, patents, and trade secret. (15)

8. a) Differentiate between laws and ethics. Why is computer crime hard to define and hard to prosecute? Explain. (10)
   b) Explain virus signature with an example. Differentiate between virus and worm. (10)
   c) Explain buffer overflow with an example. How does buffer overflow affect code and data? Discuss with necessary figures. (15)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2012
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define cryptosystem. Discuss the ingredients of a cryptosystem briefly. (08)
   b) Discuss the characteristics of asymmetric cryptosystem. Make a comparison between block and stream ciphers. (10)
   c) Discuss the basic protocol of public-key algorithm so that it can be used for digital signatures. Does it satisfy the characteristics of signature? (10)
   d) Why and how one-way hash function is associated with digital signature? (07)

2. a) How RSA keys are generated and what are the limitations of RSA encryption algorithm? Explain with example. (10)
   b) Define 'primitive element' with example. How is it related to ElGamal cryptosystem? (10)
   c) What is meant by malleability? Show that RSA and ElGamal are malleable cryptosystem. (10)
   d) Explain triple DES mechanism. (05)

3. a) Define cryptographic hash function. Discuss some properties of it. (08)
   b) How ElGamal is a probabilistic cryptosystem? Explain. (08)
   c) Define symmetric key cryptosystem. Explain the steps of AES encryption algorithm. (12)
   d) What is meant by 'confusion' and 'diffusion'? Give an overview of DES algorithm. (07)

4. a) Explain the terms: i) denial-of-service (DoS) attack and ii) distributed denial-of-service (DDoS) attack. (08)
   b) How is trade secret protection vanished by 'reverse engineering'? Explain. (08)
   c) Discuss the dimension of reliability and integrity of database. What are the requirements for database security? Explain them. (11)
   d) Define sensitive data. Discuss the steps of interaction among a user, a trusted front end and a DBMS. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are the parameters that can make a computer network vulnerable? Discuss them briefly. (10)
   b) What do you mean by security and computer security? "Security measures must balance between security and productivity"- Justify the statement. (10)
   c) Discuss the threats to electronic mail. (07)
   d) Define risk analysis. Discuss the steps of risk analysis. (08)

6. a) Discuss about risk exposure and risk leverage. (10)
   b) How can fence register be used to provide extra benefit in relocation method? What are the advantages of two pair base and bound register? (13)
   c) Discuss tagged architecture with pictorial view. (12)

7. a) Discuss about exhaustive attack and one time password. (10)
   b) How does buffer overflow affect code and data? Discuss elaborately. (15)
   c) What is computer virus? Differentiate virus and worm. How is virus attached? (10)

8. a) How does virus gain control? Discuss with pictorial view. Differentiate copyright, patents, and trade secret. (10)
   b) What are the qualities of virus? What is virus signature and polymorphic virus? (10)
   c) Differentiate laws and ethics. "Computer crime is hard to define and hard to prosecute"- Justify the statement. (15)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2011
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define digital signature. Write some properties of digital signature. (09)
   b) How RSA algorithm is used for digital signing? Explain. (08)
   c) Define the terms: cryptosystem, cryptanalysis. Discuss the benefits of asymmetric cryptosystem. (10)
   d) What is meant by homomorphic encryption? Explain with an example. (08)

2. a) Explain ElGamal encryption algorithm with an example. (13)
   b) What is meant by 'confusion' and 'diffusion'? Give an overview of DES algorithm. (09)
   c) Explain AES encryption algorithm. (09)
   d) Give an example of transposition cipher. (04)

3. a) In RSA algorithm, how $ed = 1$? [Where e and d have their usual meanings]. (07)
   b) What are the goals of security? Explain them briefly. (06)
   c) What are the parameters that can make a computer network vulnerable? Discuss them. (13)
   d) Discuss the requirements for database security. (09)

4. a) What is the purpose of encryption in a multilevel secure database management system? (08)
   b) Define sensitive data. What are the factors that can make data sensitive? Discuss them. (12)
   c) Explain the terms: i) denial-of-service (DoS) attack and ii) distributed denial-of-service (DDoS) attack. (09)
   d) What is cryptographic checksum? Explain with example. (06)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) "Security measures must balance between security and productivity"- Justify the statement. (08)
   b) Define risk analysis. Discuss the steps of risk analysis. (10)
   c) What is security planning? Write down the issues of security planning. (10)
   d) How does operating system offer protection at several levels? (07)

6. a) ~~What is~~ How does virus gain control? Discuss with pictorial view. (10)
   b) What are the qualities of virus? What is virus signature and polymorphic virus? (12)
   c) "Computer crime is hard to define and hard to prosecute"- Justify. (13)

7. a) Discuss about exhaustive attack and one time password. (10)
   b) How buffer overflow affects code and data? Discuss elaborately. (15)
   c) What is Computer Virus? Differentiate between Virus and Worm. How Virus is attached? (10)

8. a) Write short notes on- (i) Risk exposure (ii) Risk leverage. (10)
   b) How fence register can be used to provide extra benefit in relocation method? What is the advantage of two pair base and bound register? (12)
   c) Discuss the tagged architecture with pictorial view. (13)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2010
Department of Computer Science and Engineering
CSE 4215
Information Security and Control

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is public key encryption? How does it differ from private key encryption? (10)
   b) Define homomorphic encryption. RSA and ElGamal cryptosystems are homomorphic – explain. (12)
   c) Why is RSA encryption algorithm deterministic? In RSA, how $ed=1$? [Where e and d have their usual meanings]. (13)

2. a) ElGamal encryption algorithm is probabilistic – explain. (11)
   b) Explain the steps of AES encryption algorithm. (12)
   c) Define hash function. What are the applications of hash function in cryptography? (07)
   d) Explain triple DES mechanism. (05)

3. a) What are the requirements for the security of database? Explain them precisely. (10)
   b) Define sensitive data. What are the factors that can make data sensitive? Discuss them. (12)
   c) What are the parameters that can make a network vulnerable? Among them, explain 'anonymity' and 'sharing'. (13)

4. a) Explain RSA digital signature algorithm with an example. (12)
   b) What is the limitation of existing internet system? Explain denial-of-service (DoS) attacks. (10)
   c) Define authentication. What are the threats to e-mail? Explain. (13)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Make a comparison between copyrights and patents. (09)
   b) How trade secret protection is vanished by 'reverse engineering'? Explain. (08)
   c) How does operation system offer protection at several levels? (08)
   d) Explain paging combined with segmentation. What are the advantages and disadvantages of this technique for protecting in general purpose operating system? (10)

6. a) "Information, Security, and Control are three main ingredients of a secure system" – justify the statement. (09)
   b) What are the complementary goals of information security? Describe them vividly. (10)
   c) What is a threat in a computing system? Describe different types of threats with proper example(s). (10)
   d) Differentiate between AES and DES. (06)

7. a) What is data security? How does it differ from network security? (08)
   b) What is buffer overflow problem? How it affects in security implication of a system? (08)
   c) Differentiate between virus and worm. (06)
   d) What is virus gain control? Explain with proper figure(s). (08)
   e) How a virus can be identified? (05)

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8. a) What is directory system? Describe its advantages and disadvantages with appropriate figure(s). (12)
   b) How does military security model fulfill need–to–know rule and security requirements? (14)
   c) What is lattice? How is military security model represented by lattice? (09)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Briefly describe the role of different components of an image processing system. (06) [CO1]
b) A lunar rover captured a 4-bit image of moon-surface under poor lighting condition, then it transmitted multiple 6×6 RLC patches. You are given an RLC image of a patch and a PDF. Your task is to adjust the pixel intensity according to the given PDF, then stretch the contrast of the output/obtained image to its maximum possible range. (20) [CO3]
RLC image: <3,3>, <4,2>, <3,3>, <6,4>, <5,4>,
<3,5>, <7,2>, <4,5>, <6,4>, <7,4>
PDF:

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	0.06	0.07	0.1	0.15	0.2	0.15	0.1	0.07	0.05	0.05

c) Prove that 'convolution in time domain corresponds to multiplication in frequency domain'. (09) [CO2]

2. a) Explain the impact of wrong choice of order for contra-harmonic mean filter. (05) [CO2]
b) All the MRI images captured by an MRI machine contains periodic noises of different frequencies within the band (6 to 10). Design a butterworth filter to visualize the noise pattern. (10) [CO4]
c) Given a grayscale 4×4 image of $f(x,y)$ and a quantization matrix Q. Apply JPEG algorithm and compute the RMS error of compression-decompression. (20) [CO3]

$$f(x,y) = \begin{bmatrix} 52 & 55 & 61 & 66 \\ 70 & 61 & 64 & 73 \\ 63 & 59 & 55 & 90 \\ 67 & 73 & 78 & 85 \end{bmatrix} \quad Q = \begin{bmatrix} 16 & 11 & 10 & 16 \\ 12 & 12 & 14 & 19 \\ 14 & 13 & 16 & 24 \\ 14 & 17 & 22 & 29 \end{bmatrix}$$

3. a) Compute the Golomb code for the following representation: (10) [CO3]
i) $G_{12}(-15)$
ii) $G^2_{exp}(13)$
b) Apply pruning on the given binary image (Fig: 3(b)) using the structuring elements ($B1 \sim B8$) where- (15) [CO2]

$B1$=

x	0	0
1	1	0
x	0	0

$B5$=

1	0	0
0	1	0
0	0	0

$B2, B3$ & $B4$ are 90° rotated version of B_{i-1} from $B1$
$B6, B7$ & $B8$ are 90° rotated version of B_{i-1} from $B5$

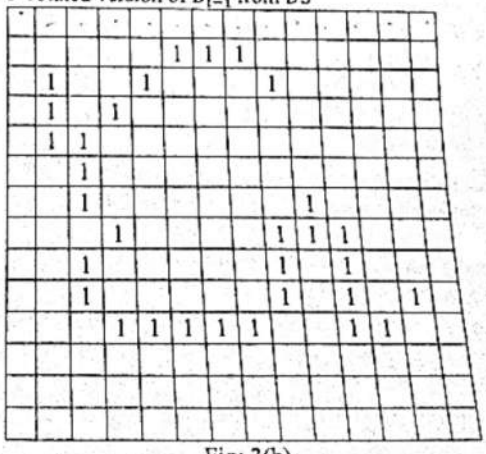

Fig: 3(b)

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c) Develop a mathematical model for degradation due to uniform linear motion. (10) [CO4]

4. a) Find the frequency domain filter $H(u,v)$ that works as a Laplacian filter in spatial domain, hence design a Wiener filter using $H(u,v)$ taking $SNR = 10$. (12) [CO3]

b) Explain the procedure of morphological reconstruction with its application. (10) [CO1]

c) A satellite captured an image of earth when one side of the earth is exposed to sunlight and other side remained dark, resulting a non-uniform illumination. Further, the memory of the satellite was lit by the cosmic rays causing an addition of very dense impulse noise to the stored image. What are the appropriate filters that can be used one after another to remove the impulse noises then to suppress the illumination? (13) [CO4]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Perform a full convolution on the following image using the given kernel. Be aware of the input and output indices. Use border replication padding while convolving. (10) [CO1]

	0	1	2	3
0	5	6	8	2
1	3	1	0	2
2	7	8	3	9
3	3	4	5	1

image

	-1	0	1
0	1	2	-2
1	-1	0	1

kernel

b) Prove that convolution with a 2D separable filter can be accomplished by performing two one-dimensional convolutions. (09) [CO2]

c) Suppose you want to segment the following image using normalized graph cut method proposed by Shi-Malik. Represent the image as a weighted graph considering 4-connected pixels and $\sigma_F = 4$; $\sigma_s = 1.5$, where symbols have their usual meaning. (16) [CO4]

8	4	9	25
5	11	22	28
9	2	27	30
29	27	28	31

6. a) Consider a grayscale image where foreground and background pixels are normally distributed with $\mu_f = 150, \mu_b = 50, \sigma_f = \sigma_b = 15$ and foreground pixels are one-third of total pixels. Find a minimum error threshold to separate the foreground from background. (09) [CO2]

b) For the following binary image, compute first two invariant *Hu* moments. Equations given below are for your convenience, where symbols bear usual meanings. (15) [CO3]

y \ x	0	1	2	3	4
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	1	1	0
3	1	1	1	1	0
4	0	0	1	1	0

$$\emptyset_1 = \eta_{20} + \eta_{02}; \emptyset_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$$

$$\eta_{pq} = \frac{\mu^{pq}}{\mu_{00}^{\gamma}}; \gamma = \frac{p+q}{2} + 1$$

$$\mu_{pq} = \sum\sum (x - \bar{x})^p (y - \bar{y})^q f(x,y); \bar{x} = \frac{m_{10}}{m_{00}}; \bar{y} = \frac{m_{01}}{m_{00}}$$

$$m_{pq} = \sum\sum x^p y^q f(x,y)$$

c) Briefly explain Harris-Stephens corner detection method to find interest points from an image. Why is this method not covariant to image scaling? (11) [CO2]

7. a) What does it mean by homography transform of an image? Explain how can you estimate homography matrix using direct linear transform from two images. (12) [CO2]

b) Following figure shows a RGB image with a white and a yellow region, where values in the parenthesis means RGB channel values. Compute edge magnitude and orientation at the diagonal line separating the regions using- (16) [CO3]

(i) Di Zenzo-Cumani color edge operator.

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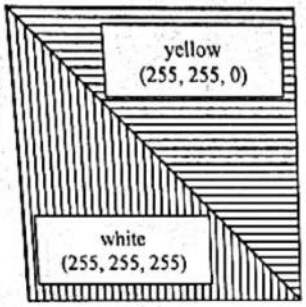
(ii) First convert the image to grayscale then apply monochromatic edge operator. Use simplest possible derivative operator.



c) State the pros and cons of histogram based texture descriptors with proper example. (07) [CO3]

8. a) How can you modify the Lucas-Kanade optical flow estimation method to handle reasonably large motion? Explain with necessary illustration. Do you need image warping? If so, why? (13) [CO3]

b) The following table shows some features extracted from some binary images along with their known classes. If you would like to use them as training set of your Naïve Bayes classifier, then, estimate the classifier parameters. Also, determine the class of the test sample (0.5, 0.42, 260, 0.59). (16) [CO4]

Form factor	Roundness	Compactness	Eccentricity	Class
0.455	0.327	276	0.587	Star
0.469	0.416	268	0.26	
0.48	0.34	262	0.64	
0.453	0.29	277	0.75	
0.78	0.66	161	0.46	Hexagram
0.79	0.58	159	0.66	
0.82	0.54	153	0.67	
0.84	0.56	148	0.62	

[Hint: Estimate μ_{ji}, σ_{ji} for features x_j and class c_i]

c) Explain how histogram intersection can be used as a distance measure for feature descriptor matching. (06) [CO1]

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2023
Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

TIME: 3 hours | FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates the course outcomes.

SECTION A
(Answer **ANY THREE** questions from this section in Script A)

1. a) For the following two signals $f(x)$ and $g(x)$, find $f(x) \star g(x)$, where $\star$ is a convolution operator. (07) [CO1]

$$f(x) = \begin{cases} 1 & 0 \le x \le 1 \\ 0 & otherwise \end{cases} \qquad g(x) = \begin{cases} 1/2 & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$$

b) You are assigned to smooth the following image while preserving the edges. You planned to use a bilateral filter of size 3×3, and Gaussian function for both spatial kernel and range kernel. If you use $\sigma_s = 0.6$, and $\sigma_r = 15$ for those kernels, then show the spatial, range, and bilateral kernel for the encircled position of the following image. (16) [CO2]

$$\begin{bmatrix} 10 & 12 & 76 & 70 \\ 6 & \textcircled{5} & 66 & 68 \\ 4 & 25 & 55 & 62 \\ 8 & 9 & 71 & 58 \end{bmatrix}$$

c) For the given filter, determine the DTFT $H(u, v)$ of the filter (assuming the origin is at the center), and sketch the one-dimension profiles $H(u, 0)$ and $H(0, v)$. (12) [CO3]

$$\begin{bmatrix} -1 & -3 & -1 \\ 0 & 0 & 0 \\ 1 & 3 & 1 \end{bmatrix}$$

2. a) Show that Laplacian sharpening is a special case of an Un-Sharp Mask (USM) filter. (07) [CO3]
b) Use required mathematical formulation to show that the between class variance used by Otsu's thresholding method can be written as (14) [CO1]

$$\sigma_B^2(k) = \frac{(m_G P_1(k) - m(k))^2}{P_1(k)(1 - P_1(k))}$$

c) Estimate the optical flow for the following successive video frames using Lucas–Kanade method. Use masks suggested by Horn-Schunck to compute necessary partial derivatives. Also use a 2×2 spatial coherence window, and show the estimated result for encircled position only. (14) [CO2]

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 2 & \textcircled{9} & 9 & 8 \\ 1 & 9 & 9 & 10 \\ 0 & 9 & 8 & 9 \end{bmatrix}$$
Frame at t

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 2 & 10 & 9 \\ 1 & 1 & 9 & 8 \end{bmatrix}$$
Frame at $t + 1$

3. a) Geometric transformation of the following image is done by an inverse polynomial mapping function such as: $Y = WP(X)$, where, (13) [CO3]

$$X = [x_1 \quad x_2]^T, Y = [y_1 \quad y_2]^T,$$
$$P(x) = [1 \quad x_1 \quad x_2 \quad x_1^2 \quad x_1 x_2 \quad x_2^2], \text{ and}$$
$$W = \begin{bmatrix} 0.1 & 1 & 0 & 0.01 & -0.01 & 0.01 \\ 0.4 & 0 & 1 & 0.02 & -0.01 & 0.04 \end{bmatrix}$$

Determine output intensity at (1, 2), if bilinear interpolation is used during transformation.

x_2 \ x_1	0	1	2	3	4
0	100	105	108	109	102
1	99	104	109	100	103
2	200	204	205	102	105
3	201	106	101	103	111
4	206	208	204	200	100

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b) Let two interest points are described by some feature descriptors. Provide at least two matching ideas, with necessary equations, that can be used to decide whether those interest points are same or not. Also mention the pros and cons of those matching ideas. (12) [CO4]

c) Write Niblack's adaptive thresholding algorithm. Mention its use-cases. (10) [CO1]

4. a) During Hough transform based line detection, generally we get similar vote count in a small neighborhood. Explain what strategy can be applied to locate the peak voting position in the accumulator array. (06) [CO4]

b) Apply a 3×3 vector median filter on the location $(2, 2)$ of the given RGB image. (13) [CO2]

$$R = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 3 & 2 & 1 & 2 \\ 3 & 0 & 2 & 4 \\ 2 & 1 & 0 & 1 \end{bmatrix} \quad G = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 3 & 1 & 0 \\ 3 & 0 & 2 & 2 \\ 1 & 2 & 2 & 3 \end{bmatrix} \quad B = \begin{bmatrix} 3 & 3 & 1 & 2 \\ 2 & 3 & 0 & 1 \\ 1 & 0 & 2 & 1 \\ 3 & 1 & 2 & 2 \end{bmatrix}$$

c) Use DiZenzo-Cumani operator to get the edge strength and orientation for the following color image: values inside the parenthesis means (r,g,b) color vector. Also mention the importance of the operator compared to monochromatic edge detection technique for color image. Show result only for the center pixel. (16) [CO3]

$$\begin{bmatrix} (3,2,1) & (2,1,2) & (2,3,0) \\ (1,3,1) & (3,0,1) & (2,2,3) \\ (1,1,0) & (0,2,4) & (1,1,1) \end{bmatrix}$$

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

a) Define digital image processing. Briefly discuss the fundamental steps in digital image processing. (10) [CO1]

b) Consider the image segment shown below. (12) [CO2]

	3	1	2	1 ← q
	2	2	0	2
	1	2	1	1
p →	1	0	1	2

Compute length of the shortest-4, shortest-8 and shortest-m paths between pixels p and q where, $v = \{1, 2\}$. If a particular path does not exist between these two points, explain why.

c) Consider the following 3-bit image. (13) [CO3]

i) Generate the equalized image by using global histogram equalization processing.

ii) Plot the histogram of the processed image.

2	3	3	5	7
5	4	3	1	6
1	5	2	4	7
2	3	6	4	1
6	5	3	2	4

6. a) Define morphological gradient. Discuss when do we use top-hat and bottom-hat transformation in grayscale image. (10) [CO2]

b) Let, you are asked to enhance dark areas of an image while leaving the bright area as unchanged as possible. Propose strategy with necessary formulation to handle the problem. Use example to explain that the proposed method works. (14) [CO3]

c) Briefly discuss the following morphological image processing operations with necessary equations. (11) [CO3]

i) Convex hull.

ii) Pruning.

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2022
Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

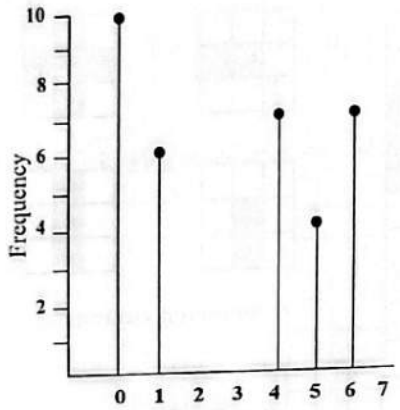
TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

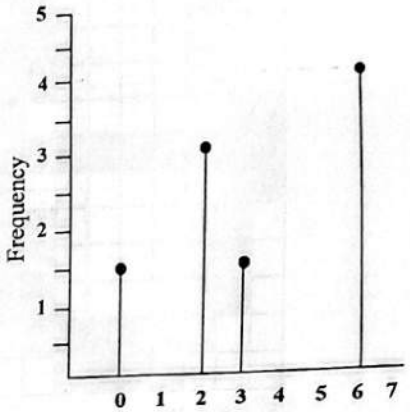
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Construct a modified image using histogram specification. The histograms of original and target images are given below. (20)



(a) Original Image



(a) Target Image

b) Use 8-connectivity and m-connectivity to find the shortest path between pixels (1, 1) with value 10 and (8, 8) with value 100, if V = {0, 10, 50, 65, 100, 106}. (10)

10	55	60	65	05
106	65	56	102	200
112	0	50	50	150
30	45	65	60	120
45	106	10	65	100

c) Let, you have computed the 2D DFT of a given image. The magnitude spectrum ranges from 0 to 10^5, but you want to display the image as an 8-bit gray scale image. For this purpose, you apply base-10 log transformation to the spectrum. What must be the normalizing constant of the transformation? (05)

2. a) Following is a 5×5 image with 5 distinct gray levels: (25)

18	18	4	12	12
10	10	4	12	4
10	4	12	4	15
12	10	15	15	10
15	12	4	10	4

Compare the efficiency of compressing the image using Huffman encoding and Golumb coding with parameter, $m = 5$.

b) Explain why transformation function to perform image complement approximation is different for each component in HSI model but similar for all components in RGB model. (10)

3. a) Consider a line of intensity values, {100, 129, 145, 56, 99} If it is quantized with 4 higher bit planes, compute the error and signal-to-noise ratios for the quantized data. (08)

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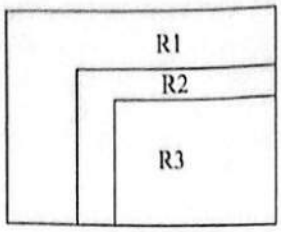
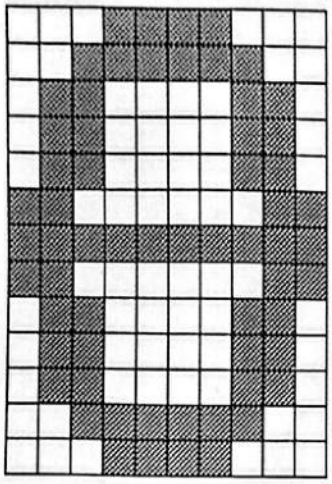
b) Explain which type of histogram equalization would perform better visual improvement for the following image. (12)

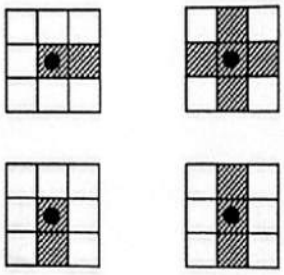


Pixel values of region, R1 = 127, R2 = 129, R3 = 130.

c) Figure below shows a binary image containing Greek letter θ, and four structuring elements. What morphological operations(s) would you use to obtain the number zero (0) from the letter θ? Consider that you can use any of the given structuring elements as many times as you wish. Provide a reason for your choice of operation(s) as well as the structuring elements. (15)

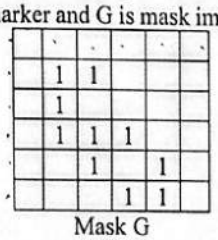


Binary image

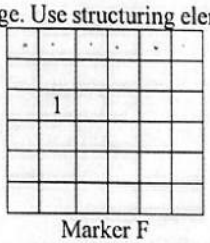


Structuring elements

4. a) Sketch the result of the following operations: (10)
$D_G^{(n)}(F)$ where F is marker and G is mask image. Use structuring element of 3×3 with all 1's.



Mask G



Marker F

b) Use necessary diagrams to show the conceptual relationship between the RGB and HSI color models. (12)

c) Explain with necessary mathematical formulation, how you can apply a linear filter on a color image represented in HSV color space. Also mention the need of saturation weighted hue component. (13)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) State the property of a linear operation. With necessary mathematical formulation, show that convolution is a linear operation. (10)

b) You are given the following spatial domain filter. (15)

0.15	0.35	0	−0.35	−0.15

i) Compute Fourier spectrum and phase of this filter,

ii) What do you think the effect of applying this filter to an image will be? Justify your answer.

c) Construct a homomorphic filtering function using a Butterworth high pass filter to correct nonuniform illumination. (10)

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6. a) Consider the image histogram given below. (18)

Intensity level	0	1	2	3	4	5
Frequency	14	11	4	12	16	7

Use maximum entropy-based thresholding technique to find a global threshold to segment the image.

b) State the necessity of non-maximum suppression stage used in Canny edge detection algorithm. Provide an idea to get edge orientation by avoiding trigonometric function that can be used in the Canny's algorithm. (12)

c) "Median filter may reduce the number of occupied bins in a histogram." – justify the statement. (05)

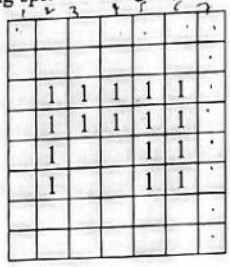
7. a) A 100×100 image is warped applying the following affine transformation matrix T. Mention the warped image size. Let, you are currently at pixel (50, 65) in the warped image, find its intensity if inverse mapping technique and bilinear interpolation were used during warping. Consider the following image segment of the input image. (16)

$$T = \begin{bmatrix} 1 & 0.6 & 5 \\ 1.2 & 0.8 & 10 \\ 0 & 0 & 1 \end{bmatrix}$$

y \ x	36	37	38	39
10	60	65	70	72
11	66	70	80	88
12	80	90	95	120
13	90	92	110	150

image segment

b) Segment the following image using spit and merged procedure. (12)



c) Explain aperture problem in optical flow estimation. (07)

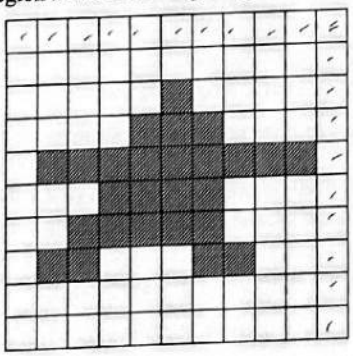
8. a) Let you are going to restore a noisy image having only additive noise. For the restoration purpose you construct the following histogram of a flat region of the image. (12)

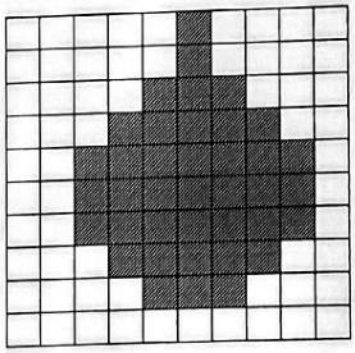
Intensity level	0	1	2	3	4	5	6	7
Probability	0	0	0.210	0.404	0.277	0.102	0.022	0.003

i) Determine the type of noise present,

ii) Provide an estimation of the parameters: mean and standard deviation of the noise density function.

b) Find the region descriptors: form factor, compactness, and circularity to describe the binary region in the following images. (12)





c) Derive the brightness consistency equation to estimate motion field as a form of optical flow. Suggest an idea to overcome its limitation. (11)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4127
Image Processing & Computer Vision

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

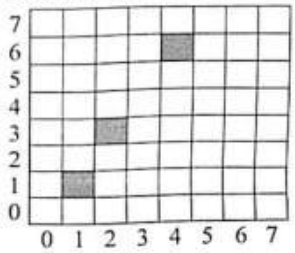
1. a) Derive the equation for Laplacian of a 2D Gaussian function $G = A \cdot e^{-\frac{x^2+y^2}{2\sigma^2}}$. Sample the LoG function to get a convolution kernel with $A = \sigma^2 = 1$. Scale the kernel values to the nearest integer. Use appropriate kernel size to support the function. (17)

b) Determine the 4-point 1D DFT matrix. Use it to find the 2D DFT of the following image. (13)

$$\begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & -1 \\ -1 & 2 & 1 & 0 \\ 2 & -1 & 1 & 1 \end{bmatrix}$$

c) Explain an implementation idea for hysteresis thresholding used in Canny edge detection method. (05)

2. a) Use Hough line transformation for the following edge image to detect the line. Use an appropriate accumulator array with $d\theta = 30°$, $d\rho = 1$. θ spans from $0°$ to $180°$, ρ spans from -4 to 8. Show vote count in the array and determine the line equation. (18)



Edge Image

b) Use weighted median filter to restore the following 4-bit image. Show results only for the middle row, use the given weight matrix. (10)

4	3	1	2	6
2	7	3	5	0
9	15	6	0	3
12	4	10	8	7
11	6	2	7	6

Image

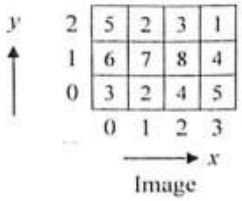
1	2	1
2	3	2
1	2	1

Weight Matrix

c) If $x_0(t) = at/T$ and $y_0(t) = bt/T$ denotes the time varying components of motion in x and y directions, then show that uniform linear motion between image and the sensor during image acquisition can be modeled by the following degradation function. Here, symbols have their usual meanings. (07)

$$H(u,v) = \frac{T}{\pi(ua+vb)} \sin[\pi(ua+vb)] e^{-j\pi(ua+vb)}$$

3. a) What is the importance of image warping in computer vision? Warp the following image with the given transformation matrix, T. Use backward mapping and bilinear interpolation during the process. Mention the size of the warped image and determine the intensity only at pixel (2, 3). (16)



Image

$$T = \begin{bmatrix} 0.43 & -0.25 \\ 1 & 1.73 \end{bmatrix}$$

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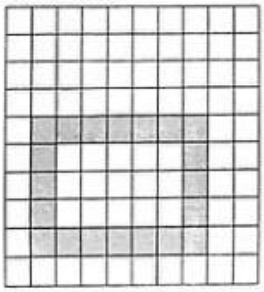
b) State the maximum entropy based image thresholding algorithm with necessary mathematical derivation. (16)

c) Draw signature for the following binary image. (03)



Binary Image

4. a) State the pros and cons of histogram based texture descriptors with proper example. (08)

b) For the following image, determine the Gray Level Co-occurrence Matrix (GLCM) with displacement $d = [1, 1]$. And thereby, measure the uniformity and homogeneity of the matrix. (17)

y \ x					
	0	1	2	4	1
	1	4	0	2	3
	1	1	3	2	0
	3	2	0	3	1
	4	3	2	4	2

c) Derive the equations used by Lucas-Kanade to estimate motion field from a video sequence. (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Illustrate the 'Golomb Coding' technique to encode non-negative integer in data compression. Calculate the Golomb Codes for the following representation: (20)
   i) $G_{11}(37)$
   ii) $G_{12}(-15)$
   iii) $G^2_{exp}(13)$

b) Explain false contouring effect in digital image using the concept of sampling and quantization. (08)

c) What measure can be taken to improve the brightness discrimination provided that a weber ratio is given? (07)

6. a) Explain the application of pseudo-color processing for highlighting explosives contained in a luggage using proper transformation functions. (12)

b) Write down the conditions for pre-color-component processing and vector-based processing to be equivalent with proper example. (08)

c) Derive the CMY intensity mapping function $s'_i = kr'_i + (1 - k)$ using its RGB counterpart where $0 < k < 1$. (10)

d) Prove, erosion and dilation are duals of each other with respect to set complementation and reflection. (05)

7. a) Write down the technique for filtering the hue component in HSI color space provided that the saturation is low and hue value is quite unstable. (12)

b) 'It is possible to save a digital image with roughly 50% less storage than the original image using bit plane slicing.' – Justify the statement. (08)

c) Illustrate the objective fidelity criteria for quantifying the nature of loss in image compression. (10)

d) Mention the drawbacks of 'Convex Hull' algorithm. (05)

8. a) Illustrate the automated hole filling algorithm for binary image based on morphological reconstruction. (12)

b) Suppose, you are asked to enhance dark areas of an image that may contain hidden features while leaving the light area as unchanged as possible. Now, apply 'local enhancement by contrast manipulation' to solve this problem using histogram statistics. (15)

c) If histogram equalization is applied twice to an image, that is, if it is applied to the result of histogram equalization, will the second application change the image or not? Explain why or why not? (08)

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2020
Department of Computer Science and Engineering
CSE 4127
Image Processing & Computer Vision

TIME: 1.5 hours FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) Consider the following two gray scale images (Image *A*, Image *B*) of size $N \times N$. Although they look quite different, their histograms are identical. If each of the images are blurred with the Mask *M*, then (10)
   i) Would the resultant histograms still be the same? Explain.
   ii) Calculate the approximate histograms of the resultant images.

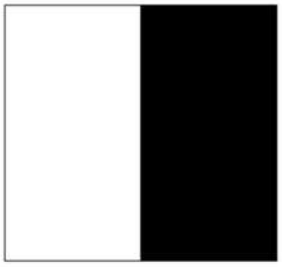


Image: *A*

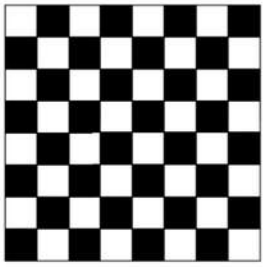


Image: *B*

$$\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$$

Mask: *M*

b) Let an image *A* has gone through an intensity transformation function $S = T(r) = cr^{\gamma}$, where c, γ are constants to get the output image *B*. Some pixel intensities of *A* and *B* are shown below. (13)
   i) Find the missing input intensities P, Q.

	4		15
		Q	
P		16	

$\longrightarrow T(r) \longrightarrow$

	12		87
		105	
22		96	

Image: *A* Image: *B*

c) Do you think the discrete histogram equalization technique is generally able to produce a flat histogram? Justify your answer. (07)

2. a) Suppose an image with intensities in the range [0,1] has the PDF $P_r(r)$ as shown in Fig. i. Assume continuous quantities and find the transformation in terms of r and z that will transform the intensity levels of the image so that they will have the specified $P_z(z)$ as shown in Fig ii. (12)

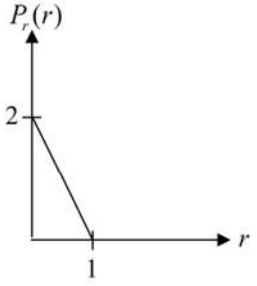


Fig. i

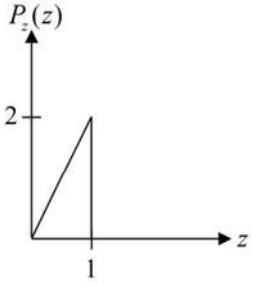


Fig. ii

b) Explain convex deficiency. Is it possible to reduce the hit-or-miss transform to simple erosion? Justify your answer. (12)

c) What does it mean by sampling and quantization in digital image acquisition? (06)

3. a) Consider the following 4×8 image. (15)

21	169	169	243	21	95	132	95
243	169	132	95	109	169	109	132
95	21	132	169	169	95	109	109
95	132	169	21	95	243	132	169

i) Compress the image using Huffman coding.
ii) Calculate the compression ratio and the percentage of the original fixed-length representation that is removed as the coding redundancy.

b) Calculate the morphological thinning on image E using P as the initial structuring element and show the outcome of each step separately. (15)

0	0	0	0	0	0	1	1
1	0	0	0	0	0	1	1
1	1	1	1	1	0	0	0
1	1	1	1	0	1	1	0
1	1	1	1	1	1	0	0
1	1	1	0	1	1	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Image: E

0	0	0
×	1	×
1	1	1

Structuring Element: P

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

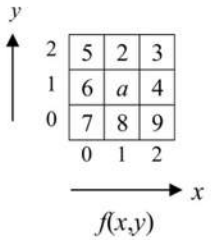
4. a) Let a Gaussian function $G(x,\sigma)=e^{-x^2/2\sigma^2}$ and its derivative G' is sampled at 5 points symmetric to the origin with $\Delta x=1,\sigma=1$. The sampled values of G and G' are used as $1D$ separable kernel to convolve an image in x and y direction, respectively. Find (14)
i) The x and y directional $1D$ kernels; scale and round the values to integer.
ii) The $2D$ kernel from $1D$ kernels.
iii) Note the benefits of using separable $2D$ kernels.

b) Consider the following 4-bit image corrupted with impulse noise. To remove the noise (16)
i) Use a contraharmonic mean filter of size 3×3 with appropriate order Q.
ii) Use a suitable order-statistics filter of size 3×3.
Show results only for the middle row.

7	6	8	5	4
3	10	9	8	11
8	x	11	12	1
5	8	2	10	7
9	3	4	15	14

Here, $x=\begin{cases}0; & \text{if your roll is an odd number}\\ 15; & \text{Otherwise}\end{cases}$

5. a) Let you are assigned to warp the following image $f(x,y)$ with the transformation matrix T. You are also constrained with the conditions that the transformation should be done using backward mapping technique having bilinear interpolation and the warped image should be large enough to contain all the processed samples. If so (14)
i) Find the size of the warped image.
ii) Find the intensity at pixel (1,3) of the warped image.

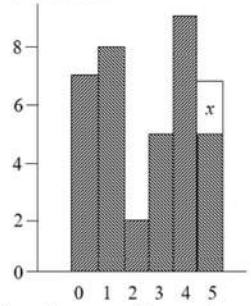


$f(x,y)$

$$T=\begin{bmatrix}1 & -0.5\\ 0.5 & 1\end{bmatrix}$$

Here, $a=$ last digit of your roll + 1.

b) An image intensity distribution shown by the following bar chart. Find a global optimal threshold for the image using Otsu's method. (16)



Here, $x =$ sum of last two digits of your roll number.
Therefore, $frequency(5) = x + 5$.

6. a) Let you have planned to describe an image texture as an Entropy of Gray Level Co-occurrence Matrix (GLCM). For the given image below (15)
   i) Find the GLCM with displacement $d = [1,0]$.
   ii) Determine the measure of Entropy.

0	1	4	2	2
3	4	3	0	1
1	1	4	2	0
3	2	4	3	1
4	3	0	2	2

b) Using a 2×2 window, estimate the optical flow using Lucas-Kanade method for the following two successive frames of a video sequence. Show result for (1,1) position only. (15)

	0	1	2	3	→ x
0	1	1	0	1	
1	1	9	9	9	
2	1	9	10	9	
3	0	9	9	10	
↓ y					

Frame at t

	0	1	2	3	→ x
0	1	1	0	1	
1	0	*a*	1	0	
2	1	1	9	9	
3	0	1	9	9	
↓ y					

Frame at $t+1$

Here, $a = \begin{cases} 0; & \text{if your roll is even} \\ 1; & \text{Otherwise} \end{cases}$

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Specify image compression. 'Huffman Coding can be used as a symbol encoder in an image compression system'-illustrate the statement. (12)
   b) Explain false contouring with necessary examples using the concept of sampling and quantization. (08)
   c) Consider the image segment shown below: (10)

4	2	3	2 (q)
3	3	1	3
2	3	2	2
(p) 2	1	2	3

Complete length of the shortest-4, shortest-8, and shortest-m paths between pixels p & q where, v = {1, 2}. If a particular path does not exist between these two points, explain why?
   d) Interpret the significance of shading correction in digital image processing. (05)

2. a) Construct a modified image using histogram specification where histogram of original and target images are given below: (20)
original histogram:

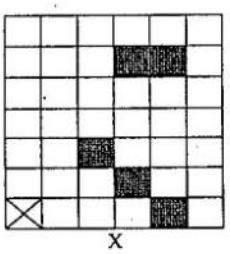
Gray level	0	1	2	3	4	5	6	7
No. of Pixels	8	10	10	2	12	16	4	2

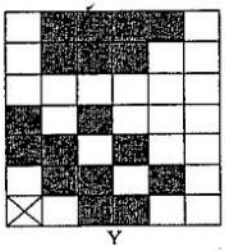
Target histogram:

Gray level	0	1	2	3	4	5	6	7
No. of Pixels	0	0	0	0	20	20	16	8

   b) Assuming continuous intensity values, suppose that an image has the intensity PDF $P_r(r) = {}^{2r}/_{(L-1)^2}$ for $0 \le r \le (L-1)$ and $P_r(r) = 0$ for other values of r. find the transformation function that will produce an image whose intensity PDF is $P_z(z) = {}^{3z^2}/_{(L-1)^3}$ for $0 \le z \le (L-1)$ and $P_z(z) = 0$ for other values of z. (10)
   c) 'Local Histogram processing is necessary to enhance details over small areas in an image'-justify the statement. (05)

3. a) Image X and Y are shown below with crossing in the left bottom showing their correspondence. (12)


X


Y

Design an appropriate structure element such that Y can be obtained after dilating X. (Black for 1 and white for 0).
   b) 'Textural segmentation is crucial in finding a boundary between two regions based on textural content'-justify the statement with necessary figures. (11)
   c) Detect edge on the following image using erosion and show the outcome of each step separately. (12)

0	0	1	1	1	1	0	1	1	1	0
0	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	0
1	1	1	1	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	0
0	0	0	0	0	1	1	1	0	0	0

0	1	0
1	1	1
0	1	0

Structuring Element

4. a) State and explain the data redundancies that can be identified in an image for the exploitation of image compression. (09)
   b) Define Hue and Saturation. Calculate the HSI value of the following RGB point: (11)
   (155, 210, 75)
   c) What will happen if you change the value of H channel of an HSI image while keeping the other channel's value constant? (05)
   d) Illustrate the concepts: (i) Color slicing and (ii)Tone correction (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Compare between image processing and computer vision. Explain some typical challenges in solving a computer vision problem. (12)
   b) Consider the 1D image below: (08)

a	b	c	d	e	f	g

   i) Suppose, you apply a width-3 mean filter twice. Give an expression for the center pixel after the filter has been applied.
   ii) Define a single 1D kernel that, when applied only once to the image, will produce the same results.
   c) Find the unsharp mask first and then the sharped image for the following image: (15)

2	8	6
5	4	1
7	3	9

   Use a 3x5 mean filter as a blurring kernel and do wrap around padding where required.

6. a) Design a Bulterworth Low-pass Filter (BLPF). What happens when you choose a higher order BLPF to smooth an image? When would you prefer BLPF than Gaussian Low-pass Filter? (12)
   b) Show that the Fourier transform of a shifted function is unaltered except for a linearly varying phase factor. (08)
   c) Verify the soundness of the statement: 'Laplacian with a -8 in the center yields sharper results than the one with -4 in the center'. (10)
   d) Explain estimation of image degradation function by mathematical modeling. (05)

7. a) Segment the image shown below with a global threshold computed by Otsu's method. (13)

1	2	0	12	14
2	3	4	8	13
9	2	3	15	7
3	2	2	11	13
1	2	0	14	12

   Mention some limitations of global thresholding.
   b) Rotate the image shown below by backward mapping method with $\theta = 30^{\circ}$. (12)

1	5	2
6	3	4
7	9	5

   Use bilinear interpolation and allow the image to grow to fully contain the rotated image.
   c) Find the 4-point basis matrix of Discrete Fourier Transform (DFT) and then use it to find the DFT of [1 -1 1 -1]. (10)

8. a) Define image representation and description. Measure the entropy and uniformity of the following 3-bit image: (13)

1	2	0	7	5
6	2	3	4	3
4	0	2	1	5
4	7	6	2	4
5	6	4	3	3
5	4	1	2	4

   b) Explain the basic principle of Hough line detection mechanism. (09)
   c) What is optical flow? Explain Lucas-Kanade method of calculating optical flow. (13)

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TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Compare and contrast between image processing and computer vision. Mention some applications of them. (08)

   b) Use Laplacian operator to sharp the following image. Use necessary border reflection padding. (15)

$$\begin{bmatrix} 1 & 2 & 5 & 4 \\ 5 & 2 & 5 & 5 \\ 2 & 1 & 6 & 3 \\ 2 & 4 & 7 & 6 \end{bmatrix}$$

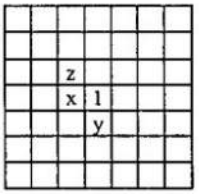
   c) What is periodic noise? Design a band reject and a notch reject filter to remove such noise from image. Also discuss their pros and cons. (12)

2. a) Given an input image of size 7x7 shown below, was filtered using 3x3 adaptive median filter with maximum allowed size of 5x5. (12)

3	3	4	3	3	3	0
3	0	0	0	0	0	3
3	4	0	0	4	4	3
4	5	7	7	0	0	3
3	3	6	0	0	7	0
3	0	A	3	3	5	3
3	4	3	3	0	0	4

input image

Adaptive median filter →



output image

Answer the following questions with reasoning:
   i. What is the value of the pixel A in the input image?
   ii. What are the values of pixels x, y and z in the output image?

   b) Explain why and how non-maximum suppression is performed in Canny edge detection algorithm. (10)

   c) Let a 3-bit grayscale image is given below: (13)

0	5	2
6	1	7
3	4	7

Resize the image by using bilinear interpolation and backward mapping to obtain 3x5 image.

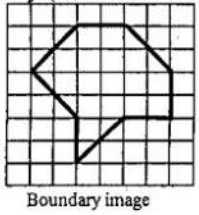
3. a) Show that $\nabla^2 G = \frac{1}{\pi\sigma^4}\left[1 - \frac{x^2+y^2}{2\sigma^2}\right].e^{-\frac{x^2+y^2}{2\sigma^2}}$, where G is the Gaussian function and ∇^2 is the Laplacian operation. (08)

   b) What is Thresholding? List the steps of Otsu's Thresholding algorithm. (11)

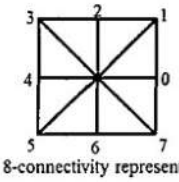
   c) Compare the first order and second order derivative based edge detection methods. (10)

   d) State and prove the separability property of 2D DFT. (06)

4. a) For the following boundary (in counter-clock-wise direction) (09)



Boundary image



8-connectivity representation

compute the 8-connected
   i. chain code.
   ii. chain code that is invariant to rotation.
   iii. chain code that is invariant to both starting point and rotation.

b) Consider two consecutive frames of a video shown below. (14)

y \ x	0	1	2	3	4
0	3	15	6	4	0
1	4	4	12	1	2
2	4	14	5	5	8
3	6	1	13	2	3
4	10	7	11	9	12

Frame - 1

y \ x	0	1	2	3	4
0	4	6	5	3	0
1	2	3	15	6	4
2	3	4	5	12	1
3	10	4	13	5	5
4	9	6	1	13	2

Frame-2

Estimate the optical flow of the center pixel using Lucas-Kanade method. Use 3x3 window for spatial coherence.

c) Explain inverse filtering process to restore a noisy image. What is the shortcoming of inverse filter? How does Wiener filter overcome such shortcoming? (12)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is digital image and digital image processing? What are the fundamental steps in digital image processing? (10)

b) Define blind spot and brightness adaption. Briefly describe how to characterize brightness discrimination using appropriate example. (08)

c) What do you mean by image compression? Briefly describe about Discrete Cosine Transformation of image compression. (12)

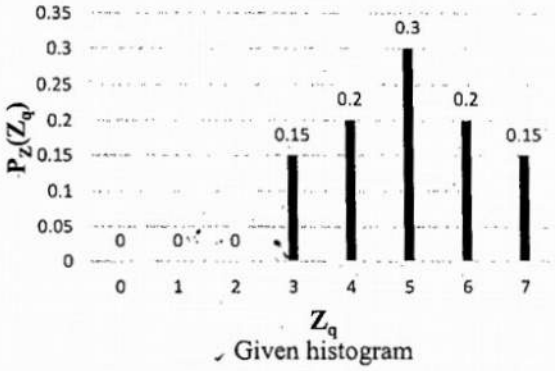
d) What are the changes occur in an image by changing values in spatial domain and quantization (Gray value)? (05)

6. a) Define contrast. Design a mapping function to enhance the contrast of an image having very bright pixels. (11)

b) Briefly describe Power-law transformation for digital image processing and also show the image appearance of different values of gamma. (08)

c) The following table shows the information of a 3-bit 64x64 image. (16)

Pixel Value	Frequency
0	790
1	1023
2	850
3	656
4	329
5	245
6	122
7	81



Given histogram

Draw the original histogram, normalized histogram and finally show the matching histogram with the given histogram.

7. a) Let a 4-bit, 4x4 image passes through an intensity transformation function given by $s = T(r) = \alpha \log_2(1+r) + \beta$, where α, β are constants. Few pixels are available in the input and output images, as shown below. (10)

	3		3
9		15	B
	A		
1			3

input image

$\rightarrow$ T(r) $\rightarrow$

	5		5
C		11	9
3	8		
D	9		5

output image

What are the values of the pixels A, B and C, D in the input and output images respectively?

b) Define Pseudo color image processing. Discuss about intensity slicing with appropriate diagram. (10)

c) Write down the procedure to compute pruning of a binary image. (05)

d) Consider the following 5x5 8-bit image with grayscale range of [0, 9]. (10)

$$\begin{bmatrix} 0 & 5 & 7 & 7 & 5 \\ 7 & 2 & 6 & 2 & 6 \\ 6 & 9 & 7 & 7 & 0 \\ 6 & 6 & 1 & 7 & 6 \\ 9 & 3 & 2 & 4 & 8 \end{bmatrix}$$

Compress the image using Huffman encoding techniques and also discuss the efficiency of this compression.

8. a) Consider the following binary image A. (12)

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Image A

Show the result of applying the following operations on image A with a 3x3 structuring elements.

i. $B = (A \oplus S) \cap A^C$

ii. $B = (A \ominus S) \cup A^C$

iii. $B = (A \oplus S) \cup A^C$

b) How can you detect how many template images are there in a full image using hit or miss transformation algorithm? Briefly describe with appropriate example. (11)

c) Define Hue and Saturation. Write down the process of conversion of a pixel value from HSI color space to RGB color space. (12)

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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Relate and differentiate between computer vision and image processing. Explain the possible challenges one has to face in solving a computer vision problem. (10)
   b) Define convolution. Show by using an example that $f \star e = f$, where f is an image, e is a unit impulse, and $\star$ is a convolution operator. (10)
   c) Sharp the following image using unsharp mask. Use a 3×3 Gaussian blurring kernel with $\sigma = 1$ and do zero padding where necessary. (15)

$$\begin{bmatrix} 1 & 2 & 4 & 5 \\ 5 & 2 & 5 & 5 \\ 1 & 1 & 3 & 6 \\ 2 & 4 & 6 & 7 \end{bmatrix}$$

2. a) What is the necessity of transforming an image into frequency domain? Let, $F(u, v)$ is a [M, N] point Discrete Fourier Transform (DFT) of an image defined by $f(x, y)$. Then show that $F(u, v) = F(u + pM, v + qN)$, where p, q are arbitrary constant. (10)
   b) Describe Canny edge detection algorithm in detail. (12)
   c) Design a Butterworth Lowpass Filter (BLPF) to smooth an image. How can you choose the cut-off frequency D_0 to pass α percent of image power? Also explain why the lower order BLPF is more preferable than higher order one. (13)

3. a) Define order statistic filter. Use a 3×3 alpha-trimmed mean filter with $d = 4$ to filter noises from the image presented below. (13)

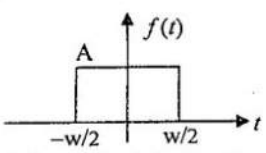
$$\begin{bmatrix} 2 & 3 & 5 & 8 & 7 \\ 8 & 12 & 4 & 6 & 3 \\ 5 & 7 & 6 & 3 & 9 \\ 4 & 1 & 4 & 8 & 3 \\ 6 & 7 & 8 & 14 & 17 \end{bmatrix}$$

   b) What is periodic noise? How can you remove such noise from an image? (05)
   c) Explain inverse filtering process to restore a noisy image. What is the shortcoming of inverse filter? How does Wiener filter overcome such shortcoming? (12)
   d) What is image segmentation? Mention some applications of it. (05)

4. a) Find a global threshold for the following image. What are the problems of using a single global threshold to segment an image? How can these be resolved? (13)

$$\begin{bmatrix} 1 & 2 & 1 & 0 & 12 & 14 \\ 2 & 3 & 4 & 8 & 13 & 12 \\ 9 & 2 & 3 & 15 & 13 & 7 \\ 3 & 2 & 2 & 10 & 11 & 13 \\ 1 & 2 & 0 & 5 & 12 & 14 \\ 2 & 3 & 6 & 8 & 15 & 13 \end{bmatrix}$$

   b) Perform Fourier transform of the function shown in the following figure. The transformation equation is given by $F(u) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi ut} dt$. (08)



   c) Define motion field and optical flow. Explain the Lucas-Kanade optical flow estimation method stating their key assumptions. (14)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is a digital image? What are the stages through which an image passes in a image processing system? Briefly explain each stage. (10)
   b) Briefly discuss about Weber law and optical illusions. (09)
   c) What is mean by sampling and quantization? Write down the transformation matrices of image Scaling, Rotation and Translation. (08)
   d) Define image compression. Briefly describe any one of the image compression techniques. (08)

6. a) Define bit plane slicing for piecewise linear transformation and histogram equalization. Calculate each plane for the following 4-bit image. (10)

$$\begin{bmatrix} 11 & 13 & 3 & 5 & 8 \\ 8 & 6 & 0 & 11 & 10 \\ 9 & 0 & 2 & 9 & 7 \\ 7 & 12 & 6 & 13 & 14 \\ 3 & 4 & 10 & 0 & 2 \\ 8 & 0 & 15 & 5 & 1 \end{bmatrix}$$

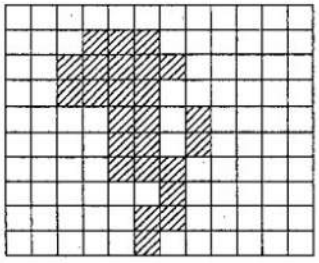
   b) Consider the following 5×5 image with grayscale range of [0, 9]. (17)

$$\begin{bmatrix} 0 & 5 & 7 & 7 & 5 \\ 7 & 2 & 6 & 2 & 6 \\ 6 & 9 & 7 & 7 & 0 \\ 6 & 6 & 1 & 7 & 6 \\ 9 & 3 & 2 & 4 & 8 \end{bmatrix}$$

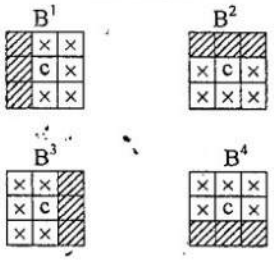
      i) Plot the histogram of the image.
      ii) Perform histogram equalization and then show the equalized image and plot histogram again.
      iii) Histogram matching is a useful contrast manipulation technique that transforms an image's histogram to match with that of another image. Clearly describe how can you achieve it.

   c) What do you mean by contrast stretching? Explain a simple non-linear contrast stretching method. (08)

7. a) Define full color image processing. Discuss about intensity slicing with appropriate diagram. (08)
   b) Define convex hull. Draw the convex hull of the following binary image. (17)



Binary imge: A



Structure elements: c is the center

   c) Define morphological gradient. When do we use top-hat and bottom-hat transformation in grayscale image? (10)

8. a) Define pigments. Write down the process of conversion of pixel value from RGB color space to HSI color space. (16)
   b) Prove that in continuous domain, probability distribution function of equalized image is always uniform. Discuss it with appropriate example. (07)
   c) Consider a portion of a binary image as shown below. Perform the following operations on the image. (12)
      i) Opening
      ii) Closing

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Binary image

$$[\,\textcircled{1}\quad 1\quad 1\,]$$

Structuring Element (SE)

The circle indicates the center of SE

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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is computer vision? How is image processing related to computer vision? Mention some challenges of computer vision problem. (10)

b) Define histogram equalization. Consider the 5×5, 3 bits/pixel image shown in Fig. 1(b). (18)

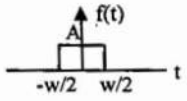
$$\begin{bmatrix} 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \\ 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 4 & 5 & 3 \\ 4 & 3 & 2 & 6 & 7 \end{bmatrix}$$

Fig. 1(b)

i) Perform histogram equalization to the image.
ii) What will happen if we apply the same process on the resultant image of step (i).
iii) Explain why the discrete histogram equalization does not, in general, yield a flat histogram.

c) Perform Fourier transform of the function shown in the following figure follows from (07)

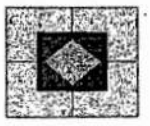
$$F(\mu) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi\mu t}dt$$

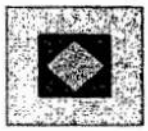


2. a) Prove that the convolution of an image in spatial domain is equal to multiplication in frequency domain. (13)

b) What does it mean by low pass filter? Design an Ideal Low-Pass Filter (ILPF). Explain why there is a ringing effect on the image that is smoothed by ILPF. (10)

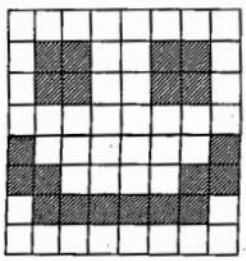
c) Consider the following figure, when the image on the left was filtered using a smothing filter, the result was image on the right. The filter used was one of these of size 3×3: (i) mean filter; (ii) ideal low-pass filter; (iii) Gaussian low-pass filter; (iv) median filter. For each of the four possible filters listed above, give at least one reason why you think it was or was not, the filter actually used. (12)


original image

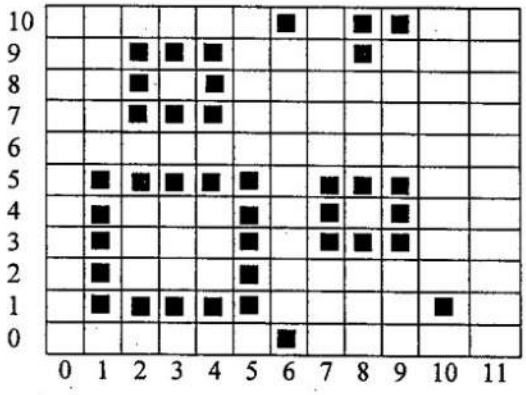

filtered image

3. a) Sketch and explain a model for image restoration process. How can you estimate the image degradation functions? (12)

b) What is the importance of image segmentation? Segment the binary image shown in following figure. Using split-and-merge technique. Also, show the quad tree corresponding to the segmentation. (12)



c) If we apply the Hough transform on the following image, what would be the maximum value for the accumulator cell in the (P,Q) space? What is the corresponding (P,Q) value? (11)



4. a) Explain how the k-means clustering can be used for image segmentation. What are the weaknesses of k-means clustering algorithm? How can they be overcome? (13)
   b) What is motion field? Explain a direct method for estimation of motion field stating the key assumptions and deriving the necessary equations. (15)
   c) State the steps of calculating Ostu's threshold for an image. (07)

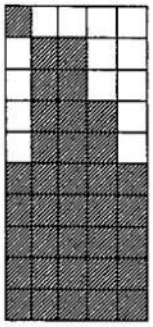
SECTION B

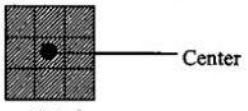
(Answer **ANY THREE** questions from this section in Script B)

5. a) What is digital image processing? What are the fundamental steps in digital image processing? (09)
   b) Discuss about different neighborhood and connectivity of a pixel with appropriate example. (12)
   c) Define blind spot and brightness adaption. State some example of optical illusions. (06)
   d) What do you mean by image compression? Briefly describe any one of the image compression techniques. (08)

6. a) What is discretization? How are images presented in computer? (07)
   b) What is the property of an adaptive filter? Write a pseudocode for adaptive median filter. Instead of median filter, when will you prefer the adaptive one? Explain why. (12)
   c) State and explain the data redundancies that can be identified in an image and can be exploited for image compression. (10)
   d) Explain how human visual system perceives a color. (06)

7. a) Define convex hull. Write down the drawbacks of convex hull and how this can be resolved? (07)
   b) Define Thinning. Draw the skeleton of the following binary image. Is it possible to recover the original image from the skeleton? (15)


Set A



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   c) Why pruning is necessary? Write down the equations to compute pruning of a binary image. (06)
   d) Define Erosion and Dilation for Gray-scale image with appropriate example. (07)

8. a) When we use top-hat and bottom-hat transformation in grayscale image? (04)
   b) Define Hue, saturation and intensity. Write down the process of conversion of a pixel value from RGB space to HSI space. (18)
   c) Define pseudo color image processing. Discuss about intensity slicing with appropriate diagram. (08)
   d) How can you apply a spatial filter on a full color image? Explain. (05)

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Department of Computer Science and Engineering
CSE 4111
Machine Learning

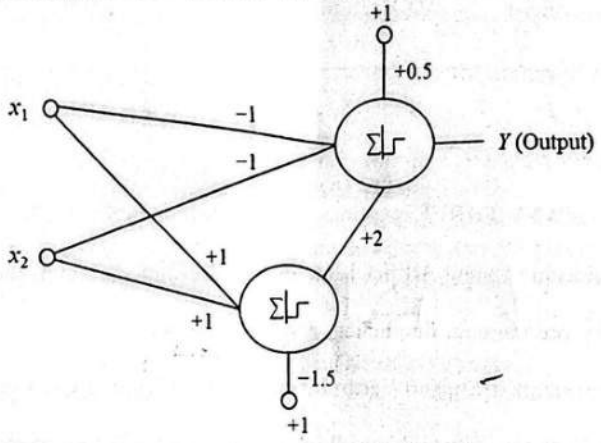
TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) "In Machine Learning (ML), explicit programming is not performed to solve a particular problem" – justify this statement. (08) [CO1]
   b) Briefly describe the basic learning types. Arrange the learning types in order of the increasing feedback from the environment; provide justification of the ordering. (09) [CO2]
   c) Differentiate learning and generalization in the context of ML. Which one mostly indicates the performance of any ML model and what techniques are employed to achieve the indicative? (09) [CO3]
   d) Does the following neural architecture solve 2 input non-linear problem? Justify your answer considering the inputs (0,0), (0,1), (1,0), and (1,1) at x_1 and x_2. (09) [CO3]



2. a) Describe importance of bias weight value in artificial neuron construction. "Changing only the bias weight value can change the operation of a neuron" – justify the statement for appropriate logic gates. (08) [CO1]
   b) What is the condition to use a function as activation function in artificial neural network? Graphically represents two different forms sigmoidal functions and their derivatives. Explain similarity and dissimilarity between the functions. (09) [CO2]
   c) What is the local gradient in Back Propagation (BP) learning algorithm? "Local gradient may be zero even error at neural network remain" – justify the statement with an appropriate activation function. (09) [CO2]
   d) Draw diagram of RBF networks with approximate parameter values to solve AND, OR, and XOR logic operations for two inputs. (09) [CO4]

3. a) What are the various ways to produce diverse networks for ensemble construction? Which one is more effective? Why? (07) [CO1]
   b) Briefly describe independent, sequential, and simultaneous training of component networks in ensemble construction. Mention prominent ensemble methods for different ensemble training methods. (10) [CO2]
   c) Consider two computation systems: (i) multiple processing units and (ii) single high power unit. What are the preferable ensemble methods for the systems individually and why? (10) [CO4]
   d) Identify the learnable parameters for the below Convolutional Neural Network (CNN) to classify numeral images having size 28×28 pixels. Also, count the learnable parameters. (08) [CO3]

$$I_{28\times28} \rightarrow \{6K1_{5\times5}C1_{24\times24} - S_{2\times2}6S1_{12\times12}\} \rightarrow \{6K2_{5\times5}C2_{8\times8} - S_{2\times2}6S2_{4\times4}\} \rightarrow W_{096\times10} \rightarrow O_{10}$$

where symbols hold usual meanings.

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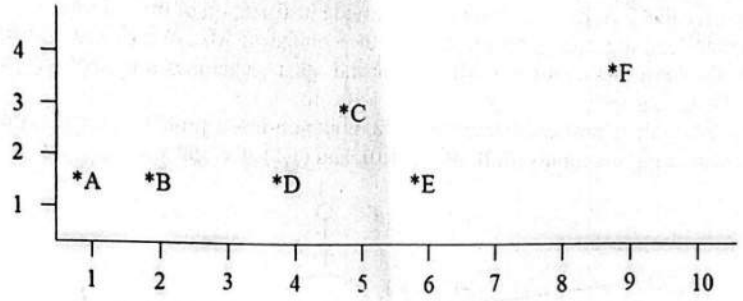
4. a) Discuss advantages and disadvantages of Decision Trees relative to Neural Networks for classification on the basis of important issues (e.g., data type, compatibilities, sample complexities, model interpretabilities). (10) [C
   b) Consider the following table where attribute "Play Tennis" represents class label. Construct a Decision Tree using the table measuring information gain and then generate rules from the Decision Tree. (10) [CO3

Sl. No.	Outlook	Temp	Humidities	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Rain	Mild	High	Weak	Yes
4	Rain	Cold	Normal	Weak	Yes
5	Rain	Cold	Normal	Strong	No
6	Sunny	Mild	Normal	Strong	Yes

   c) Briefly discuss strength and weakness of K-means clustering. (07) [CO2]
   d) Draw a dendrogram considering six data points shown below; no need exact distance calculation. (08) [CO3]



SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain the relationship among Artificial Intelligence, Machine Learning, and Deep Learning. (06) [CO1]
   b) What is meant by pre-training, fine-tuning, and transfer learning? Give example of each case. (09) [CO1]
   c) What are support vectors in Support Vector Machine (SVM)? How does Linear SVM work? (12) [CO2]
   d) What is curse of dimensionality problem? Why is it an issue in ML? Briefly explain a dimensionality reduction technique. (08) [CO2]

6. a) Differentiate between heuristic and meta-heuristic with examples. (06) [CO1]
   b) "Meta-heuristics are designed to solve a problem more quickly and efficiently than classic methods" – justify it. (10) [CO2]
   c) Explain exploration and exploitation with suitable example. (10) [CO1]
   d) "Crossover and mutation are not always beneficial" – is it true? If so, justify with a numeric example. (09) [CO2]

7. a) Write simple Genetic Algorithm (GA). Why is choosing proper fitness function important in GA? Explain with example. (10) [CO2]
   b) Suppose, you want to minimize $f(x) = x^2$ where $-5 \leq x \leq 5$ using GA. Assume the required initial values, and consider MSB as sign bit and rest of the bits are magnitude during encoding. Now, perform calculation for one generation by hand. (12) [CO4]
   c) Write an algorithm for Enhanced Edge Recombination Operator with proper example. (13) [CO3]

8. a) What is Swarm? Write properties of a Swarm Intelligence (SI) system. (05) [CO1]
   b) Explain Particle Swarm Optimization (PSO) technique. What are the drawbacks of this technique? (12) [CO2]
   c) Explain Ant Colony Optimization (ACO) to solve Traveling Salesman Problem (TSP). (10) [CO4]
   d) Which bees perform exploration and exploitation in Artificial Bee Colony (ABC) optimization? What is significance of parameter "limit" in ABC optimization? (08) [CO2]

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N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define machine learning and discuss its different types. (06) [CO1]
   b) Derive the Lagrangian dual objective function and formulate the equations that uniquely characterize the solution to both the primal and dual problems for the Support Vector Machine (SVM). (14) [CO3]
   c) Explain how Directed Acyclic Graph (DAG) SVM addresses the drawback of the 1-vs-1 SVM approach, where the number of SVMs increases super-linearly with number of classes, in data classification. (08) [CO2]
   d) Discuss the relationship between Gini Impurity and Information Gain. (07) [CO2]

2. a) How does a separating hyperplane that perfectly classifies all of the training observations lead to sensitivity to individual observations? Explain in detail. (10) [CO2]
   b) Can SVM be over trained? What can be the effect of over training? How can over-fitting be detected for a SVM model? (07) [CO2]
   c) Consider the following two hyperplanes: (08) [CO3]

$$5 + 2x_1 + 5x_2 \quad \text{(for classifier 1)}$$
$$5 + 20x_1 + 50x_2 \quad \text{(for classifier 2)}$$

   Compute the distance error function and identify which classifier built with these hyperplanes will have better performance.
   d) 'The maximal margin hyperplane for SVM depends directly on only a small subset of the observation' – justify the statement. (10) [CO1]

3. a) Explain Occam's razor principle. In the context of decision tree learning, when we prioritize simplicity over complexity, do you believe Occam's razor adequately justifies the algorithm's inclination toward simpler hypotheses? (10) [CO1]
   b) Discuss the impact of outliers on the performance of a maximum margin classifier. How can the classifier be sensitive to outliers and what techniques can be employed to address this issue? (12) [CO2]
   c) Consider the following table where attribute 'Status' represents class label. Find the root attribute of the decision tree using Gain Ratio as the attribute selection measure. (13) [CO3]

Department	Age	Status	Salary
Sales	31–35	Senior	Medium
Sales	26–30	Junior	Low
Marketing	36–50	Senior	High
Secretary	26–30	Senior	Medium
Secretary	36–50	Senior	High
Sales	36–50	Junior	High
Marketing	31–35	Senior	Medium
Marketing	26–30	Junior	Low
Secretary	31–35	Junior	Medium
Secretary	26–30	Junior	Low

4. a) Discuss the empty space phenomenon and demonstrate the working principle of kernel Principle Component Analysis (PCA) in dimensionality reduction for linearly inseparable input data. Under what conditions does kernel PCA function similarly to standard PCA? (11) [CO1]
   b) What is inductive bias in decision tree learning and which type of inductive bias is used in BFS-ID3 algorithm? Illustrate how BFS-ID3 algorithm exhibits the bias 'shorter trees are preferred over longer trees' with detailed explanation. (12) [CO2]

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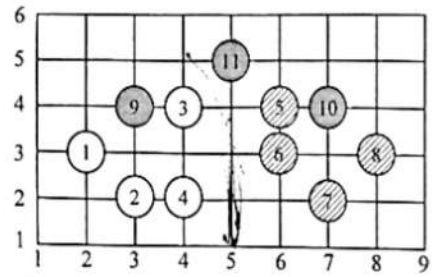
c) Consider the following training data: (12) [CO3]



Let $y_A = -1, y_B = +1$ be the class indicators for both classes where

$A = \{(1), (2), (3), (4)\}$ and $B = \{(5), (6), (7), (8)\}$

i) Just using the above standing plot, specify which of the points should be identified as support vectors.

ii) Draw the maximum margin line which separates the classes (you don't have to do any computations here). Write down the normalized normal vector $w \in \mathbb{R}^2$ of the separating line and the offset parameter $b \in \mathbb{R}$.

iii) Consider the decision rule: $H(x) =< w, x > +b$. Now determine the class for the points (9), (10) and (11).

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Suppose, we want to fit a line $y = mx + b$ through the points $(1, 2), (2, 3), (3, 4), (4, 5)$ and $(5, 6)$. Assume, initial values of m and b are 0.1 and 0.2 respectively. Calculate the initial error using mean squared error loss function and then apply a gradient descent step. Use learning rate, $\alpha = 0.01$ and calculate the error of fit again after the gradient descent step. (12) [CO3]

b) Demonstrate automatic differentiation (AD) using a suitable example. How does AD differ from numerical differentiation? (07) [CO1]

c) Explain why Convolutional Neural Network is preferred over Multi-Layer Perceptron for image related tasks. (04) [CO2]

d) Explain how artificial neural networks can be used to solve following problems. (12) [CO4]

i) Duplicate/ near duplicate image identification.

ii) Anomaly detection.

iii) Dimensionality reduction.

6. a) Briefly explain the idea of transfer learning. (05) [CO1]

b) What is exploration and exploitation in swarm intelligence algorithms? Explain how these features are provided in following algorithms. (16) [CO1]

i) Genetic algorithm (GA).

ii) Particle swarm optimization (PSO).

iii) Ant colony optimization (ACO).

iv) Artificial bee colony (ABC).

c) What are the disadvantages of Roulette Wheel selection? (03) [CO3]

d) Create a child using Cycle crossover and another child using Enhanced Edge Recombination for the following parents. (11) [CO3]

Parent 1	1	2	3	4	5	6

Parent 2	5	6	1	3	4	2

7. a) "PSO cannot be used for discrete optimization problem" – justify this statement. (05) [CO1]

b) Perform one iteration of PSO to minimize the function $f(x, y) = (x - 2)^2 + (y - 3)^2$ in the range $-10 \leq x, y \leq 10$. Parameters and initial values are given below where the symbols have their usual meanings. (10) [CO3]

Parameters:

$N = 2, \omega = 0.5,$
$C_1 = 1.5, C_2 = 1.5$
$r_1 = 0.3, r_2 = 0.7$

Initial values:

Particle 1: Position $(x_1, y_1) = (5, 5)$,
Velocity $(v_{1x}, v_{1y}) = (0.1, 0.1)$.
Particle 2: Position $(x_2, y_2) = (3, 1)$,
Velocity $(v_{2x}, v_{2y}) = (0.2, 0.2)$.

c) Write down the equations of transition probability and pheromone update of ACO. Also, describe the effects of key parameters. (10) [CO1]

d) Explain the employee bee phase, on looker bee phase and scout bee phase in ABC optimization algorithm. (10) [CO1]

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8. a) Explain how clustering can be used for image color quantization. (04) [CO3]
   b) Explain how to calculate and interpret Silhouette score. (06) [CO2]
   c) What is the problem of random centroid initialization in case of k-means clustering? Explain how k-means++ or GA can be used to solve this problem (explain both approaches). (11) [CO2]
   d) Using agglomerative hierarchical clustering create a dendrogram for the data shown in following table. (14) [CO3]

Student_ID	Marks
1	15
2	12
3	33
4	25
5	40

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B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What do you mean by "Machine" and "Learning"? What are their individual features? In what prospects both the terms combine in Machine Learning (ML) subject? (07)
   b) "Training appropriate neural network architecture with appropriate training set is necessary to develop an appropriate model for a given task" – Justify the statement for a logic gate function case. (08)
   c) Draw the minimal neural architectures for the following three input (A, B and C) logic gate operations. (12)
      i) $NN1 = (AB)'C$ ii) $NN2 = ABC'$
      iii) $NN3 = (A + B)'C$ iv) $NN4 = (A + B)C'$
   d) Differentiate learning and generalization in the context of machine learning. Which one is more desirable and why? What are the measuring techniques of those? (08)

2. a) Why is a single neuron unable to solve XOR problem? Solve XOR problem with Radial Basis Function (RBF) network considering cluster centers (0, 1) and (1, 0) with appropriate radius of each center. (08)
   b) Write down the roles of activation function in artificial neural network? Compare two different forms of sigmoidal functions through graphical representations of Input vs Output and derivatives. (08)
   c) What is the naming significance of "Back Propagation (BP)" algorithm? What is the effect of learning rate in BP? How momentum parameter speedup training in BP? (07)
   d) What is the goal of neural network ensemble construction? Explain an ensemble method for enough data and an ensemble method for limited data. (12)

3. a) Why non-deterministic operation is preferable over deterministic operation in population-based optimization algorithms? How non-deterministic feature assures in Genetic Algorithm (GA), Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO)? (12)
   b) Explain velocity and position updates of particles in PSO mentioning corresponding equations in continuous function space. How the terms may transfer to solve Traveling Salesman Problem (TSP)? (09)
   c) Why both exploration and exploitation operations are necessary in an algorithm for better performance? (06)
   d) How individual parameters of PSO velocity update equation influence exploration and exploitation? (08)

4. a) What do you understand by the search terms Global Search, Local Search and Blind Search? Compare GA with a prominent Swarm Intelligence (SI) method on the basis of the search methods. (13)
   b) Explain the working principle of GA with a flowchart. Briefly explain importance of different operators and parameters in GA. (08)
   c) Suppose two TSP solutions are 1 – 2 – 4 – 3 – 6 – 5 and 1 – 5 – 3 – 4 – 2 – 6 where numeric values are city indexes. Generate an offspring from the solutions using Enhanced Edge Recombination algorithm. (08)
   d) Explain the fundamental differences between ACO and PSO to establish interaction among individuals. (06)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is 'K' in the K Nearest Neighbor (KNN) algorithm and why is the odd value of 'K' preferred over even values in the KNN algorithm? (05)

   b) We can easily find K Nearest Neighbors using Euclidean distance and then classify an instance. Then, why do we need K Dimensional tree (K-D Tree) for KNN? (05)

   c) Suppose, a satellite image can be segmented using four categories: water body, grass land, bare land and semi grassy area. Choose a suitable algorithm to segment the image and explain the process. (05)

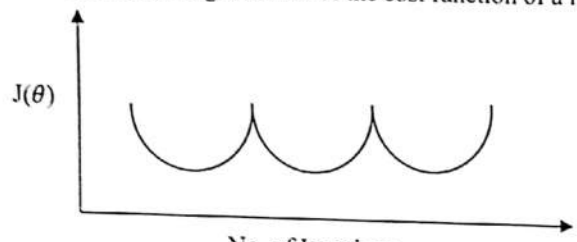
   d) Consider the following cost function and hypothesis for linear regression. Derive the normal equation, $\theta = (X^T X)^{-1} X^T Y$ to get the values of vector θ. (15)

$$j(\theta) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

$$h_\theta\left(x^{(i)}\right) = \theta_0 + \theta_1 x_1^{(i)} + \cdots + \theta_n x_n^{(i)}$$

   How do you calculate θ when $X^T X$ is non-invertible?

   e) Consider the following outcome of the cost function of a linear regression model. (05)



   Is there any issue with the outcome? Explain the issue and solution (if any).

6. a) The cost function of logistic regression is given below: (15)

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(h_\theta\left(x^{(i)}\right)\right) + \left(1 - y^{(i)}\right)\log\left(1 - h_\theta\left(x^{(i)}\right)\right)\right]$$

   Prove that, $\frac{\partial}{\partial\theta_j}J(\theta) = \frac{1}{m}\sum_{i=1}^{m}\left[h_\theta\left(x^{(i)}\right) - y^{(i)}\right]x_j^{(i)}$.

   b) Is there any similarity between a logistic regression model and a single neuron of a neural network (with sigmoid activation function)? Explain (if any). (07)

   c) What do you mean by overfitting and underfitting? Write down a way to minimize both the issues. (07)

   d) Suppose, one of your friends is planning to build a Machine Learning (ML) model to predict the diseases of many patients from various symptoms. He wants to sell the product to several hospitals. He has only 400 instances of data from 300 different patients. The dataset has both categorical and continuous features. There are only 5 diseases for prediction. Which ML model would you suggest to your friend and why? (06)

7. a) Consider the following dataset where attribute 'Salary' represents class label. (14)

Department	Age	Status	Salary
Sales	25..30	Junior	Low
Sales	31..35	Senior	Medium
Marketing	31..35	Senior	High
Admin	25..30	Junior	Medium
Admin	31..35	Senior	High
Admin	36..40	Senior	High
Marketing	36..40	Senior	High
Sales	25..30	Junior	Low
Marketing	36..40	Senior	Medium
Marketing	25..30	Junior	Low

   Find the root attribute of the Decision Tree (created from the dataset) using Gain Ratio as the attribute selection measure.

b) Consider the following problem. (14)

Minimize $f(x, y) = (x - 1)^2 + (y - 3)^2$

Subject to $x + y \leq 2, y \geq x$

Use the following formula for the minimization problem and find the minimum values for x and y.

Lagrangian, $L = f(x) + \sum_{i=1}^{p} \lambda_i \cdot h_i(x)$.

KKT constraints are as follows: $\frac{\partial L}{\partial x_i} = 0, i = 1,2, \cdots\cdots, d.$

$\lambda_i \geq 0, i = 1,2, \cdots\cdots, p.$

$h_i(x) \leq 0, i = 1,2, \cdots\cdots, p.$

$\lambda_i \cdot h_i(x) = 0, i = 1,2, \cdots\cdots, p.$

c) The learning task in Support Vector Machine (SVM) can be formulated as the following constrained optimization problem. (07)

Minimize $\mu(W, b)$

Subject to $y_i(W \cdot x_i + b) \geq 1, i = 1,2,3, \cdots\cdots, n.$

Write down the Lagrangian and KKT constraints for this problem.

8. a) Consider the case of a binary classification with a training data of 6 tuples as shown in the following table. (12)

X_1	X_2	Y	λ_i
0.38	0.48	+	65.52
0.59	0.71	−	65.52
0.82	0.31	−	65.52
0.75	0.90	−	0
0.19	0.59	+	0
0.41	0.35	+	0

The lagrange multipliers λ_i for each training tuple are solved using quadratic programming and the first three tuples are support vectors. Determine W and b to find the Maximum Margin Hyperplane of linear SVM.

b) Briefly explain the curse of dimensionality problem. (06)

c) Consider the following dataset with 2 dimensions. (17)

Dimension 1	Dimension 2
2.5	2.4
2.2	2.9
1.9	3.0
2.7	2.5
2.0	1.6
2.1	1.1
1.0	1.8
1.9	2.7

Using Principal Component Analysis (PCA), reduce the dataset to one dimension. Explain each step with necessary calculations.

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CSE 4111
Machine Learning

TIME: 1.5 hours

FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) Differentiate learning and generalization in the contexts of machine learning. Which one is more desirable and why? (08)

   b) Define classification, regression and logistic regression in short. Differentiate between the concepts: classification and regression; and regression and logistic regression. (10)

   c) Using K-Means clustering, cluster the following objects for $K = 2$.

Object	Attribute 1	Attribute 2
A	10	150
B	150	10
C	X	Y
D	150	150

The attribute values of object C are $X = Y =$ (Last three digits of your Roll Number). (12)

2. a) Suppose bias weight of an artificial neuron is $-X$, where X is the last three digits of your Roll Number. Sketch two input AND logic and OR logic by setting appropriate values of other weights of the neurons. (10)

   b) What are the significant properties of XOR in the context of machine learning? Is it possible to construct XOR logic with a neuron taking input from developed AND logic and OR logic? Do it if possible; otherwise explain the reasons for inability. (10)

   c) What is the local gradient in Back Propagation (BP) algorithm? "Local gradient may be zero even error remains" – Justify the statement with an appropriate activation function. (10)

3. a) Why dimensionality reduction is important in machine learning? How does principal component analysis reduce dimensionality? (10)

   b) What is Bayes theorem? Shortly describe Naive Bayes Classifier. (10)

   c) Consider the label data points in the following table. (10)

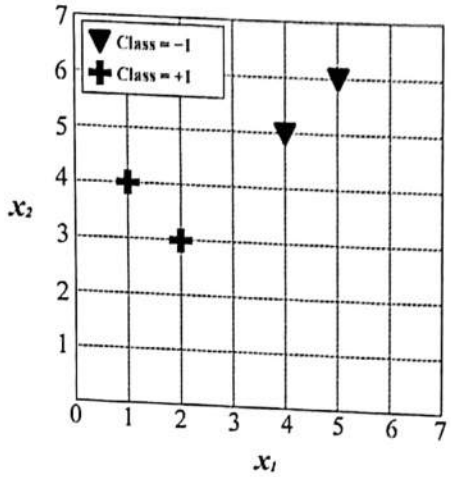
X_1	X_2	Label
05	15	+
10	30	−
30	40	+
40	100	−
80	130	+
120	250	−
Y_1	Y_2	?

Identify the label of data point (Y_1, Y_2) using KNN for $K = 3$, where Y_1 = Last three digits of your Roll Number and $Y_2 = 2Y_1$.

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Suppose you are training on a tiny dataset with 4 points shown in the following figure. This dataset consists of two examples with class label +1 (denoted with plus), and two examples with class label −1 (denoted with triangle). (10)



i. Find the weight vector w and bias b. What is the equation corresponding to the decision boundary?

ii. Circle the support vectors and draw the decision boundary.

b) Consider the following training dataset, where A, B, and C are three binary attributes, and D is a binary class label. (20)

Instance	A	B	C	D
1	0	0	0	0
2	0	0	1	0
3	0	1	0	1
4	0	1	1	1
5	1	0	1	1
6	1	0	1	1
7	1	1	0	0
8	1	1	0	0

i. What is the expected information (entropy) required to classify the training dataset?

ii. Using ID3 algorithm, build a decision tree for this classification problem.

iii. Use the built decision tree to classify two samples below.

Instance	A	B	C	D
9	1	1	1	?
10	1	0	0	?

5. a) "In PSO, each and every particle changes position in each iteration even the particle is the global best (G)" – do you agree with the statement? Justify your answer with appropriate assumption in velocity update equation. (09)

b) Is it possible to find out the population size from the Roulette wheel given below? Explain. (07)

c) Apply the Enhanced Edge Recommendation algorithm on the following two chromosomes to produce two children. Show all the steps. (14)

Parent 1	4	2	1	3	5	6
Parent 2	4	3	1	2	6	5

6. a) "Pheromone evaporation pushes ACO convergence" – Is it true? Justify your opinion. (10)

b) What two requirements should a problem satisfy in order to be suitable for solving it by GA? (05)

c) Assume 4 cities {A, B, C, D}, which are represented by a fully connected graph. The following tables represent the pheromone levels on each edge of the graph and the distances between each city (assume the pheromone levels and distances are symmetric). (15)

Table-1

Pheromone Levels				
	A	B	C	D
A				
B	0.25			
C	0.11	0.98		
D	0.34	0.54	0.67	

Table-2

Distance				
	A	B	C	D
A				
B	12			
C	10	6		
D	8	15	3	

Assume an ant started its journey at city A and has travelled to city C.

i. What are the probabilities that the ant will travel to cities A, B and D? [Let $\alpha = 1, \beta = 1$]

ii. Assume the ant completes its tour using the route ABCD. What will be the pheromone levels on each edge once they have been updated? [Let Q = 100 and evaporation parameter = 0.5]

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours

FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What do you mean by learning? In biological system, what are the elements and how they response for learning? (08)
   b) Draw computational model of a single neuron. Solve different binary logics with the model. (09)
   c) What are the different types of memory? What are their significances and uses? (09)
   d) Calculate the weight set (W) of a Hetero-Associative memory for the following samples: (09)

Pattern, p	Source s(p)	Target, t(p)
1	1 0 0 0	1, 0
2	1 1 0 0	1, 0
3	0 0 1 1	0, 1
4	0 0 0 1	0, 1

   Finally verify the system with appropriate samples.

2. a) What do you understand by linear problem and non-linear problem in the context of machine learning? Why solving non-linear problem is challenging? Briefly explain with an appropriate example. (07)
   b) "An additional dimension (input) may transform a non-linear problem to linearly separable"- justify the statement with proper example. (08)
   c) Why training may not perform after certain iteration even though error remain in training a neural network using Back-Propagation? Explain with appropriate example. (10)
   d) What is the condition to use a function as Activation Function in neural network? Graphically represent two different forms of sigmoidal functions and their derivatives. Explain similarity and dissimilarity between the functions. (10)

3. a) What are the basic differences between Radial Basis Function (RBF) network and Multi Layer Perceptron (MLP)? Briefly describe training steps of RBF networks. (09)
   b) Why ensemble of neural networks is essential when an MLP solve any arbitrary function? What are the conditions for ensemble construction? (06)
   c) What are the different techniques for training individual networks in ensemble construction? Mention different ensemble methods for each type. (08)
   d) Describe an ensemble method which uses sampling of individual patterns. Is such technique suitable for ensemble of Decision Tree? Explain briefly. (12)

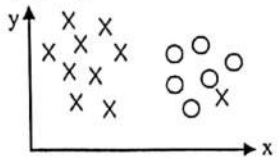
4. a) What do you mean by Swarm Intelligence (SI)? Briefly describe common features of SI based algorithms. (05)
   b) Explain importance of exploration and exploitation properties in SI algorithms. Describe exploration and exploitation matter in Particle Swarm Optimization (PSO). (10)
   c) In Ant Colony Optimization (ACO), how an ant complete a tour in solving Travelling Salesman Problem (TSP)? (08)
   d) What do you understand by the terms Global Search, Local Search and Blind Search? Does GA follows the search methods? Explain your justification with techniques included in GA. (12)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are support vectors in Support Vector Machine (SVM)? How does Linear SVM work? Explain. (10)
   b) How can SVM classify non-linearly separable data? Explain with an example and proper figure. (07)
   c) Can SVM be over trained? What can be the effect of over training? How will you detect over fitting of a SVM model? (08)

d) Can you classify the data points (x, 0) depicted in the following figure using LSVM?



Explain the procedure of finding the correct hyper plane for the data points.

6. a) Define "Occam's Razor Principle". Explain its effect on Decision Tree. (07)
   b) "Entropy measures the impurity of training data instances"-is this statement true or false? Explain with proper reasoning. (08)
   c) Formulate a decision tree based on the following experience table where Buys-computer is set as target class. (20)

Records	Age	Income	Student	Credit-Rating	Buys-computer
1	<= 30	High	No	Fair	No
2	<= 30	High	No	Excellent	No
3	31 ... 40	High	No	Fair	Yes
4	> 40	Medium	No	Fair	Yes
5	> 40	Low	Yes	Fair	Yes
6	> 40	Low	Yes	Excellent	No
7	31 ... 40	Low	Yes	Excellent	Yes
8	<= 30	Medium	No	Fair	No
9	= 30	Low	Yes	Fair	Yes
10	> 30	Medium	Yes	Fair	Yes
11	<= 30	Medium	Yes	Excellent	Yes
12	31 ... 40	Medium	No	Excellent	Yes
13	31 ... 40	High	Yes	Fair	Yes
14	> 40	Medium	No	Excellent	No

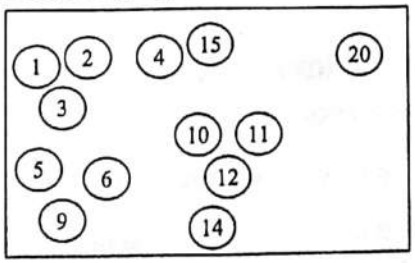
7. a) Why Pre-pruning and Post-pruning are necessary for decision trees? How does error propagate in decision tress? (10)
   b) How do you tackle the curse of dimensionality? (05)
   c) Write short notes on: (i) Data condensation, and (ii) Data squashing. (08)
   d) How does Principle Component Analysis construct the principle components? Point out the steps briefly with an example. (12)

8. a) Consider the data in the following table: (15)

ID	Height	Age	Weight
1	5	45	77
2	5.11	26	47
3	5.6	30	55
4	5.9	34	59
5	4.8	40	72
6	5.8	36	60
7	5.3	19	40
8	5.8	28	60
9	5.5	23	45
10	5.6	32	58
11	5.5	38	??

Predict the weight of ID 11 based on their height and age using KNN.

   b) What are the strengths of K-means algorithm? (05)
   c) Differentiate between divisive clustering and agglomerative clustering. (08)
   d) Draw a dendrogram from the following figure: (07)



KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Machine Learning? Briefly discuss some practical applications of Machine Learning. (05)
   b) Explain McCulloch-Pitts model of artificial neuron with appropriate sketch. Describe significance of its individual components. (08)
   c) Explain the difficulties of solving XOR problem with single neuron. Solve XOR problem through manipulation by hidden neurons for AND and OR logic. (10)
   d) How does the cross-validation technique evaluate machine learning algorithms? Explain. (06)
   e) "The choices of training data and target concepts are two important issues in designing a machine learning system"– justify the statement with appropriate examples. (06)

2. a) What is the key difference between traditional algorithms and machine learning algorithms? (05)
   b) Write down the back propagation (BP) weight update rule for both hidden layer neurons and output layer neurons where sigmoidal and tanh functions are used as activation function in output and hidden layers, respectively. (10)
   c) How momentum parameter speed up training in BP? (06)
   d) What are MLP and RBF network? Write down the differences between MLP and RBF network. (06)
   e) Draw sketch diagram of a RBF network and put parameter values to solve XOR problem. (08)

3. a) Why ensemble of several neural networks is necessary when a network with hidden layer may approximate any arbitrary function? (05)
   b) What do you mean by independent, sequential and simultaneous training of component networks in ensemble construction? Mention prominent ensemble methods with each type of component network training. (10)
   c) Describe Bagging and AdaBoost ensemble methods with graphical representation. Compare the methods in different issues such as, training set size, ensemble decision and so on. (10)
   d) Describe an ensemble method which promotes diversity altering error of individual network. Is such technique suitable for ensemble of Decision Tree? Explain. (10)

4. a) What do you understand by Swarm Intelligence (SI)? Write down common features of SI based algorithms. (05)
   b) What do you mean by exploration and exploitation? Explain exploration and exploitation tradeoff in Particle Swarm Optimization (PSO). (08)
   c) "In PSO, each and every particle changes position in each iteration even the particle is the global best (G)"– are you agree with the statement? Justify your answer with appropriate assumptions in velocity update equation. (08)
   d) Write down the equations of transition probability and pheromone update of Ant Colony Optimization (ACO). Describe effects of key parameters in ACO. (09)
   e) Describe significance properties of ACO with respect to other population based algorithms. (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define GA. Explain the working principle of GA with a flowchart. (07)
   b) "Crossover is explorative and Mutation is exploitative"– justify the statement with respect to GA. What are the effects of too high and too low Mutation rate? (10)
   c) Write down the similarities and dissimilarities between GA and PSO. (06)
   d) Apply the Enhanced Edge Recombination algorithm on the following two chromosomes to produce two child. Show all steps. (12)

Parent 1:	4	2	1	3	5	6
Parent 2:	4	3	1	2	6	5

6. a) What is the problem of using normal crossover and mutation operators in GA for solving Travelling Salesman Problem (TSP)? Briefly explain Insert Mutation operator for TSP.
   b) What is Decision Tree? Write down the steps of Decision Tree construction. (07)
   c) What is Information Gain? How does Information Gain handle continuous-valued attributes? (07)
   d) Consider the following table where attribute 'Status' represents class label. Find the root attribute of the Decision Tree using Gain Ratio as the attribute selection measure. (12)

Department	Age	Status	Salary
Sales	31··35	Senior	Medium
Sales	26··30	Junior	Low
Marketing	36··50	Senior	High
Secretary	26··30	Senior	Medium
Secretary	36··50	Senior	High
Sales	36··50	Junior	High
Marketing	31··35	Senior	Medium
Marketing	26··30	Junior	Low
Secretary	31··35	Junior	Medium
Secretary	26··30	Junior	Low

7. a) Distinguish between Supervised and Unsupervised learning. (06)
   b) What is the problem associated with K-means clustering algorithm? How K-mediod clustering method solve this problem? Explain. (12)
   c) What are the advantages of Hierarchical clustering over Partitioning method of clustering? (05)
   d) Consider the following 5 data points. (12)

   (1, 1), (3, 2), (7, 9), (8, 7), (9, 8)

   Find the clusters using Agglomerative clustering and use Complete Link proximity as distance measure. Show all steps.

8. a) What is the problem of DBSCAN algorithm? Discuss. (05)
   b) Explain core-distance and reachability-distance with graphical representation. (06)
   c) Briefly explain the curse of dimensionality problem. (06)
   d) Consider the following dataset with 2 dimensions. (18)

Dimensions 1	Dimensions 2
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
2.5	3.0
2.7	2.5
2.0	1.6
1.0	1.1
2.5	2.6
2.1	1.9

Using Principal Component Analysis (PCA), reduce the dataset to one dimension. Explain each step with necessary calculations.

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2017
Department of Computer Science and Engineering
CSE 4211
Machine learning

TIME: 3 hours FULL MARKS: 210

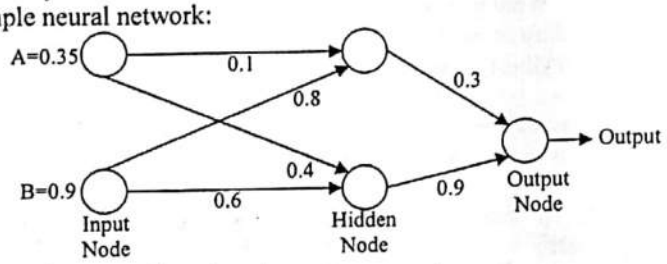
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Machine Learning? Briefly discuss some practical applications of Machine Learning. (08)
   b) Distinguish between supervised and unsupervised learning with example. (07)
   c) How do you differentiate regression and classification problems in Machine Learning? Give some practical examples. (10)
   d) Briefly explain the curse of dimensionality problem. (10)

2. a) What do you mean by feed forward network? (05)
   b) Consider the simple neural network: (10)



   (i) Perform a forward pass on the network.
   (ii) Perform a backward pass on the network
   (iii) Perform a further forward pass and comment on the result.
   Use sigmoid function as activation function.
   c) Explain in difficulties to solve XOR problem with single neuron. Solve XOR problem through manipulation by hidden neurons for AND and OR logic. (10)
   d) Draw minimal neural models for the following two logics having inputs A, B and C. (10)
   (i) NN1=(A+B)C' (ii) NN2=ABC'

3. a) What is the effect of learning rate in BP? How momentum parameters speed up training in BP? (10)
   b) What are the basic differences between RBF network and MLP. (08)
   c) What is local gradient in BP training procedure? For a neural network with two hidden layers, write local gradients for each layer. (10)
   d) Differentiate learning and generalization? Which one is more desirable and why? (07)

4. a) What is the purpose of mathematical optimization? Define following terminologies with respect to mathematical optimization: (i) Objective function (ii) variables/parameter (iii) Constraints (iv) Constrained and unconstrained optimization (v) language multiplier. (07)
   b) How to solve an unconstrained optimization problem using Gradient descent. Explain in detail. (08)
   c) Solve the following constrained optimization problems: (i) Given the objective function (20)

$f(x,y) = x^3 + y^3$ *subject* to

$g_1(x,y) = x^2 - 1 \geq 0$

$g_2(x,y) = y^2 - 1 \geq 0$

Find the optimum value.

(ii) A company has determined that its production is the cobb-douglas function, $f(x,y) = x^{2/3}y^{1/3}$ where x=number of labour hours, y=number of capital units, the budget constraint for the company is 100X+100Y=400000. What is the maximum production?

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is meant by Bio-inspired computing? Write some the popular Bio-inspired computing algorithm. (10)
   b) Why do we prefer heuristic approaches over classical methods to solve optimization problems? Explain with example. (10)
   c) Explain the trade-off criteria for deciding whether to use a heuristic for solving a given problem. (08)
   d) Define Genetic Algorithm (GA). Write simple GA. Write nature to computer mapping. (07)

6. a) Explain exploitation and exploration with examples. (10)
   b) Why do we not use one point crossover to solve TSP? Explain. (07)
   c) State the differences between roulette wheel selection and tournament selection. (08)
   d) Solve traveling salesman problem using particle swarm optimization. (10)

7. a) What is Swarm Intelligence (SI)? What are the properties of SI system? (10)
   b) Write PSO algorithm. What problems are arise when PSO is used in multimodal problems? (10)
   c) Write working procedure of MPSO. (05)
   d) Differentiate between Gbest swarm and Lbest swarm. (10)

8. a) Describe the important of transition probability in ACO algorithm. (10)
   b) "Pheromone evaporation pushes ACO convergence." is it true? Justify your opinion. (07)
   c) Write ACO algorithm. (08)
   d) Explain the employed bee phase, on looker bee phase and scout bee phase in ABC optimization algorithm. (10)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2015
Department of Computer Science and Engineering
CSE 4211
Machine Learning

TIME: 3 hours FULL MARKS: 210

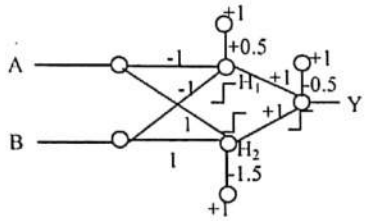
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

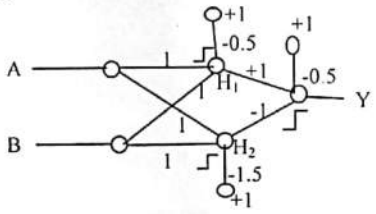
(Answer **ANY THREE** questions from this section in Script A)

1. a) Explain McCulloch-Pitts model of artificial neuron with appropriate sketch. Describe significance of its individual components. (09)
   b) Describe importance of bias weight in artificial neuron. "Only changing the weight value of bias operation of a neuron may change"-justify the statement for appropriate logic gates. (09)
   c) Draw single neuron model for following logics: (08)
   i) $y = x_1' x_2'$, ii) $y = x_1' x_2$, iii) $y = x_1 x_2$, and iv) $y = (x_1 x_2)'$.
   d) Describe shortly the three fundamental and practical issues of learning theory. (09)

2. a) Look the following two neural networks model. (10)



(i)



(ii)

For the model tabulate the outputs of H_1, H_2 and Y for inputs (A and B) values 0 or 1. Are the model performs any logic gate operation? Which ones?
   b) What is the condition to use a function as activation function in ANN? Graphically represents two different forms sigmoidal functions and their derivatives. How derivatives indicate the stability of ANN training? (10)
   c) What is the naming significance of "Back Propagation (BP)" algorithm? Describe basic steps of BP. (07)
   d) What is local gradient in BP training procedure? For an ANN with two hidden layers, write local gradients of each layer. (08)

3. a) What is the effect of learning rate in BP? How momentum parameters speed up training in BP? (05)
   b) What are basic differences between Radial Basic Function (RBF) network and MLP? (05)
   c) Why hybrid training algorithm is required for training RBF networks? Draw sketch diagram of a RBF network and put parameter values to solve XOR problem. (08)
   d) What are the properties and usefulness of the followings: (07)
   i) Local Addressable Memory, ii) Content Addressable Memory, and iii) Associative Memory.
   e) Calculate the weight set (w) of a Hetero- Associative Memory for the following samples: (10)

Pattern	Source, s(p)	Target, t(p)
1	(1 0 0 0)	(1 0)
2	(1 1 0 0)	(1 0)
3	(0 0 0 1)	(0 1)
4	(0 0 1 1)	(0 1)

Finally verify that the system work correctly.

4. a) Differentiate learning and generalization. Which one is more desirable and why? (05)
   b) Why ensemble of classifiers is necessary when a MLP able to capture any arbitrary function? What is the condition for better ensemble construction? (05)
   c) Among the studied ensemble methods, whose are able to take facility of multi-processing system? How? (07)

d) Describe an ensemble method which promotes diversity through artificial patterns. (08)
e) What is support vectors in support vector machine (SVM)? How SVM may classify non linearly separable data? Explain with appropriate examples. (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Why we prefer nature-inspired heuristic methods although they might not give exact solution for a particular optimization task? (05)
b) Why selection is necessary in Genetic Algorithm (GA)? Write down the differences between Rank based selection and Tournament selection. (12)
c) Write down the importance of using mutation in GA. Also write down the effects of mutation rate in GA. (07)
d) Is it possible to find out the population size from the Roulette wheel given below? Explain. (06)

Fig. 5(d)

e) What is Elitism? Why is it important in GA? (05)

6. a) "In Particle Swarm Optimization (PSO) every particle updates position in each and every iteration even it is the global best"-justify the statement from the velocity and position update equations. (08)
b) Write down the working procedure of Attractive Repulsive PSO and Multiswarm PSO. (08)
c) What is the effect of Lifespan controller in ALC-PSO? How Lifespan of a leader is adaptively tuned? Explain. (12)
d) Write down the similarities and dissimilarities between GA and PSO. (07)

7. a) Describe the importance of transition probability in selecting the next city in TSP using Ant Colony Optimization (ACO). (10)
b) "Pheromone evaporation pushes ACO convergence"-is it true? Justify your opinion. (08)
c) Explain the Employed bee phase, onlooker bee phase and scout bee phase of Artificial Bee Colony Optimization (ABCO) algorithm. (12)
d) Explain differences of Biogeography-Based Optimization (BBO) from ACO and PSO. (05)

8. a) How does Cuckoo Search (CS) algorithm provide exploration and exploitation? Explain. (08)
b) How does Bacterial Foraging Optimization Algorithm (BFOA) find the optimum solution? (05)
c) What is the benefit of using OPTICS clustering algorithm? How this algorithm works? Explain. (08)
d) Define the core point and border point. Consider the data points in the following figure Fig. 8(d) with radius where Minpts = 6. Find out the core points and border points. (06)

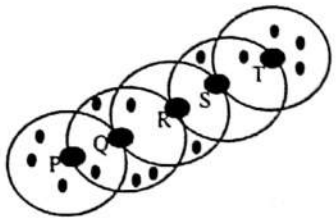


Fig. 8(d)

Is P and T density reachable from each other?

e) Consider a set of records that have three attributes namely salary, age and vacation. How CLIQUE algorithm clusters data objects? Explain. (08)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2024
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours | FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates the course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is meant by Natural Language Processing (NLP)? How does it differ from Machine Learning (ML)? (07) [CO1]
   b) Why are NLP tasks sometimes become difficult for a machine to do? Explain with examples. (12) [CO2]
   c) Differentiate between sentence-level classification and token-level classification with suitable examples. (06) [CO5]
   d) Differentiate between linear regression and logistic regression. Which one is appropriate for binary text classification problem? Give a proper justification. (10) [CO2]

2. a) How does pre-training is related to fine-tuning? Explain. (07) [CO1]
   b) "TF-IDF has better feature representation than BOW."– Is it true? If so, justify. (08) [CO3]
   c) "FFNN is not suitable for processing sequential data." – Explain. (08) [CO2]
   d) Suppose, you need to solve a 3-class text classification problem. How do you perform the task deploying Bi-LSTM? Explain with proper figure. (12) [CO5]

3. a) Draw vanilla transformer architecture and explain its main parts. (06) [CO1]
   b) What are the main functions of query, key, and value vectors in transformer architecture? (06) [CO1]
   c) Discuss pre-training and fine-tuning procedure of BERT model. (15) [CO5]
   d) "BERT can be used as a generalized understanding model." – Explain. (08) [CO1]

4. a) How does BERT is utilized to extract features from the text? Explain with detailed diagram. (12) [CO5]
   b) What is sentiment analysis? How can BERT be used to deal with the sentiment analysis task? (08) [CO1]
   c) Discuss machine translation by vanilla transformer model. (08) [CO2]
   d) What is co-reference resolution? Explain with example(s). (07) [CO1]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define corpus. What are the properties of a corpus? (05) [CO1]
   b) "Patterns are greedy." – Why? Design a regular expression to answer the following statement: "Find a computer with at least 6 GHz and 16 GB RAM within 800$," (10) [CO1]
   c) You have two strings "brief" and "drivers". Figure out whether "drive" is closer to "brief" or "drivers" using minimum edit distance algorithm. (14) [CO2]
   d) Define stop words. What are the applications of stop word detection in NLP tasks? (06) [CO1]

6. a) Define N-gram language model. How can you compute the probability of a sentence of n words P(w), $w = w_1 w_2 \ldots\ldots w_n$? How can you calculate the maximum likelihood by bigram probabilities? (10) [CO1]
   b) Consider the following corpus shown in the following figure. What is the most probable next word? (15) [CO3]

<s> I like Henry </s>
<s> I like college </s>
<s> Do Henry like college </s>
<s> Henry I am </s>
<s> Do I like Henry </s>
<s> Do I like college </s>

i) "<s> Do __" applying bigram probability model?
ii) "<s> Do I like __" applying bigram probability model?
iii) "<s> Do I like __" applying trigram probability model?

c) Define smoothing. How can you smooth the language model by interpolation? How to choose the value of 'λ' to maximize the probability? (10) [CO3]

7. a) What is the difference between tokens and embeddings? Explain the pipeline of creating embedding model. (08) [CO4]

b) Consider the following grammar G shown in the following figure. Is the string "The flight includes a meal." in L(G)? Answer it using CYK algorithm. (15) [CO3]

S → NP VP
NP → Det N
VP → V NP
V → includes
Det → the | a
N → meal | flight

c) Given a sequence of ice-cream observations (1 1 3) and an HMM λ = (A,B) in the following figure. Find the best hidden weather sequence Q (like H H H). (12) [CO2]

A		B	
start → HOT	0.25		
start → COLD	0.75		
HOT → HOT	0.60	P(1|HOT) =0.20 P(2|HOT) = 0.35 P(3|HOT) = 0.45	P(1|COLD) =0.50 P(2|COLD) = 0.40 P(3|COLD) = 0.10
HOT → COLD	0.40		
COLD → COLD	0.55		
COLD → HOT	0.45		

8. a) "POS tagging is a disambiguation task" – Justify the statement with example. (10) [CO1]

b) Consider the sentence: "It/PRP would/MD be/VB better/JJR to/TO book/? a/DT ticket/NN via/IN internet/NNP." The word "book" is often used as VB or NN. Given the probabilities below, find the right POS tag for the word "book". (12) [CO3]

P(VB|TO) = 0.67, P(NN|TO) = 0.02, P(book|VB) = 0.50,
P(book|NN) = 0.50, P(DT|VB) = 0.02, P(DT|NN) = 0.06.

c) For Hidden Markov Model (HMM) POS tagging, using the following formula, find the equation of calculating tag transition probabilities. (13) [CO2]

$$\hat{t}_1^n = \underset{t_1^n}{argmax}\, P(t_1^n | w_1^n)$$

CamScanner

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2023
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates the course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define corpus. Write a regular expression to find an answer for the following statement. (08) [CO1]
"A computer with 6 GHz and 2 TB storage within 500$".
b) What is lemmatization and stemming? How is lemmatization done? (06) [CO1]
c) How does Byte Pair Encoding (BPE) work for tokenization? Use the following training corpus to tokenize the test word "hugs". (14) [CO1]
"This is the hugging face course. It shows tokenizer algorithms".
d) Make a classification of the types of spelling errors and their solution approaches. (07) [CO1]

2. a) You have two strings *St*1 and *St*2. Develop an algorithm to go from *St*1 to *St*2 with minimum cost of edit. Use insert, delete cost = 1 and update cost = 2. Explain your algorithm using example as *St*1 = "INTENTION" and *St*2 = "EXECUTION". (15) [CO3]
b) Define language model. "Predicting something is difficult". How does *N*-gram language model predict next word? Explain with suitable example. (14) [CO2]
c) How is language model evaluated? (06) [CO2]

3. a) Define text classification. "Naïve Bayes is a form of generative classifier" – justify the statement. (08) [CO3]
b) How does Naïve Bayes learn? Using information from the following table, find the class of the test sentence. (12) [CO3]

	Document	Class
Training	The movie is amazing	Positive
	I do not like the film	Negative
	Great acting and engaging	Positive
	Terrible movie and not recommended	Negative
Test	The acting is superb	?

c) What is parsing in NLP? What are the differences between syntactic and semantic parsing? Explain. (05) [CO3]
d) How does CYK algorithm work? For the following grammar G, is "$baba$" in $L(G)$? (10) [CO3]

$$S \rightarrow AB|BC$$
$$A \rightarrow BA|a$$
$$B \rightarrow CC|b$$
$$C \rightarrow AB|a$$

4. a) "You have a true word w and a noisy channel that modifies w into w'". How can you find the w from w'? Explain. (10) [CO5]
b) How can you define a word as a vector? How can you measure the similarity between two words u and v? Show the similarity of the words from the following table. (15) [CO5]

	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

c) What is absolute discounting? Explain what are its advantages. (10) [CO5]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is closed and open class of Parts of Speech (POS)? Explain with example. (05) [CO2]
   b) Explain different type of taggers and describe Hidden Markov Model (HMM) POS tagging using example. (10) [CO1]
   c) Explain Viterbi algorithm for POS tagging with example. (10) [CO1]
   d) How do conditional random fields enhance POS tagging through feature incorporation and why are features challenging to integrate into HMMs? (10) [CO1]

6. a) What is relation extraction? Explain this concept using pattern-based relation extraction and supervised learning algorithm. (10) [CO2]
   b) Explain semi-supervised relation extraction via bootstrapping. (08) [CO2]
   c) What are temporal lexical triggers? How do we detect events? (06) [CO2]
   d) How is the *tf-idf* score computed for terms in documents and how can it be used to calculate the cosine similarity between a query and documents using BM25? Demonstrate for the following dataset. (11) [CO5]

   Assume, $|d| = 4$, $|d_{avg}| = 3$, $k = 1.3$, $b = 0.75$.

   Query: sweet love
   Doc 1: Sweet sweet nurse! Love?
   Doc 2: Sweet sorrow
   Doc 3: How sweet is love?
   Doc 4: Nurse!

7. a) Explain the evaluation of information retrieval systems and describe IR-based factoid question answering with a diagram. (08) [CO5]
   b) How can BERT-based models be effectively trained and evaluated for answer span extraction in information retrieval-based question answering systems? (09) [CO5]
   c) What is entity linking? Explain the process of entity mention detection using mathematical formulas and a diagram. (08) [CO5]
   d) What is knowledge-based question answering? Describe Watson's question answering system with the help of a diagram. (10) [CO5]

8. a) How does attention mechanism work in machine translation and why is it important for improving translation quality? (09) [CO4]
   b) Explain hourglass metaphor in text-to-speech with proper diagram. (08) [CO4]
   c) What is the difference between tokens and embeddings? Explain the pipeline of creating embedding model. (08) [CO4]
   d) Why is beam search necessary? Explain beam search with example. (10) [CO2]

Done

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2022
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define corpus. What are the properties of a corpus? (05) [CO1]
   b) "Patterns are greedy" – why? Design a regular expression to answer the following statement: "Find a computer with at least 6 GHz and 512 GB RAM for less than BDT 4000". (10) [CO1]
   c) Define subword tokenization. Explain the BPE algorithm for subword tokenization. Use the following corpus to explain the algorithm.
   "low low low low low lowest lowest Newer Newer Newer Newer wider wider wider new new"
   Test Set: newer, lower. (15) [CO2]
   d) Define stop words. What are the applications of stop word detection in NLP tasks? (05) [CO1]

2. a) You have two strings "brief" and "drivers". Figure out whether "drive" is closer to "brief" or "drivers" using minimum edit distance algorithm. (14) [CO2]
   b) What is *n*-gram language model? Find the uni-gram, bi-gram, and tri-gram from the following corpus.
   "This is Big Data AI Book". (06) [CO3]
   c) Why do you need smoothing techniques for *N* gram model? Explain the following smoothing techniques: (15) [CO3]
      i) Add-1 smoothing.
      ii) Stupid back off.
      iii) Kenser-Ney smoothing.

3. a) Give a formal definition of supervised text classification. How can you differentiate generative and discriminative classifier? (05) [CO4]
   b) What is "Bag of words" model? What are the assumptions of Bag of words model? How does Bag of words applied for multinomial Naïve Bayes Classifier? Using the following training set find the class of the test set. (15) [CO4]

		Document	Class
Training	:	Just plain boring	–ve
		No surprise and few laughs	–ve
		Very powerful	+ve
		Most fun film of the summer	+ve
Test	:	Predictable with no fun	?

   c) When is "Accuracy" not a good metric? Explain with example. (07) [CO3]
   d) "The noisy channel is a kind of Bayesian inference" – explain the term. (08) [CO3]

4. a) What are the types of spelling errors? you wanted to type 'three' but mistakenly you typed 'there'. How can you detect such an error automatically? How to correct these types of errors? (12) [CO3]
   b) What are the common structural ambiguities of Natural Language Processing? Explain. (06) [CO3]
   c) What is partial parsing? What are the applications of partial parsing? (05) [CO3]
   d) Consider the following grammar G. Using the CYK algorithm, find whether "ababa" is L(G) or not. (12) [CO3]

$S \rightarrow AB|BC$
$A \rightarrow BA|a$
$B \rightarrow CC|b$
$C \rightarrow AB|a$

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain the large language model (LLM) with an example. Discuss computational resources and energy consumption in LLM. (10) [CO1]
   b) "Parts-of-Speech (POS) tagging is a disambiguation task" – justify the statement with example. (10) [CO2]
   c) Discuss POS tagging and tagsets for English. (10) [CO3]
   d) How difficult is POS tagging in English? (05) [CO3]

6. a) Discuss Conditional Random Fields using Bayes' rule. (10) [CO2]
   b) We have a dataset (NewG). Some information about the dataset is given below. (25) [CO2]

No. of VB	:	17619	Starting probability VB	:	0.024
No. of MD	:	11338	Starting probability MD	:	0.11
No. of NN	:	16559	Starting probability NN	:	0.86
$t_{i-1} - t_i$			$t_i - w_i$		
VB – VB	:	8583	VB – race	:	5391
VB – MD	:	6988	VB – will	:	6308
VB – NN	:	5826	VB – begin	:	7488
MD – VB	:	7984	MD – race	:	6557
MD – MD	:	4204	MD – will	:	6635
MD – NN	:	4046	MD – begin	:	6905
NN – MD	:	5006	NN – race	:	4163
NN – NN	:	4568	NN – will	:	6103
			NN – begin	:	7863

Use the dataset to answer the following questions:

i) Calculate the transition probability (A).
ii) Calculate the emission probability (B) for sentence: "race will begin".
iii) Find the tag sequence for the sentence "race will begin" using hidden Markov model.

7. a) Discuss the sequence to sequence model with attention mechanism (AM). Demonstrate the limitations without AM. (10) [CO2]
   b) Can hidden Markov models incorporate features? (05) [CO2]
   c) What is bootstrapping in information extraction? Explain the architecture for bootstrapping. (10) [CO4]
   d) How do we detect events? Write down features commonly used in both rule-based and statistical approaches to event detection. (10) [CO4]

8. a) Explain term weighting and document scoring. (05) [CO5]
   b) (12) [CO5]

Query	:	Hello dear
doc1	:	Hello my friends
doc2	:	Hello Karim, hello Rahim
doc3	:	My dear friend, Hello
doc4	:	Hello, Dear, dear

Using term weighting and document scoring, rank the document for Query.

   c) Discuss language models (LMs) to do question answering (QA). Write down the comparative analysis between LM QA models and classic QA models. (10) [CO5]
   d) What is homograph disambiguation? What problems does CMU face in this regard, and how does UNISYN address these issues? (08) [CO4]

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2021
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What are the differences between Natural Language Processing (NLP) and Text Mining? Explain. (06)
   b) What is Lemmatization and Stemming? Discuss one popular Lemmatization and Stemming algorithm with example. (10)
   c) Develop the regular expression to find all the instances of the word "there" in a text. (07)
   d) You have two strings X of length n and Y of length m. Develop an algorithm to edit the string X to Y. How many operations you need? Explain your algorithm for X = "KITTEN" and Y = "SITTIN". (12)

2. a) Define N-gram language model. How can you compute the probability of a sentence of n words P(w), $w = w_1 w_2 \ldots w_n$. How can you calculate maximum likelihood by bigram probabilities? (10)
   b) Consider the following corpus. (15)

<S> I Like Henry </S>
<S> I Like College </S>
<S> Do Henry Like College </S>
<S> Henry I am </S>
<S> Do I Like Henry </S>
<S> Do I Like College</S>

What is the most probable next word
   i) "<S> Do —" applying bigram probability model?
   ii) "<S> Do I Like —" applying bigram probability model?
   iii) "<S> Do I Like —" applying trigram probability model?

   c) Define smoothing. How can you smooth the language model by interpolation? How to choose the value of 'λ' to maximize the probability? (10)

3. a) What is Bag of words representation? How do you apply Bag of words assumption and conditional independence for naïve bayes classifier? Explain. (12)
   b) Consider the following Training dataset. (08)

item	class
Just plains boring	−
Entirely Predictable and lacks energy	−
No surprises and very few laughs	−
Very powerful	+
The most fun film of the summer	+

Find the class of the following sentence "Predictable with no fun", using naïve bayes classifier.
   c) How can you apply noisy channel for real world spelling correction? (10)
   d) "Accuracy is not a good metric when the goal is to discover something rare" –justify the statement. (05)

4. a) What are the common structural ambiguities for any NLP task? Give example(s). (08)
   b) Consider the following grammar G (13)

$S \rightarrow NP\ VP$
$NP \rightarrow Det\ N$
$VP \rightarrow V\ NP$
$V \rightarrow includes$
Det → the | a
$N \rightarrow meal \mid flight$

Is the string "The flight includes a meal." in L(G)? Answer it using CYK algorithm.

c) Define PCFG. What is the condition to be maintained by a PCFG? How can you disambiguate the parsing using PCFG? (09)

d) What are the applications of shallow parsing? (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is closed class and open class of parts of speech? Explain with example. (08)

   b) Give a short description of Viterbi algorithm. What are the applications of the algorithm? (07)

   c) Use the following table to answer the following questions. (20)

We have a dataset (NextD). Some information about the dataset is given below.	
No. of VB: 10756	Starting probability VB: 0.44417
No. of MD: 19375	Starting probability MD: 0.7836
No. of NN: 14479	Starting probability NN: 0.25762
$(t_{i-1} - t_i)$	
VB–MD: 9544	VB–race: 6272
VB–NN: 9385	VB–will: 8977
MD–VB: 9918	VB–begin: 5569
MD–NN: 9590	MD–race: 6891
NN–VB: 9065	MD–will: 6301
NN–MD: 9754	MD–begin: 4857
VB–VB: 0	NN–race: 4910
NN–NN: 0	NN–will: 4012
MD–MD: 0	NN–begin: 4617

   i) Calculate the transition probability (A).

   ii) Calculate the emission probability (B) for sentence: "race will begin".

   iii) Find the tag sequence for the sentence "race will begin" using Viterbi algorithm.

6. a) Define Named Entity Recognition. Why recognizing name entity is difficult? (07)

   b) Discuss conditional random fields (CRF). Write down the features in a CRF PoS tagger. (08)

   c) Discuss the IR-based Factoid Question Answering system with proper diagram(s). (10)

   d) Define knowledge-based question answering. Discuss four broad stages of Watson QA with block diagram(s). (10)

7. a) Write down classes of algorithms for relation extraction. Describe Relation Extraction via Supervised Learning and Feature-based supervised relation classifiers. (10)

   b) Write down three ways to extract Times and temporal Expression. Explain with examples. (08)

   c) Discuss the encoder-decoder model for machine translation with proper figures. (10)

   d) Define entity linking. Discuss neural graph-based linking with proper diagram(s). (07)

8. a) Discuss the information Retrieval question-answering system. Explain with examples. (08)

   b) Consider the following data: (12)

Query	:	Sweet love
Doc 1	:	Sweet sweet nurse! Love?
Doc 2	:	Sweet sorrow
Doc 3	:	How sweet is love?
Doc 4	:	Nurse!

   Using Term Weighting and document scoring, Rank the document for query.

   c) What is Homograph disambiguation? What are the problems of CMU? How does UNISYN overcome the problems of CMU? (07)

   d) Speech synthesis performs text-to-waveform mapping in two steps. Name and discuss the steps using Hourglass metaphor. (08)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2020
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 1.5 hours FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) "Pattern matching by regular expression is greedy" – Explain the term taking example to match the word 'There' from a string. (10)
   b) Given two strings $S1$ and $S2$ and three operations ADD, DELETE and EDIT. How many minimum operations are needed to convert $S1$ to $S2$? Derive the algorithm to answer the question. Apply the algorithm for $S1$ = "*abcdef*" and $S2$ = "*azced*". If each of the operation costs 1 unit, find the total cost to convert $S1$ to $S2$. (20)

2. a) "Minimizing perplexity is same as maximizing the probability" – Explain the term. (12)
   b) Consider the following corpus: (12)
   $\langle S\rangle abc\langle /S\rangle$
   $\langle S\rangle bac\langle /S\rangle$
   $\langle S\rangle abc\langle /S\rangle$
   Calculate the bigram probabilities $P(a|\langle S\rangle)$, $P(c|b)$ and $P(b|a)$.
   c) How do you handle the stop words for text classification? (06)

3. a) Given the following text classification for C_1 and C_2 (14)
   $PQR \rightarrow C_1$
   $PPS \rightarrow C_1$
   $PT \rightarrow C_1$
   $UVP \rightarrow C_2$
   Compute the most likely class for the text "*PPPUV*".
   b) Consider the following grammar in CNF (16)
   $S \rightarrow AB \mid BC$
   $A \rightarrow BA \mid a$
   $B \rightarrow CC \mid b$
   $C \rightarrow AB \mid a$
   Is '*baaba*' in $L(G)$? Explain your answer using CYK algorithm.

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) "POS tagging is a disambiguation task" – Justify the statement with example. (10)
   b) Consider the sentence: (08)
   "*It/PRP would/MD be/VB better/JJR to/TO book/?a/DT ticket/NN via/IN internet/NNP*".
   The word "*book*" is often used as *VB* or *NN*. Given the probabilities below, find the right POS tag for the word "race".
   $P(VB|TO) = 0.67$, $P(NN|TO) = 0.02$, $P(book|VB) = 0.50$,
   $P(book|NN) = 0.50$, $P(DT|VB) = 0.02$, $P(DT|NN) = 0.06$.
   c) What is prosody? Discuss about the three aspects of prosody. Discuss the effect of zero situation. (12)

5. a) Given a sequence of ice-cream observations (1 1 3) and an HMM $\lambda = (A, B)$ in the following figure, find the best hidden weather sequence Q(*like H H H*). (12)

A		*B*	
Start → *HOT*	0.25	$P(1 \mid HOT) = 0.2$ $P(2 \mid HOT) = 0.35$ $P(3 \mid HOT) = 0.45$	$P(1 \mid COLD) = 0.5$ $P(2 \mid COLD) = 0.4$ $P(3 \mid COLD) = 0.1$
Start → *COLD*	0.75		
HOT → *HOT*	0.60		
HOT → *COLD*	0.40		
COLD → *COLD*	0.55		
COLD → *HOT*	0.45		

b) Define the term odds for logistic regression. Show that the observation should be labeled true if $\sum_{i=0}^{N} w_i f_i > 0$. (12)

c) What are the components of Markov chain? (06)

6. a) Discuss about the steps of text to waveform mapping in speech synthesis using Hourglass metaphor. (10)

b) What is Homograph disambiguation? What are the problems of CMU? How does UNISYN overcome the problems of CMU? (10)

c) What are the use of Question Answering System (QAS)? Discuss about the main phases of QAS architecture. (10)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2019
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Why is the pattern matching by regular expression greedy? Explain with example. (09)
   b) Define text normalization. What is the difference between lemmatization and stemming? Explain with example. (10)
   c) Explain porter's algorithm with example. (11)
   d) What is types and tokens? How many types and tokens are there in the following sentence? "He stepped out into the hall, was delighted to encounter a water brother." (05)

2. a) "Finding the minimum edit distance is problem of the category 'dynamic programming' "- Explain the statement to convert one String X of length n to a String Y of length m. (12)
   b) Apply the minimum edit distance algorithm (using insertion cost = 1, deletion cost = 1 and substitution cost = 2) of "leda" to "deal". Show your work in edit distance grid. (10)
   c) Define bag of words problem. How is bag of words conditional independence assumption applied in Naïve Bayes text classification? Explain. (13)

3. a) Define Stop word. How do you handle Stop word in text classification? (08)
   b) Given the following text classification for classes C_1 and C_2 (10)
      i) PQR → C_1
      ii) PPS → C_1
      iii) PT → C_1
      iv) UVP → C_2

      Compute the most likely class for the text "PPPUV".
   c) Consider the following likelihood for a review (05)

	Pos	Neg
I	0.09	0.16
always	0.07	0.06
like	0.29	0.06
foreign	0.04	0.15
films	0.08	0.11

   What class will the Naïve Bayes assign to the sentence "I always like foreign films"?
   d) Find the context free rules and hence the context free grammar (CFG) for the following English sentences. (12)
      i) I want a morning flight.
      ii) Which flight serves breakfast?
      iii) Show the lowest fare.
      iv) The flight should be at 11 am.

4. a) Consider the following grammar in CNF (10)

$$S \rightarrow AB \mid BB$$
$$A \rightarrow CC \mid AB \mid a$$
$$B \rightarrow BB \mid CA \mid b$$
$$C \rightarrow BA \mid AA \mid b$$

   Is 'aabb' in L(G)? Justify your answer using CYK algorithm.
   b) How does Early Parser differ from CYK Parser? Explain finite state chunking with example. (10)
   c) Define PCFG? How can you disambiguate a syntactic Parsing by PCFG? (06)
   d) Explain the IR based factoid question answering using example. (09)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define Natural Language Processing (NLP). Why do we need NLP? Write some industrial applications of NLP? Derive (10)
   b) What does smoothing mean? ~~Drive~~ the equation for Laplace smoothing. (10)
   c) Given the following corpus: (07)

   I like NLP.
   I like Deep Learning.
   NLP and Deep Learning have huge area for research.

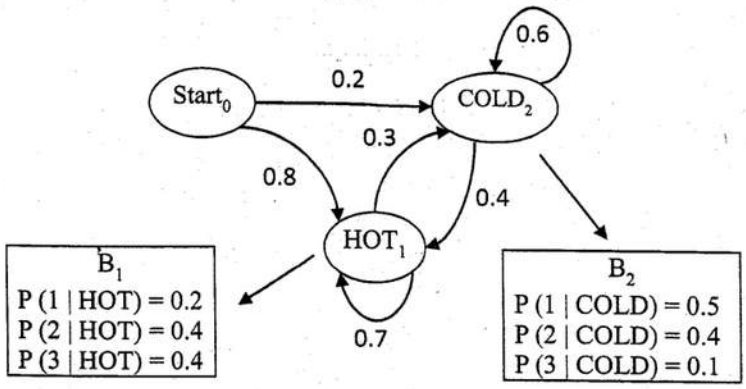
   Using Bigram Language model with add-one smoothing, what is P (NLP | like)?
   d) Define zero situation and discuss the effect of it. (08)

6. a) "POS tagging is a disambiguation task" – Justify the statement with example. (08)
   b) Define perplexity. Derive the equation of computing W with a Bigram Language model. (10)
   c) For Hidden Markov Model (HMM) Pos tagging, using the following formula, find the equation of calculating tag transition probabilities (10)

$$\hat{t}_1^n = \underset{t_1^n}{argmax}\, P\,(t_1^n \mid w_1^n)$$

   d) Consider the sentence: "Eight/CD horses/NNS will/MD race/? For/IN the/DT cup./NN." Given the probabilities below, find the right Pos tag for the word "race".(VB or NN). (07)
   P (VB | MD) = 0.0045, P (NN | MD) = 0.062, P (race | NN) = 0.048, P (race | VB) = 0.014, P (IN | VB) = 0.0012, P (IN | NN) = 0.0024

7. a) Define Markov chain. Explain the components of Markov chain. (06)
   b) For logistic regression, show that P (y = true | x) = 1 | (1 + $e^{-w.f}$) (10)
   c) Given a sequence of ice-cream observations 3 1 3 and an HMM λ = (A, B) in the following figure. (12)



   Find the best hidden weather sequence Q (like H H H).
   d) What are the statistical models for processing text and speech? Give example of fully connected & left to right HMM. (07)

8. a) Define sentence tokenization. Draw the architecture of a TTS. (07)
   b) Discuss about the steps of speech synthesis (text to waveform). Show the Hourglass Metaphor. (10)
   c) What is prosody? Discuss about the three aspects of prosody. (10)
   d) What is accent ratio? Why is accent ratio important? (08)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2018
Department of Computer Science and Engineering
CSE 4221
Natural Language Processing

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Disjunction, Grouping and Precedence for pattern matching in regular expressions? Explain with example. (10)
   b) "Pattern matching by regular expressions are greedy." – Justify the statement. (09)
   c) Design a regular expression to find all instances of the word "the" in a text. (10)
   d) Define types and tokens. How many types and tokens are there in the following sentence: (06)
   "They picnicked by the pool, then lay back on the grass and looked at the stars."

2. a) What is Lemmatization and Stemming? How is Lemmatization done? (08)
   b) What are the operations for editing one string to another? Explain. (06)
   c) Explain the algorithm to edit one string X of length n to a string Y of length m. Show the steps of your algorithm for X = INTENTION and Y = EXECUTION. (12)
   d) Discuss about the problem with Maximum Likelihood. How does the Laplace (add-1) Smoothing solve the problem? (09)

3. a) "Accuracy is not a good metric when the goal is to discover something that is rare." – Justify the statement with example. Propose a metric to solve the drawbacks of accuracy. (10)
   b) Given the following short movie reviews, each labeled with a genere, either comedy or action: (10)
      i) fun, couple, love, love ⇒ comedy
      ii) fast, furious, shoot ⇒ action
      iii) couple, fly, fast, fun, fun ⇒ comedy
      iv) furious, shoot, shoot, fun ⇒ action
      v) fly, fast, shoot, love ⇒ action

      Consider a new document D: fast, couple, shoot, fly. Compute the most likely class for D.
   c) Find the context free rules and hence the Context Free Grammar (CFG) for the following English sentences: (15)
      i) I want a morning flight.
      ii) I want a flight from Ontario to Chicago.
      iii) Show me the cheapest fare that has lunch.
      iv) Do any of these flights have stops?
      v) Which flights serves breakfast?

4. a) Consider the following grammar in CNF. (10)

$$S \rightarrow AB \mid BC$$
$$A \rightarrow BA \mid a$$
$$B \rightarrow CC \mid b$$
$$C \rightarrow AB \mid a$$

   Is '*baaba*' in L(G)? Explain your answer using CYK algorithm.
   b) Define shallow parsing. What are the applications of shallow parsing? (05)
   c) Define Probabilistic Context Free Grammar (PCFG). Consider the following PCFG. (12)

$$S \rightarrow NPVP \mid AuxNPVP \mid VP \; [0.8 \mid 0.1 \mid 0.1]$$
$$NP \rightarrow Pronoun \mid Proper\text{-}noun \mid DetNominal \; [0.2 \mid 0.2 \mid 0.6]$$
$$Nominal \rightarrow Noun \mid NominalNoun \mid NominalPP \; [0.3 \mid 0.2 \mid 0.5]$$
$$VP \rightarrow verb \mid verbNP \mid VPPP \; [0.2 \mid 0.5 \mid 0.3]$$
$$PP \rightarrow PrepNP \; [1.0]$$
$$Det \rightarrow the \mid a \mid that \mid this \; [0.6 \mid 0.2 \mid 0.1 \mid 0.1]$$
$$Noun \rightarrow book \mid flight \mid meal \mid money \; [0.1 \mid 0.5 \mid 0.2 \mid 0.2]$$
$$verb \rightarrow book \mid include \mid prefer \; [0.5 \mid 0.2 \mid 0.3]$$
$$Pronoun \rightarrow I \mid he \mid she \mid me \; [0.5 \mid 0.1 \mid 0.1 \mid 0.3]$$
$$Proper\text{-}noun \rightarrow Houston \mid NWA \; [0.8 \mid 0.2]$$
$$Prep \rightarrow from \mid to \mid on \mid near \mid through \; [0.25 \mid 0.25 \mid 0.1 \mid 0.2 \mid 0.2]$$

      i) Find the probability of the sentence "book the flight through Houston".
      ii) Using the disambiguation algorithm select the proper parse tree.
   d) What are the stages of IR based question answering? Explain. (08)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define Natural Language Processing (NLP). What are the major areas of research and development of NLP? (10)

   b) What does *n*-gram mean? Drive the equation of calculating the probability for *n*-grams model. (10)

   c) Consider the following corpus. (08)

   <s> I am Sam </s>
   <s> Sam I am </s>
   <s> I am Sam </s>
   <s> I do not like green eggs and Sam </s>

   Using a Bigram Language model with add-one smoothing, what is $P(sam \mid am)$? Include <s> and </s> in your counts just like any other token.

   d) What is absolute discounting? What is its advantages? (07)

6. a) What is closed class and open class of Part-of-Speech (POS)? Explain with example. (08)

   b) Discuss about Rule-Based POS tagging. Write the ADVERBIAL-THAT RULE. (12)

   c) For Hidden Markov Model (HMM) POS Tagging, using the following formula, find the equation of calculating tag transition probabilities. (08)

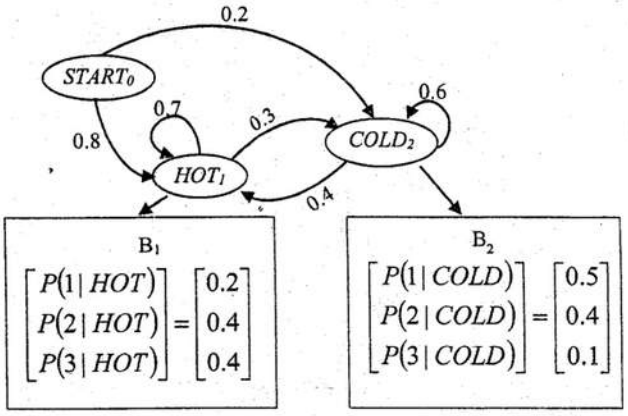
$$\hat{t}_1^n = \arg\max_{t_1^n} P(t_1^n \mid w_1^n)$$

   d) Consider the sentence: "Secretariat/NNP is/BEZ expected/VBN to/TO race/? Tomorrow/NR". The word "race" is often used as VB or NN. Given the probabilities below, find the right POS tag for the word "race". (07)

$P(NN \mid TO) = 0.00047, P(VB \mid TO) = 0.83, P(race \mid NN) = 0.00057, P(race \mid VB) = 0.00012,$
$P(NN \mid VB) = 0.0027, P(NR \mid NN) = 0.0012.$

7. a) HMM characterized by three fundamental problems. Name and discuss about the problems. (09)

   b) Given a sequence of ice-cream observations 313 and an HMM $\lambda = (A, B)$ in the following figure, find the best hidden weather sequence $Q(like\ H\ H\ H)$. (12)



   c) Define the term odds for logistic regression. Show that the observation should be labeled true if $\sum_{i=0}^{N} w_i f_i > 0$. (09)

   d) Write the three-steps of Forward Algorithm. (05)

8. a) Name and discuss about the types of TTS. (06)

   b) Speech Synthesis perform text to waveform mapping in two-steps. Name and discuss about the steps. Using Hourglass Metaphor. (12)

   c) What is Homograph disambiguation? What are the problems of CMU? How does UNISYN overcome the problems of CMU? (10)

   d) Define text normalization. Why does text normalization important for Speech Synthesis? (07)